

Program overview

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Year 2023/2024
Organization Applied Sciences
Education Master Applied Physics

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Applied Physics 2023							
Core Programme AP 2023							
Obligatory Core Modules 2023							
AP3001	Mathematical Methods for Physics	9					
AP3001-CA	MMP Complex Analysis	3					
AP3001-FE	MMP Finite Elements	3					
AP3001-PDE	MMP Partial Differential Equations	3					
WM0320TU	Ethics and Engineering	3					
Thesis Project 2023							
AP3902	Master Thesis Applied Physics	48					
AP3902-MTP	Master Thesis Project	42					
AP3902-PMT	Preparation for the Master Thesis	6					
General Advanced Physics Modules 2023							
AP3021	Advanced Statistical Mechanics	6					
AP3032	Continuum Physics	6					
AP3051	Advanced Quantum Mechanics	6					
AP3071	Advanced Electrodynamics	6					
Track Related Modules 2023							
Track Physics for Energy							
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AP3082	Computational Physics	6					
AP3141	Environmental Physics	6					
AP3211	Advanced Solid State Physics	6					
AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics	6					
AP3333	Physics of Energy Materials	6					
AP3341	Nuclear Reactor Physics	6					
AP3352	Introduction to Nuclear Science and Engineering	6					
CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3					
EE4670	PV Materials Processing and Characterization	4					
EE4680	Photovoltaic Modelling	4					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
ET4379	Photovoltaic Lab Course	4					
ME45203	Electrolysers, Fuel Cells and Batteries	4					
SET3110	Energy Storage in Batteries	4					
Track Physics for Fluids Engineering							
AE4180	Flow Measurement Techniques	3					
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AP3082	Computational Physics	6					
AP3141	Environmental Physics	6					
AP3171	Advanced Physical Transport Phenomena	6					
AP3181	Applied Multiphase Flows	6					
AP3551	Computational Multiphase Flow	6					
AP3563	Water in the Atmosphere	5					
CH3051	Applied Transport Phenomena (ATP)	4					
CH3153	Molecular Transport Phenomena	4					
CH3412	Biological Transport Phenomena	4					
CH3421	Computational Transport Phenomena	6					
ME45001	Advanced Heat Transfer	4					
ME45030	Turbulence	5					
ME45042	Advanced Fluid Dynamics	5					
ME45190	Chaos in Dynamical Systems	3					
WI4011-17	Computational Fluid Dynamics	6					
Track Physics for Health And Life							
AP3061	Acoustic, Elastic and Electromagnetic Waves	6					
AP3082	Computational Physics	6					
AP3122	Advanced Optical Imaging	6					
AP3132	Advanced Digital Image Processing	6					
AP3163	Physics of Biological Systems	6					
AP3232	Medical Imaging Signals and Systems	6					
AP3352	Introduction to Nuclear Science and Engineering	6					

AP3371	Radiological Health Physics	6	
AP3511	Biophysics	6	
AP3531	Acoustical Imaging	6	
AP3582	Medical Physics of Photon and Proton Therapy	6	
CH3412	Biological Transport Phenomena	4	
CH3764	Nuclear Medicine	4	
CH3771	Nuclear Chemistry	6	
NB4070	Soft Matter	6	
NB4160	Engineering of living systems	3	
Track Physics for Instrumentation			
AE4896	Space instrumentation	4	
AP3061	Acoustic, Elastic and Electromagnetic Waves	6	
AP3082	Computational Physics	6	
AP3091	Elementary Particles	6	
AP3113	Quantum Optics	6	
AP3122	Advanced Optical Imaging	6	
AP3132	Advanced Digital Image Processing	6	
AP3152	Optics for Lithography	6	
AP3222	Nanotechnology	6	
AP3243	Lasers and Photodetectors	3	
AP3252	Electron Microscopy Characterization of the Nanoscale	3	
AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics	6	
AP3352	Introduction to Nuclear Science and Engineering	6	
AP3382	Advanced Photonics	6	
AP3391	Geometrical Optics	6	
AP3401	Introduction to Charged Particle Optics	6	
AP3412	Experimental Techniques in Optics	3	
AP3531	Acoustical Imaging	6	
AP3701	Submm and terahertz physics and applications	3	
EE4745	Superconducting Astronomical Instrumentation	5	
ME46310	Opto-Mechatronics	4	
SC42030	Control for High Resolution Imaging	3	
SC42065	Adaptive Optics Design Project	3	
Track Physics for Quantum Devices and Quantum Computing			
AP3082	Computational Physics	6	
AP3101	The Interpretation of Quantum Mechanics	3	
AP3113	Quantum Optics	6	
AP3211	Advanced Solid State Physics	6	
AP3222	Nanotechnology	6	
AP3252	Electron Microscopy Characterization of the Nanoscale	3	
AP3261	Mesoscopic Physics	6	
AP3281	Quantum Transport	6	
AP3303	Applications of Quantum Mechanics	3	
AP3421-PR	Quantum Information Project	2	
AP3432	Quantum Hardware 1 - Theoretical Concepts	4	
AP3442	Quantum Hardware 2 - Experimental State of the Art	4	
AP3452	Quantum Error Correction	4	
AP3472	Modelling of Superconducting Devices	4	
AP3681	Fairy tales of theoretical physics	6	
CS4090	Quantum Communication and Cryptography	5	
QIST4300	Qubit Dynamics and Quantum Control	4	
QIST4310	Fundamentals of Quantum Information	4	
QIST4400	Quantum Computer Architecture	5	
General Electives 2023			
4403TGR64	Theory of General Relativity	6	
AP3652	Electronics for Physicists	3	
AP3751	Artificial Intelligence for Physicists	4	
AP3831	Systems Engineering for Physicists	3	
CS4195	Modeling and Data Analysis in Complex Networks	5	
CS4220	Machine Learning 1	5	
IFEEMCS4250	Statistical Learning for Engineers	4	
IN4049TU	Introduction to High Performance Computing	6	
LM3692	iGEM	15	
WI4201	Scientific Computing	6	
WI4260TU	Scientific Programming for Engineers	3	
WI4430	Martingales, Brownian Motion	6	
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5	
WI4771TU	Object Oriented Scientific Programming with C++	3	
Orientations AP 2023			
Orientation Research and Development (R&D) 2023			
Non-Academic Internship			
AP3911	Work Placement (Internship)	18	
Design Project			
AP3831	Systems Engineering for Physicists	3	
		15	

AP3841	Systems Engineering Company Project		
Electives			
Societal (S-list) Modules			
AS3111	ATHENS	2	
AS3132	Art, Empathy and Ethics	5	
TPM001B	Sociotechnology of Future Energy Systems	5	
TPM008A	Summer School Health Entrepreneurship	3	
TPM023B	Cost-Benefit Analysis: Theory and Application	4	
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
TPM401A	Technology Entrepreneurship and Innovation	5	
TPM402A	Technology Entrepreneurship and Health	5	
TPM403SET	Technology Entrepreneurship and Sustainability	4	
TPM404B	Technology Entrepreneurship and Internationalization	5	
TPM405A	Patent Law and Patent Policy	5	
TPM406A	Corporate Entrepreneurship and Start-ups	6	
TPM411A	Idea to Start-up IT & AI	5	
TPM412A	Idea to Start-up Health & Life Sciences	5	
TPM413A	Idea to Start-up Energy & Sustainability	5	
TPM414A	Idea to Start-up Deep Tech	5	
TPM415B	Idea to Start-up - Climate Action	5	
TPM416A	Turning Technology into Business	6	
TPM420A	Ready to Start-up	6	
TPM425B	Start-up Challenge	3	
WM0203TU-Eng	Oral Presentations	2	
WM1115TU	Dutch Elementary 1	3	
WM1116TU	Dutch Elementary 2	3	
WM1117TU	Dutch Intermediate 1	3	
WM1135TU	Advanced English for the University	3	
Orientation Study Abroad 2023			
Orientation Education 2023			
Orientation Education - Basis (Ed1)			
SL3462	Educational Sciences	6	
SL4201	Pedagogy of STEM education	4	
SL4202	Professional Learning Community	1	
SL4220	Physics Teaching Methodology	4	
SL4225	Physics Foundation Teaching Placement	15	
Orientation Education - Verdieping (Ed2)			
SL3012	Personal Professional Development	3	
SL4205	Advanced Professional Learning Community Professional Learning Community	1	
SL4301	Implementation of Education	9	
SL4320	Advanced Physics Teaching Methodology	5	
SL4325	Advanced Physics Teaching Placement	12	
Orientation Management of Technology (MoT) 2023			
1st semester MoT Modules			
MOT111A	Financial Management	5	
MOT112A	Economic Foundation	5	
MOT121A	Leadership and Technology Management	5	
MOT131A	Emerging and Breakthrough Technologies	5	
MOT141A	Research Methods	5	
MOT142A	Epistemology and Ethics of Business Analytics	5	
2nd semester MOT Modules			
MOT113A	Technology Dynamics	5	
MOT122A	Digital Business Process Management	5	
MOT123A	Inter- and Intra-Organizational Decision-Making	5	
MOT132A	Technology, Strategy and Entrepreneurship	5	
MOT133A	High Tech Marketing	5	
MOT143A	Business Analytics	5	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Master Applied Physics 2023

Director of Studies	Dr. A.J.L. Adam
Program Coordinator	Ir. A.J.W. Haket
Program Title	Applied Physics
EC Program	120
Introduction 1	Note: use the menu at the left for information on each subset of the programme.
Program Structure 1	The Applied Physics programme is a two-year master programme and comprises 120 EC. The programme has a core-orientation structure. Within this structure, there is a choice of research tracks:

Physics for Energy,
Physics for Fluids Engineering,
Physics for Health and Life,
Physics for Instrumentation,
Physics for Quantum Devices and Quantum Computing.

CORE PROGRAMME

The Applied Physics core programme comprises 90 EC and has the same structure for all tracks and students:

12 EC of compulsory modules: Mathematical Methods for Physics (9EC) and Ethics and Engineering (3EC).
12 EC of G-list modules: choose two (out of four) general advanced physics modules. These modules aim at depth as well as breadth in general physics knowledge.
12 EC of T-list modules: technical and science subjects related to the tracks. T-list modules are more specialised than G-list modules. They are representative for the research areas of the physics departments.
6 EC chosen from subjects on the G-list, T-list or the list of General, not track-related, electives.
48 EC Masters Thesis Project: to be carried out in a research section within the faculty of Applied Sciences or in an affiliated group.

ORIENTATIONS

Combining the core programme with a 30 EC orientation completes the master programme. The orientation allows for a broadening of knowledge or for additional depth. Students may opt for:

Research and Development (R&D) an Industrial Internship, a group design project, and/or 12 EC of additional electives chosen from subjects on the G-list, T-list, the list of General electives, or (with a maximum of 6 EC) Societal modules.
Education (Ed1/Ed2) get a Dutch secondary school qualification.
Management of Technology (MoT) consists of (either the first or) the second semester of the MSc MoT programme.
Study-abroad (SA) a semester, 30 EC of modules, at a foreign university; optionally including a research project.

SPECIAL PROGRAMMES

Special programmes, which imply restrictions on the Applied Physics core programme, are:

Casimir pre-PhD - a programme in collaboration with Leiden University, linked to the Physics for Quantum Devices and Quantum Computing track or to research done in the Bio-Nanoscience department. It focuses on educating students for a PhD position within the Kavli Institute of Nanoscience at TU Delft, the Leiden Institute of Physics (LION), or elsewhere.

Delft Jena special programme on Photonics - a programme in collaboration with the Friedrich Schiller University Jena, linked to the Physics for Instrumentation track. Students do the second year of the programme, including the thesis project, at the Abbe Center of Photonics of the Friedrich Schiller University in Jena.

PROGRAMME ADDITIONS

Honours Programme: This is an additional individual challenging programme of at least 20 EC on top of the regular programme for students with a better than average performance in their MSc programme (>7.5 weighted average and no study delay). Approval by the director of studies is needed. The full Applied Physics programme including the additional honours programme must be finished according to schedule.

Double degree programmes: These are three year programmes: Applied Physics Management of Technology, Applied Physics Applied Mathematics, or an individually approved combination of Applied Physics with another MSc programme. FORMAL PERMISSION TO START A DOUBLE DEGREE PROGRAMME IS ALWAYS REQUIRED IN ADVANCE !!

REGULATIONS

See <http://www.tnw.tudelft.nl/regulations> (or Brightspace) for the official regulations of the programme:
- Teaching and Examination Regulations 2023-2024 ("OER")
- TER Appendix - Programme Specifics MSc Applied Physics 2023-2024
- Rules and Guidelines of the Board of Examiners 2023-2024 ("RRvE").

Fail or Pass Regulation

The student meets the requirements for the degree audit once the following have been met:
a. a result has been earned for all required modules: a mark, a pass (v) or an exemption (vr);
b. none of the marks may be lower than 6.0.

With Honours Regulation

The designation with distinction for Masters degree audits.

1. A student can receive the designation with distinction for the Masters degree audit if the Board of Examiners decides to grant this distinction and the following requirements have been met:
a. the weighted average of the results of the courses not including the MSc final project is at least 8.0; passes (v) and exemptions (vr) will not be taken into consideration;
b. the number of credits for the courses for which a pass (v) has been earned or for which an exemption (vr) has been granted may not exceed 40 credits in total;
c. The result for the MSc final project is at least 9.0;
d. None of the marks may be lower than 7.0;

e. The students degree programme has not taken longer than two years, calculated from the date of the first examination, taking into account any study delay that falls under the Graduation Support Scheme.

2. In special cases the Board of Examiners may decide to grant the designation with distinction to a student who does not meet the requirements referred to in subsection 1 if the student in question has shown exceptional skills in the degree programme.

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Core Programme AP 2023

EC Program	90
Program Structure 1	The Applied Physics core programme comprises 90 EC and has the same structure for all tracks and students: 12 EC of compulsory modules: Mathematical Methods for Physics (9EC) and Ethics and Engineering (3EC). 12 EC of G-list modules: choose two (out of four) general advanced physics modules. These modules aim at depth as well as breadth in general physics knowledge. 12 EC of T-list electives: technical and science subjects related to the tracks. T-list modules are more specialised than G-list modules. They are representative for the research areas of the physics departments. 6 EC chosen from subjects on the G-list, T-list or the list of General, not track-related, electives. 48 EC Masters Thesis Project: to be carried out in a research section within the faculty of Applied Sciences or in an affiliated group.

Obligatory Core Modules 2023

Contact for Students	Arno Haket (arno@tudelft.nl)
Program Structure 1	The compulsory courses are Mathematical Methods for Physics (AP3001, 9EC) and Ethics and Engineering (WM0320TU, 3EC).
Options	The Board of Examiners may grant an exemption for a compulsory course to students who have successfully completed an examination in a system of higher education within or outside the Netherlands that corresponds to the examination for which the exemption has been requested in terms of content and level. Note that it is not possible to get an exemption in the MSc programme based on courses passed in your bachelor programme without doing an alternative MSc course module. ===== Students that have passed the Complex Analysis module EE2M11, AM2040, TW2040 or WI2602 in their bachelor programme at TU Delft, have two options with respect to AP3001: 1: Complete the three parts of AP3001, including the Complex Analysis part. The course material of the CA part overlaps with the BSc modules, so its less challenging but not forbidden. 2: Skip the Complex Analysis part of AP3001 and choose a different course module of at least 3EC (either a module from the MSc Applied Physics programme, a math module, a bridging module, or another MSc level module). Students that have passed the Partial Differential Equation modules WI3150TU and WI3151TU/WI4150TU, AM2070, TW2070, or WI2607 in their bachelor programme at TU Delft have two options with respect to AP3001: 1: Complete the three parts of AP3001, including the PDE part. The course material of the PDE part overlaps with the BSc modules, so its less challenging but not forbidden. 2: Skip the PDE part of AP3001 and choose a different course module of at least 3EC (either module from the MSc Applied Physics programme, a math module, a bridging module, or another MSc level module). A commonly chosen option by students who did PDE in their bachelor but not CA, is to combine Complex Analysis (AP3001-CA, 3ec) with an extended version of Final Elements (WI4014TU, 6ec; or WI4205, 6ec). The Finite Element part of AP3001 is completely complementary to (i.e. has no overlap with) bachelor and minor courses taught at TU Delft. Contact the Applied Physics programme coordinator, Arno Haket, for the administrative paperwork required for customizing your programme. Two course modules of at least 3EC can of course be combined into one 6EC module. ===== An application for an exemption based on a course done at another university must be submitted to the Board of Examiners. Documentary evidence must be included (transcript, extended course description, and if possible an example exam to verify that the level of the course is comparable to ours).

AP3001	Mathematical Methods for Physics	9
Responsible Instructor	A. Palha da Silva Clérigo	
Responsible Instructor	Dr. B. van den Dries	
Responsible Instructor	Dr.ir. Y.M. Dijkstra	
Contact Hours / Week x/x/x/x	6/6/0/0 + 2/0/0/0 tutorials	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Basic Analysis and basic knowledge of complex numbers. Separation of variables, Sturm-Liouville eigenvalue problems, Eigenfunction Expansion, Fourier Series.	
Course Contents	This course comprises three subjects: Complex Analysis, Partial Differential Equations, and Finite Element Analysis. The individual parts have their own entries in the study guide under the codes AP3001-CA, AP3001-FE and AP3001-PDE, we refer to these entries for detailed information on the contents.	
Study Goals	The aim of the course is to acquaint the student with the results and methods of the three disciplines mentioned above and to show how these are used in Physics.	
Education Method	Lectures and practical work.	
Literature and Study Materials	See the individual entries in the study guide.	
Assessment	The final grade will be the average of the grades of the three subjects with the proviso that each partial grade must be 5.0 or higher. See the individual entries in the study guide for further details.	
Special Information	See Brightspace organization 'Master Applied Physics' if you did one or more parts in your Bachelor programme.	

AP3001-CA	MMP Complex Analysis	3
Responsible Instructor	Dr. B. van den Dries	
Contact Hours / Week x/x/x/x	4/0/0/0 + 2/0/0/0 tutorials	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A good understanding of the contents of bachelor courses in mathematics: calculus, advanced calculus, differential equations.	
Course Contents	The basics of Complex Analysis: mapping behaviour of analytic functions, harmonic functions as related to analytic functions, many-valued functions and branches thereof, complex integration, the theorems of Cauchy and their applications such as series representations of analytic functions, and evaluation of (real) integrals using residue calculus. There are also some applications to Fourier and (inverse) Laplace transforms and their relations to PDEs.	
Study Goals	The aim of overarching course is to acquaint the student with the results and methods of the three disciplines to show how these are used in Physics. For the Complex Analysis part this translates to: 1. Knowledge of the basic definitions, theorems and methods of Complex Analysis. 2. Ability to solve elementary problems using standard methods. 3. Ability to solve slightly non-standard problems using combinations of methods. More concretely: - The student knows and can use topological concepts such as open, closed, interior, boundary and accumulation point; - The student can test functions for continuity, complex differentiability, analyticity and harmonicity. This includes the use of the Cauchy-Riemann equations; - The student can describe mapping behaviour of complex functions, including multivaluedness and branching; - The student knows and can use properties of standard analytic functions such as polynomials, rational functions, power functions, exp, log, Log, trigonometric functions and their inverses and hyperbolic functions and their inverses; - The student can identify singular points of functions, find their type and residue (if applicable); - The student can express a function as Taylor and Laurent series, find the domain of convergence of such series, and perform operations on such series; - The student can apply various techniques to evaluate a contour integral, including direct evaluation, use of anti-derivatives, deformation of the contour, (generalized) Cauchy Integral Theorem, Cauchy Residue Theorem; - The student can apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's inequality, path indentation, trigonometric substitution; - The student can evaluate the inverse Laplace transform of a given function.	
Education Method	Lectures (4 hours/week) and exercise class (2 hours/week). In principle on campus.	
Literature and Study Materials	UPDATE The book that we used in previous years (see below) appears to be out of print. This year, we will use two books: Main source (also used in the exercise classes): "Complex Analysis with Applications", Asmar & Grafakos ISBN 978-3-319-94062-5 (hardcopy) ISBN 978-3-319-94063-2 (e-book) Secondary source: "Applied Complex Variables for Scientists and Engineers", Kwok ISBN 978-0-521-70138-9 (hardcopy) ISBN 978-0-511-77500-0 (e-book) Both books are freely available in digital form on the website of the TU Delft Library (Go to https://tudelft.on.worldcat.org/discovery and search for the titles of the books). ----- Book used previous years: Complex variables and applications, JW Brown and RV Churchill, McGraw-Hill 9th edition 2014 ISBN13: 9781259072772	
Assessment	3-hour written on-campus exam.	
Permitted Materials during Tests	none	
Remarks	NOTE for students not enrolled in MSc Applied Physics: this module is part of AP3001, Mathematical Methods for Physics. A valid final grade and credits are only obtained for the AP3001 course and not for this module on its own.	

AP3001-FE	MMP Finite Elements	3
Responsible Instructor	A. Palha da Silva Clérigo	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	(Partial) differential equations, analysis, linear algebra, numerical analysis.	
Course Contents	Covers the use of finite-element strategies to approximate the solution to partial differential equations (PDEs). Mathematical concepts are: Minimization problems, Euler-Lagrange equations, Ritz method, Weak formulation, Galerkin's finite-element method. We will treat the application of the finite-element method to both linear and nonlinear PDEs. In this course, we will limit ourselves to the application of linear triangular elements for two spatial dimensions.	
Study Goals	The participant will be able to construct and implement a finite-element method to approximate the solution of a partial differential equation. Furthermore, (s)he will be able to analyse the accuracy and reliability of the approximation of the solution.	
Education Method	Lectures and lab work	
Literature and Study Materials	J. van Kan, A. Segal, F. Vermolen. Numerical Methods in Scientific Computing. Available as open access book by the TU Delft: https://textbooks.open.tudelft.nl/textbooks/catalog/book/70	
Assessment	Two series of take-home assignments, lab work + report, written test. Final grade = $0.8 \cdot \min(10, \text{WrittenTest}) + (\text{TakeHome1} + \text{TakeHome2})/20 + 0.2 \cdot \text{LabWork}$; with the proviso that the grade for the written test must be 5.0 or higher.	
Permitted Materials during Tests		
Remarks	NOTE for students not enrolled in MSc Applied Physics: this module is part of AP3001, Mathematical Methods for Physics. A valid final grade and credits are only obtained for the AP3001 course and not for this module on its own.	

AP3001-PDE	MMP Partial Differential Equations	3
Responsible Instructor	Dr.ir. Y.M. Dijkstra	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Calculus, ordinary differential equations, Fourier series and Fourier transforms	
Course Contents	Partial differential equations (PDEs) are the mathematical representation of many phenomena in various branches of applied physics. Therefore it is important to be able to solve this type of equations and interpret the solution in terms of the physics they describe. This course provides an introduction into several analytical solution and analysis techniques for PDEs. These analytical solution techniques are often especially suitable for analysing the general physical behaviour of solutions to PDEs and are therefore indispensable tools in addition to numerical techniques (taught in the course AP3001-FE). Hence, the way these analytical techniques may be used to gain insight into the physical interpretation of solutions to PDEs are an important part of this course. The solution and analysis techniques that will be discussed include: - Separation of variables - Method of eigenfunction expansion - Homogenisation for nonhomogeneous PDEs - Fourier transform method - Method of characteristics - Green's functions Our main examples of PDEs to apply these techniques are the linear wave and heat equations in 1, 2, and 3 dimensions. Additionally we will discuss linear and nonlinear transport equations.	
Study Goals	At the end of the course the student: 1. can explain the use of several mathematical techniques for solving and analysing PDEs, including separation of variables, eigenfunction expansions, methods for nonhomogeneous PDEs, Fourier transforms, method of characteristics, and Green's functions. 2. is able to apply these techniques to problems at the level of an introductory textbook. 3. is able to interpret the solutions obtained in terms of their physical context. 4. is able to structure the solutions to above-mentioned problems in terms of the equation, boundary and initial conditions, motivation for the use of a certain solution method, calculations, final answer, and interpretation.	
Education Method	Lectures, tutorials, homework exercises, and computer exercises.	
Literature and Study Materials	R. Haberman, Applied Partial Differential Equations (with Fourier series and boundary value problems), Pearson Prentice Hall, 5th edition, New Jersey 2013. ISBN 0321828976.	
Assessment	The PDE part of the course AP3001 will be assessed through a written exam.	
Remarks	NOTE for students not enrolled in MSc Applied Physics: this module is part of AP3001, Mathematical Methods for Physics. A valid final grade and credits are only obtained for the AP3001 course and not for this module on its own. Stand-alone courses on PDEs aimed at students outside of Applied Mathematics are offered in WI3150TU, WI3151TU. Students in Applied Mathematics can take AM2070.	

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Instructor	M. Sand	
Co-responsible for assignments	M. Sand	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).	
Elective	Yes	
Category	MSc niveau	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Thesis Project 2023	
Prerequisites	As a rule, a student should have passed all bridging/homologation modules and AT LEAST 30EC from the obligatory (12EC), G-List (12EC), and T-list (12EC) modules in the core programme before embarking upon the master thesis work. There are no requirements with regard to completed courses for starting the Preparation for the Master Thesis.
Introduction 1	Contact the programme coordinator for additional information. The topic of the Thesis Project (6ec thesis preparation, 42ec thesis work) is related to the graduation track and is done in a research section of one of the departments of the faculty of Applied Sciences, in QuTech, or in an affiliated group. Affiliated groups are: For Physics for Energy: Energy Technology section (3mE), Photovoltaic Materials and Devices group (EEMCS); For Physics for Fluids Engineering: Clouds & Climate group (CiTG), Fluid Mechanics section (3mE), Multiphase Systems section (3mE); For Physics for Instrumentation: Centre for Systems and Control - Numerics for Control & Identification / Optics group (3mE), Dynamics of Micro and Nanosystems section (3mE), Experimental Astronomy Group (EEMCS), and the Nikhef lecturers teaching in our programme. The prior approval of the Board of Examiners should be obtained if the thesis work is performed outside the mentioned departments or affiliated groups. A supervisor from one of the departments or groups is required.

AP3902	Master Thesis Applied Physics	48
Responsible Instructor	Dr. A.J.L. Adam	
Project Coordinator	Ir. A.J.W. Haket	
Project Coordinator	L. van der Elst	
Contact Hours / Week x/x/x/x	Different / start each period	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The master thesis project consists of two separate stages: a 6 EC preparation, followed by 42 EC thesis work that includes writing the report and giving the final presentation.	
Study Goals	See the study goals for the two parts, AP3902PMT and AP3902MTP.	
Education Method	See the education methods for the two parts.	
Practical Guide	Enroll yourself in the Brightspace organization "Eindprojecten Administratie TNW / Thesis Office Applied Sciences" (https://brightspace-cc.tudelft.nl/course/1826/eindprojecten-administratie-tnw--thesis-office-applied-sciences) for all information regarding the thesis project, step-by-step guide, application form, grading criteria, etc.	
Assessment	The first part (AP3902PMT, 6 EC) is assessed and graded with a pass. The second part (AP3902MTP, 42 EC) leads to a mark between 1 and 10. The mark for AP3902 is determined by the mark obtained for AP3902MTP (provided a pass has been obtained for AP3902PMT). Note: if the first part is skipped it will be graded with a 0, counting for 12.5% of the final grade.	
Enrolment / Application	The thesis project will be reviewed according to the grading scheme on the Thesis Office Brightspace organization. Students are obligated to register the thesis project at the thesis office after completing the preparation part, prior to the start of the Masters Thesis Project work. Download the thesis application form from Brightspace organization 'Thesis Office Applied Sciences'. The application form has to be signed by your responsible supervisor. The supervisor is responsible for approving the thesis project. The programme coordinator must confirm that you can start with the project: once you and your supervisor have signed the application form, please mail a digital copy of the application form and the report of the Preparation of the Master Thesis part to the programme coordinator, Arno Haket (A.J.W.Haket@tudelft.nl) for approval. He'll send the form to the Thesis Office for further processing.	
Schedule	The expected time spent on the Preparation for the Master Thesis corresponds to 6 EC (1 month FTE); usually it is spread over a longer period of time, parallel to other courses. During the preparation stage meetings with the responsible supervisor are required for discussing the different issues. The time planning for the Master Thesis Project work is based on 7 months FTE, the equivalent of 42 EC. The time taken to complete courses the student still has to do, exams, regular holidays, and work have to be taken into account to arrive at the intended end date on which the final version of the thesis should be handed in. The time planning also takes into account known periods of absence of the supervisor and contains agreements on ways of communication and/or substitution during these periods. Absence of the supervisor cannot be used to justify delay. Neither does an impending publication, or more/better results. The time schedule is binding and contains a green light meeting two weeks prior to the end date for handing in the final version of the thesis. The schedule also contains the dates for two intermediate evaluations that are meant to give feedback to the student on her/his performance. Proposed timeslots are after a period equal to three months FTE (ca 40%) and five months FTE (ca 70%). The learning outcomes - as stated in the masters thesis grading scheme - should be used for these intermediate evaluations of the student.	

AP3902-MTP	Master Thesis Project	42
Responsible Instructor	Dr. A.J.L. Adam	
Project Coordinator	L. van der Elst	
Project Coordinator	Ir. A.J.W. Haket	
Contact Hours / Week x/x/x/x	Different / start each period	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	a. Practical and theoretical work in one of the research groups including interpretation and evaluation of the results. b. Oral and written presentation of the work.	
Study Goals	At the end of the project, should be able to - read and understand the international research literature; - work independently on an academic level in a R&D environment; - plan a substantial research project and meet deadlines; - work in a (possibly interdisciplinary) and multicultural team of experts and supporting technicians; - present his or her results in English by means of a presentation to an expert audience and a scientific report.	
Education Method	Individual Project carried out within a Research Group.	
Prerequisites	You may start your Master Thesis Project work if you: · have passed all bridging/homologation modules or other obligations from the bachelor programme; · have passed at least 30EC of the 36EC obligatory, G-List, and T-list modules in the core programme; and · have made a project plan with your thesis supervisor (Preparation for the Master Thesis, AP3902-PMT), and · have handed in the signed thesis registration form together with the project plan at the thesis office. There are no requirements with regard to completed courses for starting the Preparation for the Master Thesis.	
Assessment	The assessment of the student is based on the learning goals of the masters thesis project rather than on whether the planned research outcome has been realized. The final assessment consists of one mark, based on: - Scientific knowledge and understanding - Method and approach - Report - Presentation and defense - Competences A detailed grading scheme is available in Brightspace organization "Eindprojecten Administratie TNW / Thesis Office Applied Sciences" (https://brightspace-cc.tudelft.nl/course/1826/eindprojecten-administratie-tnw--thesis-office-applied-sciences). On behalf of the Board of Examiners, the responsible supervisor composes a suitable assessment committee. The committee consists of at least three examiners, all members of the scientific staff of research groups that contribute to the master's degree programme. The members of the assessment committee come from at least two different sections of the degree programme, at least one of the members belongs to the teaching staff of the degree programme (of a course other than the thesis project), and at least one of them is a full professor or an associate professor with 'ius promovendi'. Thesis projects carried out in affiliated groups outside the Faculty of Applied Sciences shall be assessed by a committee including at least one examiner from the Faculty of Applied Sciences. The thesis office reminds the responsible supervisor and student in time to have the green light meeting. If the responsible supervisor is confident that the student will pass the defence, the exact time and date of the final presentation and thesis defence are fixed and the members of the assessment committee confirmed at this meeting. Both the date/time and the committee are determined by the thesis supervisor after hearing the student. In principle the defence takes place within two weeks after the end date for handing in the final version of the thesis. The supervisor reports the date of the defence and the composition of the assessment committee to the thesis office. In exceptional cases, the board of examiners may be involved in setting the date and time. If the responsible supervisor is in doubt whether the student will pass the defence, he/she should discuss with the other members of the assessment committee whether to give green light or not. In the case that, in spite of the effort, the student is seriously failing, the responsible supervisor can decide at the green light meeting to cancel the planned defence if he/she is confident that the student won't pass the defence. The supervisor needs to deliver a written argumentation for this decision to the thesis office and the programme coordinator. Prior to this decision the student must be warned repeatedly, at least at the intermediate evaluation, to give him/her a chance to improve. After a negative decision, a choice must be made in consultation with the student between a delay (with a new binding time schedule with a maximum of three months) or a definitive cancellation of the thesis project. The thesis project can be delayed only once. Cancellation implies starting a new project. Note that an impending publication or more/better results are not valid reasons for an extension. In case of unforeseen personal circumstances (illness; pregnancy and childbirth; exceptional family circumstances; physical, sensory or other functional disabilities) the student can apply for an extension of the end date (on which the final version of the thesis report needs to be handed in) to the Board of Examiners, with a statement of the academic counsellor. Students are required to contact the academic counsellor as soon as possible after the particular circumstance has occurred. The information is treated confidentially by the academic counsellor.	
Enrolment / Application	Before the start of this stage of the thesis project the signed thesis project registration form must be handed in at the thesis office via the programme coordinator, together with the project plan (see AP3902-PMT).	
Remarks	See AP3902 for general information concerning the final project and the way it is organised.	
Schedule	The time planning is based on 7 months FTE, the equivalent of 42 EC. The time taken to complete courses the student still has to do, exams, regular holidays, and work have to be taken into account to arrive at the intended end date on which the final version of the thesis should be handed in. The time planning also takes into account known periods of absence of the supervisor and contains agreements on ways of communication and/or substitution during these periods. Absence of the supervisor cannot be used to justify delay. Neither does an impending publication, or more/better results. The time schedule is binding and contains a green light meeting two weeks prior to the end date for handing in the final version of the thesis. The schedule also contains the dates for two intermediate evaluations that are meant to give feedback to the student on her/his performance. Proposed timeslots are after a period equal to three months FTE (ca 40%) and five months FTE (ca 70%). The learning outcomes - as stated in the masters thesis grading scheme - should be used for these intermediate evaluations of the	

AP3902-PMT	<i>Preparation for the Master Thesis</i>	6
Responsible Instructor	Dr. A.J.L. Adam	
Project Coordinator	Ir. A.J.W. Haket	
Contact Hours / Week x/x/x/x	Different / start each period	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This module contains the first stages of your final project. It consists of - Reading background literature necessary for the project; - Acquiring necessary skills, such as learning to work with experimental set-ups, clean room and/or computer skills, and transferable/soft skills; - Identifying areas, in collaboration with the supervisor, where extra training is necessary, for example English writing skills, presenting, a particular course, etc.; - Formulating the research goal of the project: which question will be addressed and what method(s) will be applied to arrive at the answer; - Making a planning, including anticipating pitfalls and how these will be dealt with in order to prevent delays (e.g. equipment failure, delayed delivery of materials, sharing/availability of facilities, unexpected results, etc); - Writing a project plan. The result of the Preparation for the Master Thesis is a project plan which addresses the project goal, the name of your supervisor(s) and others involved in the project, the track of the Masters programme, knowledge (e.g. courses done) and skills that you will use in the project, required training for skills that need improvement, anticipated pitfalls and how these will be dealt with, agreements about supervision (such as the meeting frequency between the student and the responsible supervisor), and a time planning. The report typically contains between 4 and 7 pages.	
Study Goals	At the end of this module you should - be aware of the project goal and plan; - have made a realistic planning; - know the pitfalls that can result in delays in the project and how to deal with these; - have an overview of the relevant literature in the field of your project; - have written a report (in English) described in the course content.	
Education Method	In this module, you complete the work by discussing the different issues with the supervisor and reading literature on your own. During the preparation stage meetings with the responsible supervisor are required for discussing the different issues. All members of the research group you are in can play a role. Depending on the project, learning particular skills (experimental and/or computer skills, or also transferable/soft skills). The expected time spent on the Preparation for the Master Thesis corresponds to 6 EC (1 month FTE); usually it is spread over a longer period of time, parallel to other courses.	
Assessment	The assessment is based on your report, and is done by the responsible supervisor. The result will be a pass or a no-pass. You can only start the main project (AP3902MTP) after you have passed this module. Mail the report to the programme coordinator, Arno Haket (A.J.W.Haket@tudelft.nl) to register the result in Osiris. Make clear that the report has been discussed with and approved by your responsible supervisor.	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

General Advanced Physics Modules 2023

Introduction 1

G-list, general advanced, physics modules aim at breadth as well as depth in general physics knowledge, following up on the bachelors programme. At least two modules must be chosen.
Although for some tracks there seems to be an obvious relation between that track and a particular G-list module, there is no restriction whatsoever on your choice of G-list modules.

AP3021	Advanced Statistical Mechanics	6
Responsible Instructor	T. van der Sar	
Instructor	Prof.dr. J.M. Thijssen	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	- Bachelor (applied) physics or equivalent. - Complex analysis	
Course Contents	This course covers the theory of statistical mechanics at the intermediate level, in particular: 1. The formalism of statistical mechanics and its relation to thermodynamics, Ensemble theory 2. Systems of non-interacting particles - classical, quantum 3. Classical systems of interacting particles - perturbation theory: Cluster expansion - Phase transitions: van der Waals, mean field theory, exact solutions - Phase transitions: scaling and renormalization 4. Fluctuations	
Study Goals	Course goals are: 1. Students are able to predict the behaviour of macroscopic, in-equilibrium systems composed of non-interacting particles using the formalism of statistical mechanics. 2. Students are able to analyze the behaviour of macroscopic, in-equilibrium systems composed of interacting particles using various approximation methods of statistical mechanics. 3. Students are able to analyze fluctuations of in-equilibrium systems, as well as the processes by which out-of-equilibrium systems evolve towards equilibrium.	
Education Method	Students prepare the material for each week before the first lecture. For this, they use lecture notes and books, and youtube videos developed by the lecturer. Some control questions requiring simple calculations or the interpretation of particular results are part of the preparation. During the first lecture of the week, the material will sometimes be summarized and students are invited to ask questions on points they did not yet fully grasp. The remaining time of the four hours per week, the students work on problems in groups, while the lecturer and the TA's provide help. Sometimes, some difficult points will be discussed by the whole group. The aim is that every student knows how to finish the problem at the end of these exercise classes for the week. The student can then work out a detailed solution at home and hand this in. The homework which is handed in counts to the final grade.	
Literature and Study Materials	The course follows Lecture notes on Statistical Mechanics by Jos Thijssen From the lecture notes, youtube videos with explanations on many parts of the lecture notes are accessible by mouse click or reading a QR code They can be found on brightspace. Additional textbooks: Statistical Mechanics in a Nutshell, by L. Peliti; Princeton University Press (2011). Other useful textbooks: M. Kardar: Statistical Mechanics of Particles Cambridge University Press, 2007 Statistical Mechanics of Fields Cambridge University Press, 2007	
Assessment	The course puts strong emphasis on problem solving skills. Homework is an important element of the course. The grade for the homework can contribute a maximum of 1.5 to the final grade according to the formula: $R = (1 - 0.015 \cdot H) \cdot E + 0.15 \cdot H$ where R is the final grade, E is the result for the exam (which may include the midterm, see below) and H is the result for the homework. There is a midterm exam. Students having taken the midterm exam can take a final on the second part of the course material only. There is also an option to take a final exam on all the course material (for students missing the midterm or scoring a bad mark there). If a student chooses to do the final exam on the second part of the material, the exam grade E is calculated according to $E = 0.35 M + 0.65 F$, where M is the grade for the midterm and F is the result of the final exam. The exam grade E for a student who has decided to the final exam on all the material is just the result of that final exam F: $E = F$ The bonus based on the problems keeps its validity for the retake following the semester in which the course was taught. The score for the problems is reset to 0 for the next time the course is taught. There is a retake in the spring. This retake covers all material, i.e. it is a 'full' exam.	
Permitted Materials during Tests	closed book exam. (Non-graphical/non programmable) calculator allowed.	

AP3032	Continuum Physics	6
Responsible Instructor	Prof.dr.ir. C.R. Kleijn	
Instructor	Dr.ing. S. Kenjeres	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	undergraduate calculus, differential equations and transport phenomena/fluid dynamics.	
Course Contents	The continuum description of the mechanics of solids and liquids is discussed in a unified approach using tensor analysis. Deformations of a body are studied and the strain tensor and the rate-of-deformation tensor are introduced. This is followed by a discussion of the stress tensor, and the equations of motion of a body are derived based on the balances of linear and angular momentum. For linear elastic solids the following topics are discussed: constitutive equation, Navier equations, energy principles. For Newtonian fluids we discuss the stress tensor, Navier-Stokes equations, and the first and second law of thermodynamics. The lecture series is then continued with a number of special topics a.o. multicomponent reacting flow (combustion), thermal convection and magneto-hydrodynamics (MHD).	
Study Goals	Being able to formulate and apply conservation laws (mass, momentum, energy) and constitutive equations (stress-strain relations, equations of state,) for linear elastic solids and Newtonian fluids. Being able to apply tensor algebra and tensor calculus in the formulation and analysis of laws describing continuum mechanical systems. Having knowledge of the relevant phenomena in multiphysics continuum mechanical systems (reacting flows, thermal convection, magnetohydrodynamics). Being able to solve simple problems in continuum mechanics (kinematics, kinetics, linear elasticity, Newtonian fluid mechanics, special topics covered in the course). Being able to identify some relevant aspects of physics of continuous media in practical situations (relevant phenomena, dimensionless numbers, regimes, simple estimates).	
Education Method	Theory presented in lectures. Practice sessions to make exercises. Assignments to be made at home.	
Literature and Study Materials	The general part of the course is based on the following book: J.N. Reddy, Principles of Continuum Mechanics, 2nd edition (2nd edition only!!), Cambridge University Press, 2018, ISBN: 978-1-107-19920-0 (hardback) For the lectures on special topics (reactive flows, heat transfer and thermal convection, magnetohydrodynamics), slides and/or notes are made available. Additional textbooks of interest: - For the lectures on continuum mechanics: J. N. Reddy, An Introduction to Continuum Mechanics Cambridge University Press; October 2007 ISBN 9780511473883 - For the lectures on reactive flows: K. K. Kuo, Principles of Combustion, 2nd Edition, John Wiley and sons, ISBN 10: 0471046892 and ISBN 13: 9780471046899 - For the lectures on Newtonian fluid mechanics: P.K. Kundu, I.A. Cohen and D.R. Dowling, Fluid Mechanics, 5th Edition, Academic press, 2012,	
Assessment	On-campus or Online exam at end of the second period. Final mark partially (6x2.5%) based on the performance in 6 optional homework assignments, if the assignment grade is higher than the exam grade. Resit exam at the end of the third period. Homework assignments are *not* valid for the resit exam, and also not for next year exams.	
Permitted Materials during Tests	Closed book on-campus or Open book online exam (depending on Covid-19 situation)	
Studyload/Week	12 hours per week (4 scheduled, 8 homework) during 14 weeks.	

AP3051	Advanced Quantum Mechanics	6
Responsible Instructor	Prof.dr. Y.V. Nazarov	
Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Quantum mechanics at bachelor (applied) physics level or equivalent.	
Course Contents	Second quantization, Magnetism, Superconductivity and Superfluidity, Quantization of fields, Dissipative Quantum Mechanics, Relativistic Quantum Mechanics.	
Study Goals	To master key concepts of advanced quantum mechanics, in particular the concept of second quantization. In particular, 1. Knowledge of the topics covered, all within the theory of advanced quantum mechanics 2. Being able to solve elementary problems addressing standard procedures of the theory mentioned in (1) 3. Being able to solve more advanced problems addressing the theory mentioned in (1), combining mathematical skills and physical insight 4. Relate the theory mentioned in (1) to experiments.	
Education Method	The course consists of lectures, problem-solving sessions with student presentations, and compulsory homework assignments.	
Literature and Study Materials	Lecture Slides. The book: Advanced Quantum Mechanics: A practical study guide, Yuli V. Nazarov and J. Danon, Cambridge University Press, 2013.	
Assessment	During the semester, the students are supposed to make a presentation, and homework. The students who did not make a presentation got a more difficult version of the written exam. The grade for homework accounts for 15% of the final grade. The written final exam includes problems to solve and multiple choice questions. There is an opportunity for an oral exam of similar structure in exceptional cases.	
Permitted Materials during Tests		

AP3071	Advanced Electrodynamics	6
Responsible Instructor	Dr. A.J.L. Adam	
Instructor	Dr. S.W.H. Eijt	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Electromagnetism at the level of the book Introduction to Electrodynamics by D.J. Griffiths, as well as basic elements of vector calculus (curvilinear coordinate systems, integration over surfaces and volumes in such systems, gradient operator).	
Course Contents	Maxwell equations. Time dependent charge and current distributions. Multipole expansion. Potentials and Gauges. Conservation Laws. Electromagnetic radiation. Electromagnetic waves in media. Electrodynamics and relativity. Liénard-Wiechert potentials and fields. Synchrotron radiation. Classical and relativistic motion of charged particles in electromagnetic fields.	
Study Goals	The student who passes this course should have a working knowledge in electrodynamics at a graduate level. The student will be able to solve modern problems in classical electrodynamics (including the use of its covariant foundation), with topics ranging from electrostatics to radiation by relativistic particles and the motion of (relativistic) particles in electromagnetic fields. At the end of the course, the student will have a broad view of the theory and modern applications of electrodynamics, and he or she is able to solve problems pertaining to the topics covered in the course.	
Education Method	Lectures, tutorials and homework exercises. Students are expected to be active in problem solving. Methods for solving the exercises are subsequently discussed in the tutorial sessions.	
Literature and Study Materials	We will be teaching based on the book of Zangwill: A. Zangwill, Modern Electrodynamics, Cambridge University Press, Cambridge, UK (2013). It is a textbook on advanced electrodynamics which combines the necessary mathematics with a clear attention to the physics. A list of the relevant chapters directly related to the lectures will be made available at the beginning of the course and at each lecture. Handouts of the material covered in the lectures will be made available on Brightspace. Some derivations will be only done on the blackboard during the lecture.	
Assessment	Assessment is based on a midterm exam (on the first part of the course materials) and a final written exam (on the second part of the course materials). There is also an option to take a final exam on all the course material (for students missing the mid-term or scoring a bad mark there). If a student chooses to do the final exam on the second part of the material, the exam grade E is calculated according to $E = 0.50 M + 0.50 F$, where M is the grade for the midterm and F is the result of the final exam. The exam grade E for a student who has decided to do the final exam on all the material is just the result of that final exam F: $E = F$ A retake exam is scheduled in the spring, that covers all the material of the course, i.e. it is a 'full' exam. Exercises are an important part of the course. A bonus can be received based on the homework exercises (max 0.7). The bonus based on the problems keeps its validity for the retake following the semester in which the course was taught. The score for the problems is reset to 0 for the next issues of this course.	
Permitted Materials during Tests	Only a scientific calculator and pencils (with various colors, but not red) are allowed	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Related Modules 2023

Introduction 1	T-list modules are more specialised than G-list modules. They are technical and science subjects related to and recommended for at least one of the tracks. They are representative for the research areas of the physics departments. You should take at least 12EC from the T-list of the track of your choice.
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Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Physics for Energy

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AP3082	Computational Physics	6
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.R. Akhmerov	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor physics.	
Course Contents	This course starts with an introduction into methods and techniques in computation that are viable today. Following this, several projects in the field of computational physics are executed by the students, who work in pairs. Projects are on molecular dynamics, (quantum) Monte-Carlo calculations, and a final project of own choice. The course projects are close to the research level.	
Study Goals	At the end of this course a student will be able to 1. describe standard physical simulation algorithms. 2. implement these algorithms in a well-structured computer code. 3. apply these algorithms to calculate physical properties and interpret the outcome. 4. verify correctness of the simulation outcome and use this analysis to improve the code. 5. organize and perform joint programming work. 6. present the results according to the requirements of scientific communication.	
Education Method	Before the start of the semester we offer an optional crash course of two days where you learn how to program in python, use version control and store codes online. The course itself is project-based. This means that it contains little formal teaching, but most of the knowledge is acquired hands-on by implementing, running and analyzing the results of computer simulations. Students are expected to learn the necessary background theory by studying literature and from contact with the lecturers. Another important element is code review, where the students read and comment on each other's code.	
Literature and Study Materials	The textbook Computational Physics by J.M. Thijssen (2nd edition, Cambridge University Press) is useful. Extensive use is made of online resources (course material, cloud storage portals like github/gitlab). In addition, papers from modern research literature are used.	
Prerequisites	The bachelor degree must be finished before the start of the course.	
Assessment	Assessment is based in the results of the three projects, consisting of a paper (first two projects) or oral report (final project), and the simulation code and development history. The grade of each project is rounded to two digit accuracy. The final grade is then computed from these grades, with the first project counting 40%, and the second and third 30% each. The assessment is based on 1. Presentation. 2. Contents (understanding of the physics/algorithms and results). 3. Structure and correctness of computer codes produced. 4. Contribution to the discussions, independence and learning abilities are judged by the instructors and also contribute to the final mark.	
Studyload/Week	The two-day crash course on programming during the spring break is optional and offered as a courtesy to the students. During the course, there will be 2 contact hours per week. The remaining workload is more or less evenly divided over the course, corresponding to around 8-10 hours of course work outside class. (6 ECTS = 168 hours; 168 hours / 15 week = 11 hours per week). To facilitate contact with instructors outside class, a forum will be implemented.	
Schedule	Optional two day crash course on programming during the spring break. 15 weeks with 2 contact hours.	

AP3141	Environmental Physics	6
Responsible Instructor	Dr.ir. D.J. Verschuur	
Instructor	Ir. R.A. Verzijlbergh	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. Z. Perko	
Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	Dr.ir. S.J.A. van der Linden	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Physics at BSc level, including basic Transport Phenomena and Fluid Dynamics, thermodynamics, calculus and differential equations; use of Python and/or Matlab.	
Course Contents	*) Fundamentals of the environmental physics, world climate and the greenhouse-model, durable energy-resources (solar-, wind-, wave-, nuclear-, biomass-energy), transport of pollution in ecosystems, Air Quality; *) Fundamentals of harnessing sustainable nuclear energy and physical models for nuclear reactors to do so; *) Land-atmosphere interactions and turbulent transport processes. Fundamentals of dimensional analysis & generalization in environmental physics; *) Radiative transfer & Solar cell physics, PV system design & Renewable energy integration; *) Environmental noise, basic theory, noise calculation schemes	
Study Goals	To provide physicists with the knowledge and tools so that they can contribute to a more sustainable society from their own specific field of expertise: 1. Acquiring advanced knowledge of the basic physics underlying environmental issues such as: the earth energy budget; air-, water- and ground pollution; solar, wind, tidal, wave, biomass and nuclear energy technologies and environmental noise. 2. Being able to translate problems in the above areas into appropriate physical models and their relevant equations. 3. Being able to translate physical models and model equations into (e.g. Matlab/Python) computer models. 4. Being able to find (quantitative) solutions to model equations, using either theoretical analytical methods, or numerical methods. 5. Being able to quantitatively and physically evaluate, analyse and critically acclaim proposed solutions and strategies for environmental issues. 6. Being able to contribute to solutions and strategies for environmental issues.	
Education Method	Lectures and mandatory homework assignments.	
Computer Use	Homework requires computer programming in Matlab or Python	
Literature and Study Materials	Egbert Boeker and Rienk van Grondelle, Environmental Physics, Third edition, Wiley, 2011, ISBN 978-0-470-66676-0 + lecture notes and handouts	
Assessment	There is a series of mandatory (homework) Assignments. Please note that the Assignments have to be done in the prescribed periods, and that grades cannot be improved later. Lecturers can invite students for an oral assessment of their Assignments. The grade for the Assignments is the mean over all of them. At the end of the course there will be a written Exam. This Exam will be of an open-book type (see below). The final grade of the course is the average of the Exam grade and the Assignment grade. A passing grade can only be obtained if a score of at least 5.0 is obtained for the Exam and for the Assignments. The Assignments and Exam must be done in the same academic year. The validity of result of the Assignments is restricted to one academic year. Resit exam can be a written exam (if enough candidates) or via an oral assessment (by appointment). In case an on-campus exam cannot be provided, there will an online open-book exam followed by an oral fraud-assessment of a random subset of students. See announcements on Brightspace.	
Permitted Materials during Tests	Book, handouts and lecture notes and a calculator (may be graphic/programmable) Not permitted: anything related to the homework assignments: not the assignments, not the solutions, not your solutions, not any notes about the solutions. Also not permitted: old exams, nor their solutions.	
Studyload/Week	2 lecture hours per week (14 weeks), 5 series of homework of 20 hours each, 40 hours of exam preparation.	

AP3211	Advanced Solid State Physics	6
Responsible Instructor	Prof.dr. Y.M. Blanter	
Instructor	Prof. M.N. Ali	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The prerequisites of this course are: 1) Having achieved the learning objectives of BSc level Solid State Physics. 2) Being able to explain and apply BSc level Quantum Mechanics. Concepts from the Advanced Quantum Mechanics course will also be used. 3) Being able to use advanced mathematical tools, in particular the integration of functions of complex variables and of distributions.	
Course Contents	The course is organised in two parts: 1)lectures and 2)presentations. Lectures. We start with 15 lectures. 1. Introduction. Elements of band theory and thermodynamics of the Fermi gas. 2. Occupation number representation. Phonons. 3. Fermi liquids. 4. Magnetism in solids 1. 5. Magnetism in solids 2. 6. Landau theory of continuous phase transitions. 7. Scattering and transport in metals. 8. Superconductivity, phenomenology. Kondo effect. 9. Superconductivity, microscopic theory. 10. Superconductivity, Ginzburg-Landau theory. 11. Band structure theory. 12. Quantum matter 1. 13. Two-dimensional materials. 14. Topological states of matter. Presentations. The rest of the course is occupied by student presentations on scientific literature as described below.	
Study Goals	At the end of this course the students will be able to 1) Discuss modern condensed-matter physics theories and apply these tools to the analysis of the electronic properties of real materials with a particular focus on magnetic and superconducting systems. 2) Describe modern experimental techniques used in condensed-matter physics with an emphasis on spectroscopic and transport techniques. 3) Discuss, criticise and relate modern scientific literature on condensed-matter physics.	
Education Method	Lectures, presentations. Slides, notes and references to book chapters are provided.	
Literature and Study Materials	Marvin L. Cohen and Steven G. Louie "Fundamentals of condensed matter physics", Cambridge University Press (though we do not follow the book very closely). Some lecture notes will be available.	
Assessment	The total grade is calculated from the grades of a written test (T), of two presentations (P1 and P2) and of questions (Q), according to the formula $(T + P1 + P2)/3 + Q$, subsequently rounded to the closest integer or half-integer. 6 is the minimum passing grade, 5.5 is a failing grade (it will be added to OSIRIS as 5). Written test. At the end of the third quarter, a written closed book test will be scheduled. We estimate that the test preparation would require the students to spend 6 hours per lecture (including attending the lectures), reading the literature and studying the slides, making it in total (with the attendance of the test) 74 hours. The test will be given as a set of 20 multiple choice questions covering the material of all lectures. The grade for the test, T, is an integer or a half-integer (the number of correct answers divided by two). The grade of the written test must be at least 5 or higher in order to count for the final grade. If due to the COVID limitations a written test can not be held on campus, we will choose either an open book test, or an oral exam, depending on the number of students participating. Presentations. The presentations are graded as follows: each student is assigned two subjects (each a title of a presentation), accompanied by two main research papers which are related to the content of the course (but do not necessarily strictly follow the materials of the lectures) and is scheduled for two presentations (15 minutes each). A presentation is followed by questions from the audience and with oral feedback from the lecturers. Each presentation is graded according to the grading scheme (rubric available on the Brightspace) and receives an integer grade (P1 and P2). Students who are present in the audience also use the same form to provide feedback. The grades for the first presentation (P1) will be posted on Brightspace with the names of the students, to facilitate the feedback. We expect that it takes 4 hours to read each paper and 6 hours to read related literature and to prepare the presentation. Including 4 hours presence in the classroom this amounts to a minimum effort of 24 hours. It is expected that the students who want to pass the course with more than the minimum grade will also attend other presentations and ideally read the presented papers in advance. Questions. In addition to their own presentations, every student is assigned six more papers (which are scheduled to be presented by other students, three in the first round and three in the second round). The student is supposed to read these papers and prepare one question to each of the presentations of these papers. (Study load 5 hours per paper + 2 hour attendance of the presentation session = 42 hours). The questions have to be prepared even if the presentation itself is canceled or rescheduled, and need to be sent to the lecturers 24 hours before the presentation (if the presentation was rescheduled or cancelled, 24h before if was originally scheduled). They are graded as follows: - If the student did not send the question 24h before the presentation (even if the presentation was rescheduled or cancelled), or if the question is judged by the lecturers as poor, -0.25 (subtraction of 0.25); - If the student sent a good question on time but did not show up at the presentation to actually ask it, -0.1 (subtraction of 0.1); - If the student sent the question on time and showed up on the presentation to ask the question; or if the student sent the question on time and the presentation was rescheduled or cancelled, + 0.1. The sum of these grades for six questions makes the total question grade, Q (it may vary between -1.5 and 0.6).	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Schedule	Scheduling of presentations. We will schedule up to 4 presentations per day if needed. There are 14 days for presentations, and every student gives two presentations, which means all the presentations can be held in one room if there are 28 students or less taking the course, and (partially) in two rooms if there are between 29 and 56 students. The existing format of the course can not accept more than 56 students. Due to this fact, and also due to the difficulty of scheduling, we can not accept bachelor students,	

and we require commitment in the beginning that the students will follow the whole course (the students are given two weeks, at the beginning of the course, to decide whether they want to commit themselves to the full course).

Retakes and rescheduling. If a student did not show up at the written test, the retake will be scheduled individually after the last presentation of the course. It will be an oral exam, however, very similar to the written test instead of just choosing the answer, the student may be asked to explain the answer or to provide some additional information related to the question.

Students will have a chance to review the presentation schedule in advance and verify if it is compatible with their time constraints. In case of emergency, if a student can not attend their own presentation, the presentation can be rescheduled to an available slot, or individually after the course.

Questions cannot be rescheduled or retaken.

If a student failed the test or one of the presentations, they can have a retake to improve the grade if necessary, according to the above policies.

Second retakes are not possible.

AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics	6
Responsible Instructor	Dr. W.G. Bouwman	
Instructor	Dr.ir. N.H. van Dijk	
Instructor	Dr. H. Schut	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	The microscopic structure and dynamics of condensed matter in physics, chemistry and biology is obtained by a wide variety of scattering techniques. Structures are determined on length scales ranging from 10-11 to 10-5 m, dynamics on time scales from 10 -14 to 10-6 s. In this course we discuss the different neutron, X-ray and positron techniques, illustrated by a broad range of scientific and technological applications, for which recent scientific articles will be studied. After a general introduction an overview is given of the scattering theory, both for neutrons and X-rays. Then the different techniques will be discussed in more detail. Diffraction (either by X-rays or neutrons) yields information about the structure of single crystals, powders, and liquids. Small-angle scattering (SAXS and SANS) reveals the shape and arrangement of colloidal systems, micelles or polymers in a solvent. Reflectometry is used to find the composition of layered structures like proteins adsorbed at surfaces or magnetic layers in recording materials. Polarized neutrons are used in two different ways. First, for determining magnetic structures and dynamics. Second, as a special technique (spin-echo) to study the larger length scales and time scales (polymer dynamics). Inelastic scattering measures phonons in crystals, diffusion in liquids and the vibrational spectrum of molecules. Neutron sources (fission and spallation), X-ray sources (from X-ray tube, via synchrotron to XFEL) and instrumentation will be an integral part of this course. An introduction to positron annihilation (PA) will be given. Positron annihilation lifetime, Doppler Broadening techniques are particularly suited for the detection and characterisation of (open volume) defects such as vacancies, vacancy clusters and voids in e.g. metals, alloys, semiconductors and polymers. The main application of the Angular Correlation of Annihilation Radiation (ACAR) technique is the study of the electronic structure of solids, and that of defects, precipitates and nanocrystals. A visit to the neutron and positron experimental facilities at the Reactor Institute Delft will be part of the course.	
Study Goals	Being able to explain the theoretical background of neutron and X-ray scattering and positron annihilation techniques; Being able to talk about the applications of these techniques; Being able to perform basic calculations and interpretations of the techniques; Being able to read and discuss recent articles on the application of the techniques; Being able to talk in depth about one of the topics to be chosen by the candidate in combination with a recent research article.	
Education Method	The course consists of 14 times 2 hours contact on the following chapters: I. Theoretical background neutron and X-ray scattering II. Neutron experimental techniques III. X-ray experimental techniques IV. Diffraction from crystalline materials, theoretical, experimental, application V. Diffraction from amorphous materials and liquids, theoretical, experimental, application VI. Small-angle scattering, theoretical, experimental, application VII. Reflectometry, theoretical, experimental, application VIII. Polarized neutrons, theoretical, experimental, application IX. Imaging and tomography with neutrons and X-rays, theoretical, experimental, application X. Positron annihilation, theoretical, experimental, application Each 2-hours-lecture is accompanied by a set of assignments from the website https://e-learning.pan-training.eu/. This has to be completed before the next class. In the class question about the previous lecture will be discussed as well as recent scientific articles. In turn the students have to give short presentations about the assignments and about the scientific articles. Those students who have attended at least 12 of the 14 lectures and sent in at least 11 of the 13 assignments are allowed to enter the oral exam.	
Literature and Study Materials	Lecture notes. The accompanying material with the website https://e-learning.pan-training.eu/. For mathematical background we recommend the book D.S. Sivia, Elementary Scattering Theory for X-ray and Neutron Users, Oxford University Press, ISBN 978-0-19-922868-3.	
Assessment	Having done at least 11 of the 13 times homework is a requirement for taking the oral exam. The oral exam of 45 minutes determines the final grade. The student proposes an article or scientific topic connected to the course for the first 15 minutes of the exam.	
Elective	Yes	

AP3333	Physics of Energy Materials	6
Responsible Instructor	Dr. S.W.H. Eijt	
Instructor	Prof.dr. E.H. Brück	
Instructor	Dr. X. Wang	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Physics at the BSc level, including Solid State Physics, Statistical Physics, Thermodynamics, Optics, Electromagnetism, Quantum Mechanics, as well as Calculus at the BSc level.	
Course Contents	This course focuses on the fundamental and applied physics of energy materials. Physics is key to the development of sustainable energy systems, since these depend on innovative application of physical properties and new solid state materials in devices that efficiently generate, harvest, convert or store energy. Five major topics are discussed, namely the physics of (1) Magnetocaloric Materials, (2) Solar Energy Materials, (3) Fusion Energy Materials, (4) Lithium Battery Materials, (5) Hydrogen Energy Materials. The physics underlying these topics ranges from magnetism, phase transitions and (spin-dependent) electronic structure in magnetocaloric materials, detailed balance limit to the efficiency of solar cells, quantum cutting of photons and multiple exciton generation in third generation solar cells, diffusion and drift of ions and electrons in Li/Na battery nanomaterials, to the physics of materials under extreme conditions such as plasma-surface interactions in the development of suitable plasma-facing walls for ITER, the international fusion reactor. Examples of modern research will be provided, where new insights often rely on the application of advanced structure and dynamics analysis tools such as synchrotron X-ray and neutron diffraction, high resolution electron microscopy, nuclear magnetic resonance, opto-electronic spectroscopies, positron annihilation spectroscopy, and on comparison to theory based on thermodynamic models, ab-initio calculations and molecular dynamics.	
Study Goals	General: - Knowledge of the physics of energy materials in the five mentioned topics (see Course Contents). - Being able to perform basic calculations related to the physics of these topics. - Capable of applying the theory discussed in the lectures to physics problems in these topics. Project: - Being able to read and understand a selected set of scientific articles related to one of the five topics. - Being able to write a short paper (in English) on these articles. - Present and discuss (in English) the main results of the selected scientific articles to an audience of fellow physics students.	
Education Method	The course consists of weekly 2 hours lectures plus 2 hours project class. Further, each week homework assignments will be distributed. The lectures discuss the following topics: 1. Introduction and Overview of the Course, 1 lecture 2. Magnetocaloric Materials, 3 lectures 3. Solar Energy Materials, 3 lectures 4. Fusion Energy Materials, 3 lectures 5. Lithium Battery Materials, 2 lectures 6. Hydrogen Energy Materials, 1 lecture Each 2-hours-lecture is accompanied by a set of assignments. This has to be completed for the next week 2-hours project class. In the project class, questions about the previous lecture and assignment set will be discussed, and students can work in groups of two or three students (2 to 3 depending on the size of the class) on their project. Each group will study a set of two to three recent articles in one of the topics of the course. Each student will write her/his own paper of about 2000 words, either in the form of a News article (as a Journalist), a News and Views article (as an Editor) or a short review article (as a Scientist). Each group of two students will construct and present a presentation of about 15 min. based on the topic discussed in the articles.	
Literature and Study Materials	Literature including (review) articles and lecture notes for the course Physics of Energy Materials will be made available via Brightspace. Examples of articles that relate to the material discussed during the course include: - J. Lyubina, Magnetocaloric materials for energy efficient cooling, J. Phys. D: Appl. Phys. 50 (2017) 053002. - A. Smith et al., Materials Challenges for High Performance Magnetocaloric Refrigeration Devices, Adv. Energy Mater. 2 (2012) 1288. - W. Shockley and H.J. Queisser, Detailed balance limit of efficiency of p-n junction solar cells, J. Appl. Phys. 32 (1961) 510. - A. Polman, M. Knight, E.C. Garnett, B. Ehrler and W. Sinke, Photovoltaic materials: Present efficiencies and future challenges, Science 352 (2016) aad4424 - M.A. Green and S.P. Bremner, Energy conversion approaches and materials for high-efficiency photovoltaics, Nature Mater. 16 (2017) 23. - J. Knaster, A. Moeslang and T. Muroga, Materials research for fusion, Nature Phys. 12 (2016) 424. - M.A. Wagemaker, F.M. Mulder, Properties and Promises of Nanosized Insertion Materials for Li-Ion Batteries, Acc. Chem. Res. 46 (2013) 1206. Examples of books (for consultation) that relate to the material discussed during the course include: - Magnetic refrigeration near room temperature, Ekkes Brück, in Handbook of Magnetic Materials Vol 17 Chapter 4 (2007) K.H.J. Buschow, Ed. - The magnetocaloric effect and its applications, A.M. Tishin, Y.I. Spichin (IOP Publ., Bristol, 2003). - Third Generation Photovoltaics, Advanced Solar Energy Conversion, Martin A. Green (Springer Verlag, Berlin, 2003). - Fusion, An Introduction to the Physics and Technology of Magnetic Confinement Fusion, Weston M. Stacey (Wiley VCH, Weinheim, 2010). - Hydrogen as a Future Energy Carrier, Andreas Züttel, Andreas Borgschulte, Louis Schlapbach, Eds. (Wiley VCH, Weinheim, 2008).	
Assessment	Grade option #1: 40% project + 60% final exam Grade option #2: 40% project + 45% final exam + 15% homework Final grade : Max(Grade option #1, Grade option #2) A minimum grade for the final exam of 5.0 is required to pass the course.	

For the resit exam, the same rules apply as listed above for the final exam.

Depending on the number of participants, the final exam and the resit are written or oral.

**Permitted Materials during
Tests**

The exam will be open book, meaning that it is allowed to consult the lecture slides and the readers of this year's course, but no other material. A scientific calculator and pencils (with various colors, but not red) are allowed.

AP3341	Nuclear Reactor Physics	6
Responsible Instructor	Dr.ir. D. Lathouwers	
Instructor	Prof.dr.ir. J.L. Kloosterman	
Instructor	Dr. Z. Perko	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 Exam by appointment	
Course Language	English	
Expected prior knowledge		
Course Contents	This course will give you an overview of the fundamental physical processes taking place in a nuclear reactor. After having followed this course you will know about and be able to work with the following topics: 1. Nuclear reactions * various types of reactions * probability of reactions, nuclear cross sections * energy release in fission * data files 2. Fission chain reaction * criticality, multiplication of neutrons * classification of various nuclear reactors 3. Transport of neutrons * general description of relevant variables * diffusion theory description and its assumptions * homogeneous and reflected geometries 4. Energy distribution of neutrons * neutron slowing down by moderation * description of various energy ranges * resonance absorption of neutrons, temperature dependence * perturbation theory 5. Time dependent behavior of reactors * delayed neutron emission and implication for time dependence * exact description * point kinetics equations simplification * feedback mechanisms 6. Thermal hydraulic analysis of nuclear reactor cores * power generation in reactor cores * heat transfer in reactor cores * temperature distribution in reactor cores 7. Fuel burnup * fuel composition and depletion with usage * measure of fuel burnup * fuel cycle overview and nuclear waste * breeding of fuel in special reactors * recycling of fuel 8. Nuclear power reactors * existing types, principle of operation * next generation reactors Please note that this list is only meant to give you an impression about the course.	
Study Goals	After following this course the student will be able to: Define and describe the most important nuclear processes taking place in reactors (nuclear decay, neutron interactions with the nucleus) and the basic concept of nuclear energy generation (fission, fusion). List and describe the fundamental physical processes present in a nuclear reactor core (chain reaction, neutron moderation and transport, energy release from fission, heat transport by coolant). Analyse reactor designs from neutronics and thermal hydraulics points of view, determine the consequences of different design choices on efficiency, safety, and operation. Analyse simple reactor systems in both static and transient operation, compute basic parameters (e.g. neutron multiplication factor), and perform simple reactor design calculations.	
Education Method	Lectures and instructions in combination with homework exercises and design exercises.	
Literature and Study Materials	The lectures are based on the book Nuclear Reactor Analysis by James J. Duderstadt and Louis J. Hamilton (Wiley). The book can be borrowed for a small fee with the professor (as far as stock lasts)	
Assessment	Homework problems, design exercises and exam. The type of exam will be written. The resit will be oral. The homework counts for 50% of the grade. The homework exercises can be used for 2 years. The grade for the written exam needs to be at least 5.0 to pass the course.	
Elective	Yes	
Tags	Energy Physics	

AP3352	Introduction to Nuclear Science and Engineering	6
Responsible Instructor	Dr.ir. M. Rohde	
Instructor	Dr. L. van Eijck	
Instructor	Dr.ir. M.C. Goorden	
Instructor	M. Schouwenburg	
Instructor	Dr. H. Schut	
Instructor	Dr.ir. A.G. Denkova	
Instructor	A.L. Smith	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Summary	This course is designed as an introduction to the broad range of topics that comprise nuclear science, including radioactive decay, radiation dosimetry, neutron and positron beams, nuclear reactor physics and designs, nuclear waste disposal, radioactivity in health sciences, and more. The course centers on the *essential concepts* rather than in-depth theory. In other words: this course shines in its broadness of subjects and is perfect for deliberately selecting more in-depth follow-up courses such as e.g. Nuclear Reactor Physics, Nuclear Chemistry, Medical Imaging Signals and Systems and Neutrons, X-rays and Positrons for Studying Microscopic Structures and Dynamics. The lectures are given by staff from all sections within the department of Radiation Science and Technology (RST). Students should complete this course with a greater understanding and appreciation for the relevance of nuclear science and technology in today's global society.	
Course Contents	Subjects include (tentative): - The history of radioactivity - Modes of decay / de-excitation - Interactions of radiation with matter - Radiation dosimetry - Applications of research reactors - Neutron beams - Neutron scattering techniques - Positron beams - Nuclear reactors - Nuclear waste disposal - Health application of nuclear science - Medical imaging - Radiotracers	
Study Goals	Sub-goal (total % for sub-goal) : Weekly exercises(10%)/Exam(75%)/Presentation(25%)* Has insight into the nature of radiation and radioactivity and its interaction with matter (37%) : 5%/37%/0% Understands, in a broad sense, how nuclear science is/will be applied in fields of energy, health, medicine industry and others (38%) : 5%/38%/0% Is able to read, understand and present a scientific paper on a nuclear science subject (20%) : 0%/0%/20% Is able to collaborate with another student, while preparing the presentation (5%) : 0%/0%/5% * The sum is larger than 100% to indicate that the weekly exercises are not compulsory.	
Education Method	Oral lectures, guided tours and presentations given by the students on a subject related to nuclear science.	
Computer Use	Some homework exercises may require basic knowledge on Matlab, Python or Fortran (or any other programming code)	
Literature and Study Materials	J. Kenneth Shultis, Richard E. Faw: Fundamentals of Nuclear Science and Engineering Third Edition (available as e-book. Link will be provided on Brightspace); Lecture slides Homework exercises, to practice every subject of the course. The exercises are meant to train you for the written exam.	
Assessment	A written exam + a presentation of a scientific paper. All students give a presentation in couples about a paper that is related to an NSE subject. The presentation should contain sufficient physics/chemistry (hence, no general/managerial talk). Grade is P. The exam is a written exam and is closed-book. The student is allowed to make a *written* cheat sheet of 1 A4 and bring that to the exam. Grade is E. The final grade is calculated as $G = (P + 3E)/4$ The grade for the presentation can be re-used for the resit in the same academic year. If you fail for this course (either with the exam or resit), you need to redo everything in the next academic year.	
Permitted Materials during Tests	During the tests and exam: calculator + pens + written cheat sheet	
Elective	Yes	

CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3
Responsible Instructor	A.L. Smith	
Instructor	Dr.ir. S.E. Offerman	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	General chemistry, basics of chemical thermodynamics	
Course Contents	This is a 3 ECTS course composed of 28 total lecture hours, 56 self-study and a final examination. This course is designed for Chemical Engineering, Applied Physics, Materials Science & Engineering, and Sustainable Energy Technology students that are interested in developing a working knowledge of nuclear materials chemistry and the nuclear fuel cycle. The course will cover various aspects of solution and solid state chemistry as well as materials science that play a key role at each step of the nuclear fuel cycle, from the metallurgy of uranium to the disposal of spent reactor fuel or high level waste. While the physics and engineering of controlled fission are central to the generation of electricity by nuclear reactors, chemistry in general and especially materials chemistry dominate all other aspects of the nuclear fuel cycle. This course will give students a comprehensive knowledge of the traditional fuel cycle (the uranium once-through cycle) in Light Water Reactors (LWRs). It will also cover some of the proposed nuclear fuel cycles that may well carry nuclear power through the coming decades. As an outcome of the course, the students will be able to compare and contrast existing and innovative fuel cycles, and discuss the pros and cons of each. In addition, the course will cover some of the unique properties of the actinide 5f-elements. The first three of the actinide series - Th, Pa and U - occur naturally, while the elements following uranium in the Periodic Table (transuranium elements) are man-made. The actinides are essential to nuclear power generation, but also find applications in many other areas of industry, medicine and research. The course will address 5f-electronic configurations, oxidation states, redox potentials, inorganic and structural chemistry, as well thermodynamic properties of actinide compounds. The actinides behaviour in the environment and in the geosphere will be discussed, together with analytical tools for their identification. The biological and environmental hazards associated with these elements pose certain risks which must be controlled and minimized. Due to their significant role and because the final destination of transuranic elements originating from the nuclear fuel cycle is still an open issue, the actinides chemistry and physics continue to be one of the major areas of nuclear research. The materials science part of the course will cover the main features of materials microstructures and behaviour in conditions that are relevant for nuclear reactors. The various types of defects at the atomic or micrometre scale will be explained as well as their consequences in terms of energy. This forms the basis of understanding the changes that materials undergo when subjected to elevated temperature and high-energy radiation. Changes in the microstructure and the defect configurations lead to changes in properties and behaviour and can thus jeopardise the material performance, forming a threat to safety. The course is aimed at master level students with general background in chemistry, and/or materials science.	
Study Goals	1. Students can demonstrate how chemistry and materials science influence almost all aspects of the fuel cycle. 2. Students can describe the key steps involved in the traditional nuclear fuel cycle and explain the involved chemical processes. 3. Students can apply the concepts addressed in the course to discuss alternative fuel cycles, not covered in the class, and explain the fundamental differences with the presented traditional fuel cycle. 4. Students can justify, at each step of the fuel cycle, the behaviour of the different actinide elements in view of their chemical and physical properties (electronic structure, redox potentials, solubility, complexation, speciation, thermodynamics, etc) 5. Students can predict and explain the behaviour of fission products in the irradiated fuel, during the reprocessing and in geological environment. 6. Students are able to predict the environmental behaviour of various actinide species. 7. Students can discuss major analytical techniques for actinide studies. 8. Students can perform numerical evaluations to predict the behaviour of the actinide elements and fission products at each step of the fuel cycle. 9. Students are able to discuss the major features of materials microstructures and behaviour when subjected to the conditions in a nuclear reactor. 10. Students write an essay and make a presentation on a supplemental topic related to the course material, but not covered in class, in which they apply the key concepts presented during the lectures and perform critical analysis of the information.	
Education Method	Oral lectures / Supported self-study	
Literature and Study Materials	Main: Course reader Additional reading: N. Tsoulfanidis. The Nuclear Fuel Cycle, American Nuclear Society, 2013; The Nuclear Fuel Cycle: from Ore to Wastes, P.D. Wilson (ed.), Oxford University Press, 1996. The Chemistry of the Actinide and Transactinide Elements (L.R. Morss, N.M. Edelstein, J. Fuger (Ed.)), Springer Netherlands, 2011 S.A. Cotton, Lanthanide and Actinide Chemistry, John Wiley & Sons, 2006 W. Loveland, D.J. Morrissey, G.T. Seaborg. Modern Nuclear Chemistry, Wiley-Interscience, 2006; G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. Nuclear Fuel Cycle Science and Engineering, I. Crossland (ed.), Woodhead Publishing, 2012 M. Halka, B. Nordstrom, Lanthanides and Actinides, Facts on File Inc., 2011 W. Loveland, D.J. Morrissey, G.T. Seaborg, Modern Nuclear Chemistry, Wiley-Interscience, 2006. G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. William D. Callister Jr., David G. Rethwisch, Materials Science and Engineering: An Introduction, ISBN 978-0470419977, Ed. Wiley	
Assessment	The final grade is composed of the weighted average of the final exam (40%), presentation (20%) and essay (40%). Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Elective	Yes	
Tags	Chemistry	
Location	Building 58 (preferred) or Reactor Institute Delft, Radiation Science and Technology department	

EE4670	PV Materials Processing and Characterization	4
Responsible Instructor	Prof.dr. A.W. Weeber	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3 Different, to be announced	
Course Language	English	
Course Contents	The course is a follow-up of PV Technologies. Details of processing and characterization techniques of PV absorber materials and the solar cells functional coating will be discussed. Examples of lecture topics are advanced light management structures, novel surface passivating layers and contacting schemes. Background of deposition technologies such as Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD) and Atomic Layer Deposition (ALD) will be given. Furthermore, working principles of special coatings and its functionality when applied to different solar cell concepts, including tandem devices, will be presented. This course is an elective one following the solar profile. PV Basics and PV Technologies should have been concluded successfully.	
Study Goals	After this course students are able to understand processing mechanisms, material properties, and functionality and working principle of the novel functional coating applied to different solar cell concepts. Students are expected to be able to explain these working principles and have a solid background on different deposition technologies and characterization techniques, and will be able to describe the effect on the solar cells output. Students will be able to analyse and evaluate recent technology developments and the background of it, and summarize them during an oral presentation for the entire group. As we can see, these learning objectives are specific about the content, and they are phrased in terms of observable behaviour: understand, explain, evaluate, present and describe.	
Education Method	This course consists of eight 2-hour lectures. In the same week a 2 hour work session / instruction will be organized for a deeper understanding of the technologies discussed in the lectures. During the whole course the students will work on assignments which need to be submitted as reports. All assignments/reports should be assessed as sufficiently good (no specific mark). In week 9 (final schedule will be given at the beginning of the course) you will present results of a literature study on a special topic covered by the course for the entire group.	
Literature and Study Materials	Book (not obligatory, expensive): Photovoltaic Solar Energy: from fundamentals to applications by Angèle Reinders et al. Material from online course: PVCDROM, PV-manufacturing.org Slides during lectures Articles (referred to in lectures)	
Assessment	The final grade consists of the following parts: presentation (team work) (50% of final grade); written exam (individual) (50%); Both marks will be expressed on a scale of 1 to 10 and should be 5.80 or higher. Reports on the assignments shall be completed as well, and assessed as sufficiently good (no specific mark). In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
maximum aantal deelnemers	Expected number of student: Maximum of 30. Use Brightspace and inform our secretary Mark Vielvoije (m.vielvoije@tudelft.nl) for attending the course.	

EE4680	Photovoltaic Modelling	4
Responsible Instructor	Dr.ir. R. Santbergen	
Instructor	Prof.dr.ir. O. Isabella	
Responsible for assignments	Dipl.-Ing. Y. Blom	
Co-responsible for assignments	F.S. Saitta	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Modelling is an important tool for gaining deeper insight in the working principles and loss mechanisms of solar cells and photovoltaic systems. Researchers use this insight to develop more efficient solar cells and to maximize the energy yield of photovoltaic systems. In this course we will cover the entire photovoltaic modelling chain: modelling of photovoltaic materials, devices (solar cells), modules and systems. The focus will be on understanding the working principles of these models, and using them to analyse the effects of design parameters on solar cell and photovoltaic system performance. The online course material consists of two parts, each consisting of several phases: 1) PV materials and device modelling 1.1) PV materials and device characterization 1.2) optical modelling of PV devices 1.3) electrical modelling of PV devices 2) PV systems modelling 2.1) modelling of irradiance 2.2) modelling of energy yield In addition there will be software demos, given by expert users, allowing you to familiarize yourself with the advanced modelling software commonly used by PV researchers. The prior knowledge from the courses PV basics and PV systems is required. This course PV modelling will prepare you for an MSc thesis project that involves modelling of solar cells or photovoltaic systems, as well as a future career in photovoltaics research.	
Study Goals	The online course material consists of two parts: 1) PV materials and device modelling, 2) PV systems modelling. Each consist of several phases: At the end of phase 1.1 (characterization of PV materials and devices) you should: - Be familiar with the main techniques used in the characterization of PV materials and devices - Understand the importance of a correct characterization for the modelling of PV - Be able to extract useful and important information (trends and properties) from the characterization of both PV materials and devices At the end of phase 1.2 (optical modelling of PV devices) you will; - have a complete overview of the modelling approaches commonly used for photovoltaic optical modelling, and which are their strength and limitations - get a thorough understanding of different optical modelling methods and which structures they are best suited to model - improve and practice your skills on optical modelling, by carrying out activities with our own GenPro4 software At the end of phase 1.3 (electrical modelling of PV devices) you should be able to: - have a complete overview of the modelling approaches commonly for photovoltaics, both optical and electrical - get a thorough understanding of different electrical modelling methods - improve and practice your skills on optical and electrical modelling, by carrying out activities with our own advanced semiconductor analysis (ASA) software. At the end of phase 2.1 (modelling of irradiance) you should be able to: - Gather and analyse meteorological data, including the utilization of geographic information systems, - Obtain information on horizon and shading at a specific location, and derive the relevant location parameters, - Combine meteorological data and location properties to calculate the irradiance on the angle of incidence with different levels of detail, and - Optimize the orientation of a module to maximize the irradiance on the angle of incidence. At the end of phase 2.2 (modelling of the energy yield) you should be able to: - Model the operating temperature of modules, - Obtain information from module datasheet, - Combine module properties with temperature and irradiance modelling to calculate DC power (and yield), - Model the performance of inverters (DC to AC) and batteries, - Analyse load profiles, and use them to size a PV system (module and batteries), and - Calculate the AC power and energy yield of an entire system, and analyse its performance from an economic point of view. After the software demos you will be able to explain the input, working principles and output of the following commonly used software packages: - Complete EASE (extracting optical constants from spectroscopic ellipsometry measurement of PV materials) - HFSS (rigorous solving of Maxwell equation for EM-wave (light) propagation in PV devices) - Sentaurus (modelling charge transport in PV devices by solving semiconductor equations) - Radiance (PV module plane of array irradiance calculation using backward ray-tracing) - PV Syst (PV system energy yield calculation)	
Education Method	This course consists of recorded video lectures, interactive lectures and instructions. The video lectures provide you with the basic information. The interactive lectures give you the opportunity to ask questions to the teacher and discuss selected topics in depth. In addition you will work on modeling assignments. During the instruction sessions the modeling assignments will be explained and feedback will be given on your (preliminary) results.	
Assessment	Every week you make and submit modeling assignments, which are graded. The weighted average counts for 70% of the final grade. In addition, every week you write a short report about the software demo, which is graded as well. The average of this counts for the remaining 30% of the final grade. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Remarks	The on-campus Q&A session is not mandatory. Because the course material (video lectures, live demos) is given online, this course can also be taken by students who cannot physically come to the campus. Because the assessment is done through weekly assignments, without an exam, the final grade (and credits) can be obtained already before the exam period (end of October). This can be relevant for those who have not yet obtained sufficient credits to start their graduation project.
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ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Instructor	Dr. L. Mazzarella	
Instructor	Prof.dr. I.M.F. Gordon	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Note: In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Responsible Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.8) in all six technical reports is required to complete the course successfully. In case of failure of a technical report, you can use buffer sessions on week 1.7 and 1.8 to repeat the experiments. --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2023-2024. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2023-2024.	

ME45203	Electrolysers, Fuel Cells and Batteries	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena, basic fluid dynamics, and elementary differential equations. A working knowledge of electrochemistry and in particular the Butler-Volmer equation is also assumed. Ideal preparation include the courses ME45201, CH3513, or CH3222. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesize chemicals or fuels in electrochemical reactors, like an electrolyzer producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - Tobias model for gas-liquid electrolysers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and CH3222 for students interested in analytical modelling of electrochemical systems.	
Books	Electrolysers, Fuel Cells and Batteries : Analytical Modelling. (J.W. Haverkort)	
Assessment	Written exam.	

SET3110	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	A. Vasileiadis	
Instructor	T. Famprakis	
Instructor	Dr. X. Wang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	The final grade of the course consists of the following components: Written exam (100% of the final grade) In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. E.M. Kelder	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Physics for Fluids Engineering

AE4180	Flow Measurement Techniques	3
Responsible Instructor	A. Sciacchitano	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Viscous Flows	
Course Contents	Introduction to experimental analysis of aerodynamic problems. Flow visualization methods. Measurement techniques: Laser Doppler Anemometry, Hot Wire Anemometry, Pressure measurements, Optical refractive methods, Particle image velocimetry, Infra-Red Thermography. Laboratory exercise: NACA 0012 airfoil at incidence measured with HWA and PIV	
Study Goals	After this course, students will be able to: 1) Discuss the main aspects related to the conduction of aerodynamic tests in simulation facilities 2) Describe and explain the working principle of the most important and up-to-date flow measurement techniques 3) Select the most suitable measurement technique depending on the aerodynamic problem to investigate 4) Design and conduct wind tunnel experiments for aerodynamic investigation	
Education Method	Lecture + lab. exercises	
Literature and Study Materials	Course and lecture notes available on the electronic Brightspace.	
Assessment	Report and oral	
Set-up	1. Fundamentals of experimental aerodynamics 2. Introduction to measurement techniques 3. Flow visualization (classroom assignment) 4. Pressure measurements 5. Hot Wire Anemometry (lab exercise) 6. Laser Doppler Velocimetry 7. Particle Image Velocimetry (classroom assignment+lab exercise) 8. InfraRed Thermography 9. Schlieren and shadowgraphy 10. Fundamentals of data processing	

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: describe the components and configurations of wind turbines describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. identify the drivers of dynamic behaviour and calculate their typical values describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AP3082	Computational Physics	6
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.R. Akhmerov	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor physics.	
Course Contents	This course starts with an introduction into methods and techniques in computation that are viable today. Following this, several projects in the field of computational physics are executed by the students, who work in pairs. Projects are on molecular dynamics, (quantum) Monte-Carlo calculations, and a final project of own choice. The course projects are close to the research level.	
Study Goals	At the end of this course a student will be able to 1. describe standard physical simulation algorithms. 2. implement these algorithms in a well-structured computer code. 3. apply these algorithms to calculate physical properties and interpret the outcome. 4. verify correctness of the simulation outcome and use this analysis to improve the code. 5. organize and perform joint programming work. 6. present the results according to the requirements of scientific communication.	
Education Method	Before the start of the semester we offer an optional crash course of two days where you learn how to program in python, use version control and store codes online. The course itself is project-based. This means that it contains little formal teaching, but most of the knowledge is acquired hands-on by implementing, running and analyzing the results of computer simulations. Students are expected to learn the necessary background theory by studying literature and from contact with the lecturers. Another important element is code review, where the students read and comment on each other's code.	
Literature and Study Materials	The textbook Computational Physics by J.M. Thijssen (2nd edition, Cambridge University Press) is useful. Extensive use is made of online resources (course material, cloud storage portals like github/gitlab). In addition, papers from modern research literature are used.	
Prerequisites	The bachelor degree must be finished before the start of the course.	
Assessment	Assessment is based in the results of the three projects, consisting of a paper (first two projects) or oral report (final project), and the simulation code and development history. The grade of each project is rounded to two digit accuracy. The final grade is then computed from these grades, with the first project counting 40%, and the second and third 30% each. The assessment is based on 1. Presentation. 2. Contents (understanding of the physics/algorithms and results). 3. Structure and correctness of computer codes produced. 4. Contribution to the discussions, independence and learning abilities are judged by the instructors and also contribute to the final mark.	
Studyload/Week	The two-day crash course on programming during the spring break is optional and offered as a courtesy to the students. During the course, there will be 2 contact hours per week. The remaining workload is more or less evenly divided over the course, corresponding to around 8-10 hours of course work outside class. (6 ECTS = 168 hours; 168 hours / 15 week = 11 hours per week). To facilitate contact with instructors outside class, a forum will be implemented.	
Schedule	Optional two day crash course on programming during the spring break. 15 weeks with 2 contact hours.	

AP3141	Environmental Physics	6
Responsible Instructor	Dr.ir. D.J. Verschuur	
Instructor	Ir. R.A. Verzijlbergh	
Instructor	Dr. S.R. de Roode	
Instructor	Dr. Z. Perko	
Instructor	Dr.ir. B.J.H. van de Wiel	
Instructor	Dr.ir. S.J.A. van der Linden	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Physics at BSc level, including basic Transport Phenomena and Fluid Dynamics, thermodynamics, calculus and differential equations; use of Python and/or Matlab.	
Course Contents	*) Fundamentals of the environmental physics, world climate and the greenhouse-model, durable energy-resources (solar-, wind-, wave-, nuclear-, biomass-energy), transport of pollution in ecosystems, Air Quality; *) Fundamentals of harnessing sustainable nuclear energy and physical models for nuclear reactors to do so; *) Land-atmosphere interactions and turbulent transport processes. Fundamentals of dimensional analysis & generalization in environmental physics; *) Radiative transfer & Solar cell physics, PV system design & Renewable energy integration; *) Environmental noise, basic theory, noise calculation schemes	
Study Goals	To provide physicists with the knowledge and tools so that they can contribute to a more sustainable society from their own specific field of expertise: 1. Acquiring advanced knowledge of the basic physics underlying environmental issues such as: the earth energy budget; air-, water- and ground pollution; solar, wind, tidal, wave, biomass and nuclear energy technologies and environmental noise. 2. Being able to translate problems in the above areas into appropriate physical models and their relevant equations. 3. Being able to translate physical models and model equations into (e.g. Matlab/Python) computer models. 4. Being able to find (quantitative) solutions to model equations, using either theoretical analytical methods, or numerical methods. 5. Being able to quantitatively and physically evaluate, analyse and critically acclaim proposed solutions and strategies for environmental issues. 6. Being able to contribute to solutions and strategies for environmental issues.	
Education Method	Lectures and mandatory homework assignments.	
Computer Use	Homework requires computer programming in Matlab or Python	
Literature and Study Materials	Egbert Boeker and Rienk van Grondelle, Environmental Physics, Third edition, Wiley, 2011, ISBN 978-0-470-66676-0 + lecture notes and handouts	
Assessment	There is a series of mandatory (homework) Assignments. Please note that the Assignments have to be done in the prescribed periods, and that grades cannot be improved later. Lecturers can invite students for an oral assessment of their Assignments. The grade for the Assignments is the mean over all of them. At the end of the course there will be a written Exam. This Exam will be of an open-book type (see below). The final grade of the course is the average of the Exam grade and the Assignment grade. A passing grade can only be obtained if a score of at least 5.0 is obtained for the Exam and for the Assignments. The Assignments and Exam must be done in the same academic year. The validity of result of the Assignments is restricted to one academic year. Resit exam can be a written exam (if enough candidates) or via an oral assessment (by appointment). In case an on-campus exam cannot be provided, there will an online open-book exam followed by an oral fraud-assessment of a random subset of students. See announcements on Brightspace.	
Permitted Materials during Tests	Book, handouts and lecture notes and a calculator (may be graphic/programmable) Not permitted: anything related to the homework assignments: not the assignments, not the solutions, not your solutions, not any notes about the solutions. Also not permitted: old exams, nor their solutions.	
Studyload/Week	2 lecture hours per week (14 weeks), 5 series of homework of 20 hours each, 40 hours of exam preparation.	

AP3171	Advanced Physical Transport Phenomena	6
Responsible Instructor	Dr.ing. S. Kenjeres	
Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamental Physical Transport Phenomena	
Course Contents	Analytical/Numerical/Modelling Aspects of Advanced Physical Transport Phenomena (Fluid Flow, Heat Transfer and Turbulence); 1. Basic Equations of Transport Phenomena - Field Description; 2. Mathematical Methods for Solving Transport Equations (PDE, separation of variables, eigenfunctions and eigenvalues, Bessel functions, Laplace transformation, Error-Gamma functions, integral methods); 3. Transport in Stagnant Media (diffusion, moving front problems, diffusion with source terms); 4. Momentum Transport (potential flows, creeping flows, boundary layers); 5. Transport in Flowing Media (stationary transport in flows with uniform velocity, heat transfer in laminar pipe flow, natural convection); 6. Numerical Heat and Fluid Flow (discretization methods for heat conduction, convection and diffusion; differencing schemes, numerical diffusion; steady and time-dependent convection and diffusion; calculation of flow field/velocity-pressure coupling, SIMPLE algorithm); 7. Turbulence: Some Features and Rationale for Modelling (some generic types of turbulent flows and convective processes, wall-bounded turbulent flows: velocity and temperature distributions/wall functions, Reynolds decomposition, RANS); 8. Turbulence Modelling (closure problem, eddy viscosity/diffusivity models, k-e model, other two-equation eddy-viscosity models).	
Study Goals	1. to be able to identify and to mathematically define particular physical mechanisms of the complex transport phenomena; 2. to be able to specify and to analytically solve characteristic PDEs describing simplified transport phenomena; 3. to be able to discretise the system of governing transport equations by using a finite volume method by performing term-by-term analysis (time-dependent, diffusive, convective, source/sink terms) for one-, two- and three-dimensional generic cases; 4. to be able to understand basic mechanisms of turbulence and to derive the transport equations for fluctuating field variables (Reynolds-decomposition); 5. to learn characteristic classes of the turbulence-models (zero-, one-, two-equations, full-stress models); 6. to be able to computationally perform some basic generic cases of flow, heat and mass transfer (CFD) (channel flow, back-step flow, differentially heated enclosure, in both laminar- and turbulent-regimes).	
Education Method	Combination of Lectures (4 Lectures per week, covering theoretical aspects) and practical exercises (both analytical and computational/computer exercises, 2 hours per week).	
Computer Use	Computational exercises covering step-by-step solving some of basic generic cases of flow, heat and mass transfer on Linux computers (mesh-definition, specification of the boundary and initial conditions, solving, postprocessing, analysis and critical assessment of results).	
Literature and Study Materials	1. Book: "Analysis and Modelling of Physical Transport Phenomena", Hanjalic K., Kenjeres S., Tummers M.J., Jonker H.J.J., VSSD Book, ISBN-13 978-90-6526-165-8, Second Edition, December 2009. 2. Book: "Transport Phenomena", Bird R.B., Stewart W.E., Lighfoot E.N., 2nd edition, Wiley (2002) 3. Book: "Fysische Transportverschijnselen II", Hoogendoorn C.J. and van der Meer, Th.H., Delftse Uitgevers Maatschappij (1991) 3. Handouts: For computational/computer exercises a reference manual and quick start manuals will be provided.	
Assessment	Written Exam (two times per year). The exam contains three distinct parts: (1) analytical; (2) numerical discretization; (3) turbulence modeling. At least 50% of each part needs to be completed correctly to successfully pass this course.	
Permitted Materials during Tests		
Elective	Yes	
Tags	Fluid Mechanics Physics Transport phenomena	

AP3181	Applied Multiphase Flows	6
Responsible Instructor	Dr. L. Portela	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course assumes some elementary knowledge of Fluid Mechanics and/or Transport Phenomena. The level of prior knowledge required is typically acquired in a BSc course in Engineering or Applied Physics. Prior knowledge of multiphase flow is not required.	
Course Contents	1. Introduction. Overview of multiphase flow. Examples of industrial and environmental flows. Examples of flows with heat transfer and phase change. Classification of multiphase flows: separated vs. dispersed. Brief introduction to flow patterns. 2. Balance equations. Physical mechanisms involved and material behavior. Constitutive equations and mechanistic models of behavior. Interfacial interactions and forces. Analogies and differences with respect to single-phase flow. Flow parameters, non-dimensional numbers and scaling. 3. Introduction to turbulence, turbulence modeling and turbulence effects. Analogies and differences with respect to single-phase flow. 4. Separated flows and interfacial phenomena. Stability associated with interfacial flows. Interfacial waves. 5. Dispersed flows. Interaction between the dispersed and continuous phases. Momentum, heat and mass transfer. Differences and similarities between solid particles, droplets and bubbles. 6. Dynamics of single particles: solid particles, droplets and bubbles. Bubble and droplet dynamics. Bubble growth and collapse. Introduction to cavitation. Droplet dynamics and breakup. 7. Inter-particle interactions and collisions. Droplet and bubble coalescence and breakup. Collision kernel. Population balance models and simulations. 8. Quasi-1D flows, both internal and external (channel, pipe, jet, etc.). Balance equations and simplifying assumptions. Simple mechanistic models. Similarities and differences between: gas-liquid, liquid-gas, liquid-liquid, solid-liquid, and solid-gas. Flow classification and flow patterns, both separated and dispersed. Physical explanation and underlying mechanisms in the construction of a flow map. Steady and transient phenomena. 9. Stirred flows, both internal and external (bubble columns, fluidized beds, stirred tanks, ocean and atmospheric turbulence, etc.). Similarities and differences with respect to quasi-1D flows. Motion-driven, pressure-driven and gravity-driven flows. Dilute and dense flows. Flows dominated by inter-particle interactions (dense fluidized beds, bubble columns, etc.). 10. Multiphase flows with complex fluids (granular flow, slurries, agglomerates, emulsions, foams, etc.). Internal mesoscale structures. Thermodynamic and physicochemical interactions. Complex particles (with a complex structure and/or a complex particle-dynamics). 11. Heat transfer and phase change. Boiling and condensation. Radiation. 12. Industrial flow examples. Complex flow in pipelines in the oil-gas industry. Pneumatic conveying and solid-gas flows. Process equipment. Microfluidics applications. 13. Environmental flow examples. Sediment transport in rivers and the ocean. Aerosols and particulates dispersion in the atmosphere. Droplets in clouds.	
Study Goals	To give a general unified perspective on multiphase flow, with an emphasis on: (i) the understanding of the several essential physical mechanisms involved; and (ii) the formulation of complex engineering problems in terms of simple physically-based models. The students will develop the ability to: (i) formulate the problems in a structured way, starting from first principles; (ii) obtain (approximate) solutions using standard analytical and numerical techniques; (iii) interpret and critically analyse the solutions obtained; (iv) report the formulation of the problems and its solution in a clear well-structured form; and (v) sustain a critical discussion on the formulation of the problems and its solution. Also, during the course the students (i) will acquire a broad knowledge of multiphase flow; and (ii) will develop the ability to read and understand scientific literature on multiphase flow.	
Education Method	The education method is based on the combination of regular lectures, homework and interaction with the instructor outside the lectures (regular and impromptu office hours). The Learning by Doing is a key component of the education method: the homework involves both standard problems and open-ended problems, with the initiative and creativity of the students being strongly encouraged. The interaction with the instructor outside the lectures plays a key role, and it involves an iterative process, based on the Learning by Struggling philosophy.	
Literature and Study Materials	The study material consists mostly of lecture notes. Also, during the course, sections of books, on specific topics, and articles, from the scientific literature, are suggested for reading. Possible auxiliary general reference-books on Multiphase Flow are: 1. Brennen, C.E. (2005). Fundamentals of Multiphase Flow. Cambridge University Press. 2. Ghiaasiaan, S. M. (2007). Two-Phase Flow, Boiling and Condensation: in Conventional and Miniature Systems. Cambridge University Press. 3. Kleinstreuer, C. (2003). Two-Phase Flow: Theory and Applications. Taylor & Francis. 4. Crowe, C.T., Schwarzkopf, J.D., Sommerfeld, M. and Tsuji, Y. (2012). Multiphase Flows with Droplets and Particles. Second Edition. CRC Press. 5. Fan, L-S and Zhu, C. (1998). Principles of Gas-Solid Flows. Cambridge University Press. 6. Peker, S.M. and Helvacı, S.S. (2008). Solid-Liquid Two Phase Flow. Elsevier. Possible auxiliary general reference-books on Fluid Mechanics and Transport Phenomena are: 1. Kundu, P.K. and Cohen, I.M. (2008). Fluid Mechanics. Fourth Edition. Academic Press. 2. Ghiaasiaan, S.M. (2011). Convective Heat and Mass Transfer. Cambridge University Press.	
Assessment	The assessment is based on the homework proposed during the course (30%) and on the final exam (70%). The final exam consists of a written part followed by an oral discussion with the instructor. The written part is open book: the students can consult any reference they wish (including the internet) and use any tool they wish (including a laptop). The oral discussion will be arranged individually, shortly after the written exam, in consultation between the student and the instructor. It is possible to have a resit of the final exam; the format of the resit is the same as the format of the original final exam, consisting of a written exam followed by an oral discussion with the instructor.	
Permitted Materials during Tests	Any material (including laptop).	
Elective	Yes	
Tags	Fluid Mechanics Physics	

AP3551	Computational Multiphase Flow	6
Responsible Instructor	Dr. L. Portela	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	The course assumes some elementary knowledge of: (i) Fluid Mechanics and/or Transport Phenomena; and (ii) Numerical Methods. Also, the students should have elementary working knowledge of a programming language (e.g. Python, MATLAB, FORTRAN, C, C++) and should be able to write a simple computer code in a language of their choice. The level of prior knowledge required is typically acquired in a B.Sc. course in Engineering or Applied Physics. Prior knowledge of multiphase flow is not required.	
Course Contents	1. Introduction. Possibilities and limitations of computer simulations of multiphase flows. 2. Review of flow equations for single-phase flows. Introduction to turbulence and turbulence modeling. DNS, LES and RANS. Review of numerical methods for incompressible flows. 3. Project discussion and assignment. 4. Basics of dispersed multiphase flows. Heat, momentum and mass transfer between the two phases. Time and length scales. Point-particle concept. 5. Eulerian-Lagrangian DNS/LES. 6. Eulerian-Lagrangian RANS. 7. Two-fluid approach. 8. Basics of multiphase flow modeling. Particle-fluid interaction. Particle-particle interaction. Boundary conditions and particle-wall interaction. 9. Numerical aspects of multiphase simulations. 10. Complex flows and geometries. 11. Introduction to simulations with interface resolution. Immersed boundary methods, front-tracking and front-capturing. 12. Project presentation.	
Study Goals	The main goals are: (i) develop the ability to understand the essential physical mechanisms in complex fluids-engineering problems and to use this understanding to develop simple models to tackle these problems; (ii) develop the ability to transform simple models into concrete numerical algorithms and to write them in the form of a well-structured computer program; (iii) develop the ability to interpret and critically analyse the results from numerical simulations, understanding the possibilities and limitations of the models and computer programs used; (iv) develop the ability to use a computer program to solve a concrete fluids-engineering problem, involving a simple design and/or optimization. The main goals are taught in the context of multiphase flow, with an emphasis on turbulent dispersed multiphase flow, therefore, as secondary goals, the students: (i) acquire knowledge of numerical methods and modelling techniques in multiphase flow, with an emphasis on turbulent dispersed multiphase flow; and (ii) develop the ability to read and understand scientific literature on numerical methods in multiphase flow. The course involves a project, with a professional report and open presentation in the end, therefore, as secondary goals, the students: (i) improve their ability to write a well-structured clear report; and (ii) improve their ability to make a clear well-structured presentation and to sustain a critical discussion on it.	
Education Method	The education method is based on the Learning by Doing philosophy. Starting from scratch, the students develop their own computer code for (turbulent dispersed) multiphase flow. In the lectures, the physical and numerical aspects of multiphase flow are introduced, with an emphasis on dispersed turbulent multiphase flow, and scientific reading material is proposed. The students incorporate the knowledge acquired into their computer code, which expands during the course; this is done through a series of tasks that are proposed along the course, resulting in a well-structured computer code; this computer code will then be used to study a concrete multiphase flow problem. This computer code is part of a project (individual or in a group with a maximum of 3 students). This project results in a written report and an oral presentation (open to the public, with a typical length of a conference presentation). The project consists essentially of three parts: (i) development of a generic computer code for turbulent single-phase flow; (ii) development of a generic computer code for turbulent dispersed multiphase flow, where the students will choose one of two methodologies: (a) Euler-Lagrange (particle-tracking) or (b) Euler-Euler (two-fluid); and (iii) use of the computer code to study a concrete engineering and/or research problem. The particular tasks of the project and the concrete problem to be studied can vary, depending on the interest of the students; the students are strongly encouraged to present their own ideas for the project, which will then be defined in consultation with the instructor. Apart from the regular lectures, the students are also strongly encouraged to interact among themselves and with the instructor, during regular and impromptu office hours.	
Computer Use	The students should have access to a standard PC or laptop (using any operating system) with a programming language of their choice installed (e.g. Python, MATLAB, FORTRAN, C, C++).	
Literature and Study Materials	The study material consists mostly of lecture notes. Also, during the course, sections of books, on specific topics, and articles, from the scientific literature, are suggested for reading. Possible auxiliary general reference-books on (Computational) Multiphase Flow are: 1. Kleinstreuer, C. (2003). Two-Phase Flow: Theory and Applications. Taylor & Francis. 2. Crowe, C.T., Schwarzkopf, J.D., Sommerfeld, M. and Tsuji, Y. (2012). Multiphase Flows with Droplets and Particles. Second Edition. CRC Press. 3. Prosperetti, A. and Tryggvason, G. (2007). Computational Methods for Multiphase Flow. Cambridge University Press. Possible auxiliary general reference-books on Fluid Mechanics, Convective Heat and Mass Transfer, and Computational Fluid Dynamics are: 1. Kundu, P.K. and Cohen, I.M. (2008). Fluid Mechanics. Fourth Edition. Academic Press. 2. Ghiaasiaan, S.M. (2011). Convective Heat and Mass Transfer. Cambridge University Press. 3. Ferziger, J.H. and Peric, M. (2002). Computational Methods for Fluid Dynamics. Third Edition. Springer.	
Assessment	The assessment is based on a project developed during the course. The final grade takes into account: (i) the level of understanding and quality of the work performed by the student, demonstrated in: the interaction with the instructor during the course, the project report, and the discussion following the oral presentation (75%); (ii) the quality of the project report (15%); (iii) the quality of the oral presentation (10%).	
Elective	Yes	

Tags	Fluid Mechanics Numeric Methods Physics Transport phenomena
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AP3563	Water in the Atmosphere	5
Responsible Instructor	Dr. S.R. de Roode	
Instructor	Dr. A.P. Siebesma	
Instructor	I.M. Steinke	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Anyone with a BS degree in technical studies (e.g. Applied Physics, Applied Earth Sciences, Aerospace Engineering, Chemical Engineering, Civil Engineering, Mechanical Engineering) and an interest in meteorology and climate should be able to follow this course. Note that as the course has a strong focus on transports in the atmosphere some background in fluid dynamics/physical transport phenomena is strongly recommended.	
Course Contents	This course deals with atmospheric processes on all relevant scales and relevant to water: the global scale (clouds and climate), intermediate scale (deep convection, turbulence), to the smallest scales of cloud droplets and aerosols. After reviewing fundamental concepts of radiative transfer and atmospheric thermodynamics, the theory of turbulent atmospheric boundary layers will be treated. This paves the way towards addressing cloud formation processes, i.e. cloud dynamics and cloud-microphysics, and rain formation. Apart from treating the underlying principles and concepts, also attention will be devoted to the modelling and forecasting of the phenomena.	
Study Goals	1. Explain the physics and dynamics of the transport of energy, water and momentum between the surface, the atmospheric boundary layer and the broader atmospheric circulation 2. Apply simplified models of turbulence and energy and water exchange, including convection and clouds 3. Reflect on the parameterization and coupling of atmospheric processes in general circulation models and their role in current uncertainties in climate prediction 4. Assess the influence of land-atmosphere coupling, turbulence, convection and clouds on large-scale circulations (weather) and climate. 5. Hypothesize how land-atmosphere coupling, turbulence and convection and clouds are influenced by changes in weather and climate	
Education Method	~14x2 hours of lectures, including practical classes during which you will work on computer-based and regular exercises.	
Computer Use	Experience with software like Matlab or Python is very helpful to do the practical exercises.	
Literature and Study Materials	A great text book that covers most of the topics addressed in the course is "Atmospheric Science, An Introductory Survey", by John Wallace and Peter Hobbs.	
Assessment	The grade will be based on a combination of mandatory homework exercises and a written exam. The final grade will be the average of the two grades with the proviso that the grade for the exam must be 5.0 or higher.	
Permitted Materials during Tests	Calculator.	
Schedule	Scheduled together with ...	

CH3051	Applied Transport Phenomena (ATP)	4
Responsible Instructor	Dr. T.E. Burdyny	
Instructor	Dr. A.Y. Rwei	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Transport Phenomena (on BSc level) Molecular Transport Phenomena Differential Equations	
Course Contents	Overview: In many processes in (bio)chemical industrial, as well as in health and energy related applications, fluid flow, heat transfer, mass transfer, and chemical reactions interact in a complex way. These equations can relatively easily be described by differential equations, but their solution requires substantial computational requirements without providing insights into the system at hand. To reduce complexity, generic rules are taught in ATP with an emphasis on simplification and deriving valuable interpretations about the underlying operation of the system. Concepts such characteristic times, scales and regimes are dealt with to simplify problems enough to make them tangible or solvable analytically. Several techniques are also introduced for finding approximate solutions to partial differential equations of transport phenomena systems. Content: The course content is split into three sections: 1. A recap of key course material but with an Applied perspective. 2. A Methods section that teaches simplifying concepts including Scaling, Reduction in Dimensionality and Timescales, Perturbation Theory and Self-Similarity 3. An applications section that covers a set of common physical systems. This section also combines conservation laws to apply the concept of Scaling across equations.	
Study Goals	At the end of this course, the student can: - Determine approximate solutions to chemical engineering transport phenomena problems - Analyse the behavior of a transport phenomena system without an analytical or numerical solution - Identify small secondary phenomena and determine whether their inclusion or exclusion greatly impacts the system. - Reduce complex problems to simpler ones using the method of order-one scaling applied to both length and time scales.	
Education Method	Standard 2-hour lectures are followed or preceded by self-study sessions. Study Guides and teaching assistants support these self-study sessions. The lectures provide learning activities to introduce the conceptual material, and for students to test this out in example. These relate to the above learning objectives. The tutorials/self-study sessions will provide additional learning activities supporting the material learnt in class the same week. We use Brightspace forums as an outside of class hours medium for discussion and questions.	
Assessment	100% Final Exam Here the learning objectives associated to the methods and transport phenomena applications will be tested in a 3-hour exam format. Some methods will be directly tested conceptually in shorter problems, while other will be tested in an application-centered problem. All the above learning objectives can then be tested within a 3-hour examination.	

CH3153	Molecular Transport Phenomena	4
Responsible Instructor	Dr. V. Garbin	
Instructor	P. Boukany	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	- Physical Transport Phenomena (4052FYSTRY) - Physical Chemistry (4052FYSCKY) - Thermodynamics (4051CHTHEY and 4052STATHY) - Calculus I and II (4051CALC1Y and 4051CALC2Y) - Statistical methods (4052STAMEY) - Differential Equations (4052LADIFY).	
Course Contents	1. Introduction and motivation for the course. 2. Review of course prerequisites: Review of BSc transport phenomena (e.g. Ficks law, Fouriers law) and approaches to problem solving (e.g. shell balances). Continuum scale and link to molecular scale. 3. Kinetic Theory: Basic assumptions of kinetic gas theory. Mean free path. Maxwell velocity distribution. Calculating pressure from microscopic picture. Effect of intermolecular forces (van der Waals). 4. Binary diffusion: Einstein-Smoluchowski theory - Diffusion as a random walk. Langevin theory. Link to macroscopic description of diffusion. 5. Multicomponent diffusion: Limitations of Ficks first law of diffusion. Diffusion driven by a gradient in chemical potential. Generalized Maxwell-Stefan diffusion model. Multicomponent mass transport. 6. Ion transport and electrohydrodynamics: Transport of an isolated ion. Transport of non-isolated ion. Nernst-Planck transport equations. The Debye-Hückel double layer. Electro-osmotic flow. Electrophoresis for non-isolated particles. Examples of applications.	
Study Goals	1. Reproduce the basic laws of transport phenomena; solve problems involving the transport of mass, species, momentum, and energy, including setting up the problem using a shell balance, deriving the corresponding differential equation(s), formulating the initial and boundary conditions, and solving basic ordinary differential equations. 2. Use microscopic models (kinetic gas theory, random walk, Langevins theory) to derive expressions for macroscopic quantities such as pressure and diffusion coefficients; use these expressions to estimate the order of magnitude of these macroscopic quantities. 3. Explain the microscopic framework of multi-component diffusion and solve simple quasi-static one-dimensional problems using the Maxwell-Stefan equations, without and with additional forces (such as wall friction and electric fields). 4. Describe the motion of isolated and non-isolated charged particles using a molecular model; solve basic problems that involve electro-osmotic flows or electrophoresis.	
Education Method	Lectures: theory and applications. Working classes: solving problems, applying concepts to new examples.	
Computer Use	A Python code for generating random walks is provided as part of the learning materials for the working classes.	
Literature and Study Materials	Handouts including theory and exercises with (partial) solutions. Book chapter on multi-component diffusion.	
Assessment	The assessment includes a mid-term assignment in groups, and a final exam. The assignment will enable students to apply the concepts learned in the lectures and working classes to an unfamiliar problem. The grade of the assignment can count towards 20% of the course grade, and the final exam in principle counts for 80% of the course grade. In case the course grade based on both the assignment and the exam is lower than the grade of the exam itself, then the exam counts for 100% of the final grade. Students who do not secure a passing mark for the course based on the assignments and the final exam, can do a resit. In this case, the resit alone counts for 100% of the final course grade.	
Permitted Materials during Tests	The exam is closed-book. Only a non-programmable calculator is allowed.	

CH3412	Biological Transport Phenomena	4
Responsible Instructor	Dr. B. Bera	
Responsible Instructor	Dr.ing. S. Kenjeres	
Course Coordinator	Dr.ing. S. Kenjeres	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	- Concepts from a B.Sc. Transport Phenomena course - Bulk Transport: mass, energy and momentum transport - Micro-scale transport: Diffusion, Conduction, Laminar Flow - Fick's Law, Fourier's Law, Newton's Law of Viscosity - Concepts of Velocity profiles and Shear Stress profiles in Momentum Transport - Derivation of Navier Stokes Equation	
Course Contents	In this course, we will deal with: - Mass Transport in Biological Systems - Momentum Transport in Biological Systems - Energy Transport in Biological Systems - Micro-scale Flows in Biological Systems - Transport at/across Interfaces in Biological Systems	
Study Goals	After completing this course, you will be able to: - demonstrate understanding of the relevance of transport phenomena in various biological systems - Apply the concepts of mass, energy and momentum transport for various examples related to plants and human body - Analyse the features of transport in human body - Apply the concepts of micro-scale transport to relevant biological systems, both natural (blood-stream to cell) and artificial (drug delivery)	
Education Method	Lectures & Tutorials. Familiarity to contemporary relevant literature through group project.	
Computer Use	No programming involved	
Literature and Study Materials	Book: Transport Phenomena in Biological Systems by G. A. Truskey, F. Yuan, D.F. Katz (4th Edition)	
Assessment	There are two parts in the assessment: a literature review (in group) and a final written exam. 40% of the Final Grade will be from the group project literature review, and 60% of the final grade will be from the written exam.	
Permitted Materials during Tests		

CH3421	Computational Transport Phenomena	6
Responsible Instructor	Prof.dr.ir. H.E.A. van den Akker	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc level Transport Phenomena: Applied Transport Phenomena for Chemical Engineering students, or Advanced Transport Phenomena for Applied Physics students	
Summary	This elective course Computational Transport Phenomena is complementary to Molecular Transport Phenomena (about the fundamental processes at molecular level) and Applied Transport Phenomena (how to scale and reduce the model equations and introduce techniques to get approximate solutions which still reflect the relevant mechanisms and processes), all of the MSc program Chemical Engineering. For Applied Physics MSc students of the Physics for Fluids Engineering track, this course builds on the modules Advanced Fluid Dynamics and Advanced Physical Transport Phenomena. Computational Transport Phenomena is about numerical techniques for solving the Navier-Stokes and other transport equations - all being partial differential equations which do not have an exact analytical solution. The course is intended to provide some guidance as to a number of numerical issues and on simulation strategies such as size and type of the computational grid (including 2D vs 3D). The course, however, is not just a tutorial as to how to run Fluent for a variety of applications. Most of the course is devoted to the wide variety of models available for simulating turbulent flows, multi-phase flows (including free-surface flows) and micro-fluidics. If time permits, simulating chemical reactions will also be discussed. While most of the course will deal with finite volume techniques, as used in the common commercial solvers and OpenFoam, attention will also be paid to Lattice Boltzmann techniques.	
Course Contents	An introductory course on Computational Fluid Dynamics (CFD) and Computational Transport Phenomena (CTP): elementary fluid mechanics; numerical and computational aspects; turbulence & turbulence modelling; RANS vs Large-Eddy Simulations; operations and mixing & transport processes in process equipment among which cyclones; fluid-particle interaction; two-phase flows (Euler-Lagrangian vs Euler-Euler); chemical reactions	
Study Goals	Building an understanding of CFD, its promises and its limitations; acquire some experience in numerical and computational exercises; understanding turbulence and multiphase flows and their modelling; getting a good idea about the effects of all types of models on the CFD outcomes; becoming capable of interpreting and assessing CFD results	
Education Method	The course comprises regular classes (where new material is presented by the Professor) as well as hands-on workshops for exploring ANSYS/Fluent CFD software in a Studio classroom type of environment, with the help of Fluent tutorials, a TA being present for assistance. In these workshops and also in the final assignment, students work in teams of two. The emphasis in the course is on the physics and the models underpinning a software code such as Fluent as well as on the impact of boundary conditions, input parameters and numerics. Make sure you will have sufficient time during the semester to be an active participant in the course and to stay on board till the very end. You are also obliged this towards your fellow team member. Attending CLASS AND WORKSHOP is therefore COMPULSORY.	
Assessment	The grading is based on two elements: (a) the active participation in the mandatory weekly tutorials (counts for maximum 10% of the final grade), and (b) the grading of the report on the final assignment (counts for minimum 90% of the final grade). The latter report should make clear that the student has understood the course material, at least as far as relevant to the final assignment. This report should be a critical assessment (using literature data) of the simulation results carried out for the final assignment. A more detailed scheme of aspects contributing to the eventual grading of the final report is to be found on Brightspace. The effort in writing such a report is not to be underestimated! Students are allowed to improve their grades on the basis of the review by the teacher but cannot improve their grade by more than 1.5 on the scale 1-10. The lecturer may invite students for an oral assessment of their final assignment report.	
Remarks	This course will only be given if at least 10 students will participate. The lectures will be given by Professor Van den Akker, on Monday afternoons.	
Elective	Yes	
Tags	Fluid Mechanics Modelling Numeric Methods Transport phenomena	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45030	Turbulence	5
Responsible Instructor	Prof.dr.ir. J. Westerweel	
Instructor	Dr.ir. G.E. Elsinga	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	3mE Department Process & Energy	

ME45042	Advanced Fluid Dynamics	5
Responsible Instructor	Dr. A.J.L.L. Buchner	
Instructor	Dr. D.S.W. Tam	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Undergraduate level course in thermo-fluids (WB2542) Knowledge of vector analysis, multivariable and differential calculus is essential for this course (WBMT1050, WBMT2048)	
Course Contents	This course surveys the principal concepts and methods of fluid mechanics. Topics include control volume analysis for mass and momentum, the derivation of Navier-Stokes equations from mass and momentum conservation for incompressible flows, boundary conditions at fluid/fluid and fluid/solid interfaces, dimensional analysis, interfacial tension, inviscid flows, circulation and vorticity, lift and drag, boundary layers, lubrication theory, classical instabilities and waves. Classical solutions of the Navier-Stokes equations are derived and Concepts are illustrated through practical examples from engineering, geophysics and biological fluid dynamics.	
Study Goals	At the end of this class, the student is able to formulate and to use the fundamental governing equations of incompressible, laminar fluid mechanics in classical flow regimes, and to apply these to a new problem in flow technology. More specifically, the student must know how to: 1. formulate the conservation equations for mass and momentum 2. derive the governing equations of motion for an incompressible flow. (Starting from the conservation equations for mass and momentum, the constitutive equation for a Newtonian fluid, and the appropriate boundary conditions at a fluid/solid and fluid/fluid interface) 3. compute the stress tensor and the aero/hydrodynamic forces on a mechanical system 4. derive the vorticity equation and the consequence on the generation and transport of vorticity in 2D (Kelvin circulation theorem) 5. perform a dimensional or scaling analysis to identify the dominant terms in the governing equations. Also, use the non-dimensional numbers introduced in class to determine the flow regime (Reynolds, Strouhal, Froude, Bond, Capillary, Weber) 6. simplify the Navier-Stokes equations for the case of an inviscid flow (the Euler equations), a boundary-layer, a unidirectional flow, a creeping (Stokes) flow and in the lubrication limit. 7. apply appropriately the different forms of the Bernoulli equation The student must be able to derive classical solutions to the Navier-Stokes equations discussed in class. The student must be able to determine whether these classical solutions can be applied to a new problem and use them to compute the flow fields. These solutions include: 8. For ideal flows: a source, a sink, an irrotational vortex, a dipole, corner flows and the superposition of any combination of these using complex potentials (e.g. the 2D irrotational flow around a cylinder and the interpretation of Lift (Joukowski theorem) and Drag (d'Alembert Paradox)) 9. 3D axisymmetric irrotational flow around a sphere for steady and unsteady flows (added mass) 10. For viscous flows: Unidirectional laminar flows, Couette and Poiseuille flows, flow in a lubrication layer 11. Blasius solution for the boundary layer over a flat plate 12. Waves and instabilities in fluid mechanics, gravity waves, Kelvin-Helmholtz instability When facing a new problem in fluid dynamics, the student should be able to: 13. Write all the relevant boundary conditions 14. Use the appropriate formulation of mass and momentum transport (at the control volume level or at the differential level i.e. Navier Stokes equations) 15. Simplify the governing equations using a dimensional/scaling analysis 16. Solve the governing equations for simple flows, when analytically possible 17. Use well-known solutions derived in class to model flows appropriately for the new problem. 18. Model the effect of aero/hydrodynamic forces on the dynamics of a simple mechanical system	
Education Method	Lectures: 3 hours/week Recitations: 1 hour/week Office Hours: 1 hour/week	
Books	Fluid Mechanics Sixth Edition Kundu, Cohen and Dowling	
Assessment	Midterm Written Exam (35%) Final Written Exam (65%)	
Department	3mE Department Process & Energy	
Contact	email: a.j.buchner@tudelft.nl	

ME45190	Chaos in Dynamical Systems	3
Responsible Instructor	Prof.dr.ir. W. van de Water	
Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Working knowledge of ordinary differential equations, and how to solve them practically on the computer.	
Course Contents	Chaos is the seemingly erratic behavior of simple deterministic, but nonlinear dynamical systems. The systems that we are used to in Mechanics and Physics are almost always linear, as they explore tiny excursions from equilibrium. However, once systems become nonlinear, we are in for surprises. Their state can change dramatically upon variation of a parameter (bifurcations), or they may become chaotic altogether. We will first discuss the route to chaos, where we will already encounter the scaling concepts that return in the description of synchronization, chaotic fluid flow and turbulence. The route to chaos is often universal, and scaling concepts demonstrate why. Systems with no friction are called "Hamiltonian", the route to chaos in Hamilton systems follows a beautiful geometry cartoon, with striking pictures and little mathematics. We will discuss the basic concepts of chaotic behavior, such as the continued sensitivity to variation of initial conditions and the fractal structure of phase space. Then we will look at chaos and synchronization in nonlinear coupled oscillators, again from the scaling perspective. In the meantime we will have practiced bifurcation using "path following" software.	
Study Goals	With this course you will be able to recognize universal routes to chaos, understand synchronization of dynamical systems, and the application of chaos to fluid flow and turbulence.	
Education Method	8 Lectures of 2 hours. There are 4 blocks, corresponding to the 4 assignments. A block has 2 lecture hours, and 2 "studio-classroom" hours, where we will work on the assignments.	
Computer Use	We will use Matlab, but for each assignment there is a template that gets you started.	
Literature and Study Materials	Lecture notes are available	
Assessment	There will be 4 assignments where you play with chaotic dynamical systems using Matlab. The subjects of these assignments cover the course subjects. The last assignment even involves a simulation of a turbulent flow.	
Department	3mE Department Process & Energy	
Contact	w.vandewater@tudelft.nl	

WI4011-17	Computational Fluid Dynamics	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: + continuum mechanics + partial differential equations + tensor calculus linear algebra + basic concepts of the finite element method + basic concepts of iterative solvers and preconditioning The students are also expected to be familiar with programming (e.g., in MATLAB or Python).	
Course Contents	Part I: Numerical methods for the convection-diffusion equation; Stability, consistency and convergence of the numerical methods; Extensions to incompressible Navier-Stokes equations; Part II: Basic iterative solvers and preconditioners for the Poisson problem; Schwarz domain decomposition preconditioners for the Poisson problem; Extension of Schwarz preconditioners to saddle point problems (Stokes and Navier-Stokes equations)	
Study Goals	Learning objectives of the course: Upon completion, the student should be able to: Part I: 1. Derive the equations of fluid dynamics using multivariable calculus and physical conservation laws 2. Explain the singularly-perturbed behavior of solutions to the equations of fluid dynamics 3. Discretize the convection-diffusion equation and derive discretization errors 4. Choose discretizations suitable for convection-dominated flows 5. Discretize the incompressible Navier-Stokes equations Part II: 6. Construct and implement Schwarz domain decomposition methods 7. Explain the concept of the abstract theory of Schwarz methods 8. Apply the abstract Schwarz theory to one- and two-level Schwarz preconditioners to derive condition number bounds for the Poisson problem 9. Apply nonlinear and linear solution techniques to (Navier-)Stokes equations 10. Extend Schwarz methods to (Navier-)Stokes equations	
Education Method	The course comprises 14 lectures (7 lectures for Part I taught by S. Jain, 7 lectures for Part II taught by A. Heinlein).	
Course Relations	Some concepts discussed in the course WI4201 'Scientific Computing' are assumed as prior knowledge for the course WI4011-17. Even though parts are reviewed very briefly to be adapted to computational fluid dynamics problems, students are strongly recommended to take the course WI4201 before WI4011-17.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: + "The Finite Element Method" by T. J. R. Hughes + "Domain Decomposition Methods - Algorithms and Theory" by A. Toselli and O. B. Widlund + "Finite Element Methods for Incompressible Flow Problems" by V. John + "Principles of Computational Fluid Dynamics" by P. Wesseling Additional current topics, which are not covered in standard text books yet, will also be discussed. Further references will be given during the course.	
Assessment	The final grade of the course consists of the following components: - Individual assignments (50%) - Project presentation - Project report (together 50%) Final grade calculation: $(0.5 * \text{assignments}) + (0.5 * \text{presentation and report})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Individual assignments: resubmit deliverable - Presentation: resit presentation - Report: resubmit improved report To pass the course, the following is mandatory: + The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. + The projects must be completed with grades greater than 5.5. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This course focuses on the mathematical background, analysis, and motivation of numerical methods for the challenging applications of the incompressible Navier-Stokes equations and convection-diffusion equations. It follows a bottom-up approach, developing the numerical techniques from scratch rather than building a complex numerical framework for simulating realistic fluid problems, as is often done in more engineering focused courses.	
Co-Instructor	Dr.ir. S. Jain	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Physics for Health And Life

AP3061	Acoustic, Elastic and Electromagnetic Waves	6
Responsible Instructor	Dr. F. Bociort	
Responsible Instructor	Dr. K.W.A. van Dongen	
Contact Hours / Week x/x/x/x	4/4/0/0 (only the first half of the second quarter)	
Education Period	1 2	
Start Education	1	
Exam Period	2 Exam by appointment	
Course Language	English	
Expected prior knowledge		
Course Contents	Introduction to elasticity theory; acoustic waves in fluids; elastic waves in solids; electromagnetic waves; wave propagation in heterogeneous media; Fraunhofer diffraction; Fresnel diffraction.	
Study Goals	To describe a variety of wave phenomena, notably acoustic, elastic and electro-magnetic waves, by the same mathematical concept: the wave equation. To obtain insight in phenomena like scattering, refraction, diffraction and polarisation of waves. These phenomena play an important role in imaging systems based on wave propagation. To understand the relationship between observed wave phenomena and the relevant properties of the medium in which the waves are propagating. Once the forward problem is understood, quantitative inversion of recorded wave-fields, to obtain the media properties, can be undertaken. To understand the motivation for approximate descriptions of wave propagation such as the WKB method as well as the Fraunhofer and Fresnel approximations.	
Education Method	Oral lectures	
Literature and Study Materials	The course is based on the slides from Brightspace and the following books: 'Physics of Waves', Elmore and Heald, Dover 0-486-64926-1. 'Electromagnetic Waves and Antennas', S. J. Orfanidis, 2010 Recommended reading: 'Thin-film Optical Filters', H. A. Macleod, 2001	
Assessment	Two oral exams; one on acoustics and elastodynamics, and one on electromagnetics. The final grade is the arithmetic mean of the grades of each oral exam. The minimum required grade for each exam is 5.0	

AP3082	Computational Physics	6
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.R. Akhmerov	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor physics.	
Course Contents	This course starts with an introduction into methods and techniques in computation that are viable today. Following this, several projects in the field of computational physics are executed by the students, who work in pairs. Projects are on molecular dynamics, (quantum) Monte-Carlo calculations, and a final project of own choice. The course projects are close to the research level.	
Study Goals	At the end of this course a student will be able to 1. describe standard physical simulation algorithms. 2. implement these algorithms in a well-structured computer code. 3. apply these algorithms to calculate physical properties and interpret the outcome. 4. verify correctness of the simulation outcome and use this analysis to improve the code. 5. organize and perform joint programming work. 6. present the results according to the requirements of scientific communication.	
Education Method	Before the start of the semester we offer an optional crash course of two days where you learn how to program in python, use version control and store codes online. The course itself is project-based. This means that it contains little formal teaching, but most of the knowledge is acquired hands-on by implementing, running and analyzing the results of computer simulations. Students are expected to learn the necessary background theory by studying literature and from contact with the lecturers. Another important element is code review, where the students read and comment on each other's code.	
Literature and Study Materials	The textbook Computational Physics by J.M. Thijssen (2nd edition, Cambridge University Press) is useful. Extensive use is made of online resources (course material, cloud storage portals like github/gitlab). In addition, papers from modern research literature are used.	
Prerequisites	The bachelor degree must be finished before the start of the course.	
Assessment	Assessment is based in the results of the three projects, consisting of a paper (first two projects) or oral report (final project), and the simulation code and development history. The grade of each project is rounded to two digit accuracy. The final grade is then computed from these grades, with the first project counting 40%, and the second and third 30% each. The assessment is based on 1. Presentation. 2. Contents (understanding of the physics/algorithms and results). 3. Structure and correctness of computer codes produced. 4. Contribution to the discussions, independence and learning abilities are judged by the instructors and also contribute to the final mark.	
Studyload/Week	The two-day crash course on programming during the spring break is optional and offered as a courtesy to the students. During the course, there will be 2 contact hours per week. The remaining workload is more or less evenly divided over the course, corresponding to around 8-10 hours of course work outside class. (6 ECTS = 168 hours; 168 hours / 15 week = 11 hours per week). To facilitate contact with instructors outside class, a forum will be implemented.	
Schedule	Optional two day crash course on programming during the spring break. 15 weeks with 2 contact hours.	

AP3122	Advanced Optical Imaging	6
Responsible Instructor	Dr. J. Kalkman	
Instructor	Dr. K.S. Grußmayer	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	BSc level courses on analysis, systems and signals, waves, and optics. Experience in programming with Python.	
Course Contents	We explain the basic properties of advanced optical imaging systems within the framework of Fourier optics and linear systems. We treat wave propagation, diffraction, phase transformation of lenses, modulation by apertures and objects. The spatial domain optical systems description will be transformed to the Fourier domain description of coherent and incoherent optical systems. Additionally, the course will treat various advanced imaging systems such as microscopes with different contrast mechanisms, optical coherence tomography, digital holography, and adaptive optics. We will show applications of microscopy in biomedical imaging. The course consists of lectures, pen and paper assignments, and (Python) programming assignments.	
Study Goals	The learning objectives of this course are: - Students can quantify the performance of optical imaging systems by applying Fourier optics methods - Students can theoretically describe the working of advanced optical imaging systems - Students can apply computational optics methods to model and analyze imaging concepts	
Education Method	Lectures, (pen and paper) homework, and computational optics assignments.	
Computer Use	Part of the course consists of making assignments in Python.	
Literature and Study Materials	Lecture notes and online material. The book: Introduction to Fourier Optics by J. Goodman is used as reference material.	
Assessment	Final written exam, computational optics assignments, and homework. The exam grade and the assignments grade both need a grade higher than 5.0. Homework is assessed by pass-fail. A maximum of one point bonus is provided on the end grade according to the ratio #passes/6. The final grade is computed with the following formula: Final grade = 0.6*(Exam grade) + 0.4*(Average assignment grade) + (Homework). The validity of result of the computational optics assignments and homework is restricted to one academic year.	
Permitted Materials during Tests	Handwritten notes page.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-guest-student/	

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morphological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: $((0.85 \cdot EX + 0.15) + AS) / 2$ or without the bonus point from the practicals: $(EX + AS) / 2$ With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3163	Physics of Biological Systems	6
Responsible Instructor	Dr.ir. L. Laan	
Instructor	Dr. J.W. Zwanikken	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic differential equations. Not necessary to have prior knowledge of biology.	
Course Contents	This is a course intended for physicists, biologists and engineers who are fascinated by the question: "What is life?: How does a collection of molecules that are governed by the laws of physics execute all the functions necessary to be alive?" If you are a physicist and have wondered why so many physicists are entering biology, this course will hopefully answer your question in depth. If you are a biologist fascinated by physics this course will show you the basic principles and methods. You do not need prior knowledge from biology or physics (rather you need a willingness to learn), however you need to have basic knowledge of solving ordinary differential equations. You will also need to program, but if needed we will teach you basic programming skills. Specifically you will learn how to apply different fields of physics, such as statistical physics, thermodynamics and information theory to solve biological questions, such as: how does a cell know where to divide, how does a cell know its destiny, how does a cell make decisions,...? You will answer these questions using mathematical models grounded in physics, using both analytical and computational methods. The first part of the course will be mostly dedicated to mathematical and theoretical aspects, while in the second part of the course we focus more on how you can test these theories using quantitative data-analysis and biophysical techniques. The course aims to cover the following topics: cytoskeletal dynamics, material properties of living matter, noise in development and pattern formation inside cells, in combination with contemporary biophysical techniques. Students will explore results of cutting edge, modern research, where novel experimental or theoretical approaches are used to investigate central questions in biology. Students will in groups work on an assignment where they are given experimental or simulated data and are challenged to develop and answer a research question such that they can directly apply their knowledge they obtained in the course.	
Study Goals	- To be knowledgeable about how math and physics can be applied to study life at the molecular and cellular level. - To be knowledgeable about modelling the following physical processes: - noise and stochasticity for decision processes - cytoskeletal dynamics for spatial organization - pattern formation by reaction-diffusion and phase separation processes - contemporary biophysical techniques - To be able to design and interpret a quantitative experiment to address concepts listed above. - To be knowledgeable about current research literature in physics of life. - to be able to formulate, answer and critically review a research question in the field of physics of life.	
Education Method	Lectures and tutorials Peer review in written and oral form student presentations	
Literature and Study Materials	We will provide the study material in class.	
Assessment	The final grade for the course will be determined as follows: - report describing a research study you perform in groups of 3-4 students, where you (1) formulate a research question based on data provided by the lecturer and (2) analyse and interpret the data to answer your research question (20%) - peer review of 3-4 reports by other groups of students (10%) - presentation of your research study including a discussion of the peer reviews you received (20%) - Final exam (50%) In order to pass this course, - students must obtain at least a passing grade (5,8) on the final exam and an overall grade of at least 5,8. - students need to have handed in a report, peer reviews and have given a presentation.	
Permitted Materials during Tests	all material handed out to you during the course.	

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary."	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. We will provide Chapter 1 "Brief introduction to MRI physics" of the book: "Magnetic Resonance Image Reconstruction Theory, Methods, and Applications", 1st Edition - November 4, 2022, Editors: Mehmet Akcakaya, Mariya Ivanova Doneva, Claudia Prieto, Paperback ISBN: 9780128227268 , eBook ISBN: 9780128227466	
Assessment	Additional handouts wherever necessary. The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: $(A + B)/2$. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

AP3352	Introduction to Nuclear Science and Engineering	6
Responsible Instructor	Dr.ir. M. Rohde	
Instructor	Dr. L. van Eijck	
Instructor	Dr.ir. M.C. Goorden	
Instructor	M. Schouwenburg	
Instructor	Dr. H. Schut	
Instructor	Dr.ir. A.G. Denkova	
Instructor	A.L. Smith	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Summary	This course is designed as an introduction to the broad range of topics that comprise nuclear science, including radioactive decay, radiation dosimetry, neutron and positron beams, nuclear reactor physics and designs, nuclear waste disposal, radioactivity in health sciences, and more. The course centers on the *essential concepts* rather than in-depth theory. In other words: this course shines in its broadness of subjects and is perfect for deliberately selecting more in-depth follow-up courses such as e.g. Nuclear Reactor Physics, Nuclear Chemistry, Medical Imaging Signals and Systems and Neutrons, X-rays and Positrons for Studying Microscopic Structures and Dynamics. The lectures are given by staff from all sections within the department of Radiation Science and Technology (RST). Students should complete this course with a greater understanding and appreciation for the relevance of nuclear science and technology in today's global society.	
Course Contents	Subjects include (tentative): - The history of radioactivity - Modes of decay / de-excitation - Interactions of radiation with matter - Radiation dosimetry - Applications of research reactors - Neutron beams - Neutron scattering techniques - Positron beams - Nuclear reactors - Nuclear waste disposal - Health application of nuclear science - Medical imaging - Radiotracers	
Study Goals	Sub-goal (total % for sub-goal) : Weekly exercises(10%)/Exam(75%)/Presentation(25%)* Has insight into the nature of radiation and radioactivity and its interaction with matter (37%) : 5%/37%/0% Understands, in a broad sense, how nuclear science is/will be applied in fields of energy, health, medicine industry and others (38%) : 5%/38%/0% Is able to read, understand and present a scientific paper on a nuclear science subject (20%) : 0%/0%/20% Is able to collaborate with another student, while preparing the presentation (5%) : 0%/0%/5% * The sum is larger than 100% to indicate that the weekly exercises are not compulsory.	
Education Method	Oral lectures, guided tours and presentations given by the students on a subject related to nuclear science.	
Computer Use	Some homework exercises may require basic knowledge on Matlab, Python or Fortran (or any other programming code)	
Literature and Study Materials	J. Kenneth Shultis, Richard E. Faw: Fundamentals of Nuclear Science and Engineering Third Edition (available as e-book. Link will be provided on Brightspace); Lecture slides Homework exercises, to practice every subject of the course. The exercises are meant to train you for the written exam.	
Assessment	A written exam + a presentation of a scientific paper. All students give a presentation in couples about a paper that is related to an NSE subject. The presentation should contain sufficient physics/chemistry (hence, no general/managerial talk). Grade is P. The exam is a written exam and is closed-book. The student is allowed to make a *written* cheat sheet of 1 A4 and bring that to the exam. Grade is E. The final grade is calculated as $G = (P + 3E)/4$ The grade for the presentation can be re-used for the resit in the same academic year. If you fail for this course (either with the exam or resit), you need to redo everything in the next academic year.	
Permitted Materials during Tests	During the tests and exam: calculator + pens + written cheat sheet	
Elective	Yes	

AP3371	Radiological Health Physics	6
Responsible Instructor	D.S. Dihalu	
Instructor	M. Schouwenburg	
Instructor	J.J. van Kammen	
Instructor	Dr. K.R. Huitema	
Contact Hours / Week x/x/x/x	0/0/8/8 (friday) = Note that the course starts earlier and ends later than Q3.	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge		
Summary	The course provides a broad overview of modern health physics. All subjects of health physics will be discussed in depth during a series of lectures. The course also consists of a practical part which is mandatory. The final exam is a theoretic exam in which both the student's ready knowledge and problem solving skills in the field of radiation protection are tested. This course brings participants up to the level of allround radiation protection experts, who may function as a coordination expert in a facility. With sufficient results on the final exam you receive an additional nationally recognized diploma "Coordinating Expert in Radiation Protection". The diploma is required for coordinating radiation experts, but also experts in radionuclide laboratories, medical clinical physicists and doctors of nuclear medicine need the diploma for their work.	
Course Contents	Lectures and tutorials on: - Radiation, radioactivity, decay. - Radiation sources. - Interaction of radiation with matter. - Methods of radiation detection. - Radiation dosimetry. - Radiation shielding. - Biological effects of ionizing radiation. - Internal dosimetry. - Natural and man-made sources/background radiation. - Radiation protection philosophy (protection principles). - Rules and regulations; organizational, procedural and administrative tasks. - Safety measures; operational radiation protection. - Radiation protection when handling open sources. Mandatory parts: - Practical experiments. These include prior preparation at home, active participation during the experiments, documenting results, reporting including discussion and conclusion/giving advice. All these aspect of the practical work will be monitored and scored during the course. - Several written assignments during the course. - Special modules on: Laser Safety, Communication, Organization & Task and Responsibilities, Risk Analysis and Ethics.	
Study Goals	This course will teach you how to work safely with sources of ionizing radiation. It is a broad introduction into the field of radiation protection and the application of ionizing radiation in science, medicine and industry. Because of this broad coverage of the subject it consists of theoretical parts, a mandatory practical part and special modules on topics like laser safety, communication, organisational aspects and risk analyses.	
Education Method	Oral lectures (classes), tutorials/instructions, assignments, lab experiments	
Literature and Study Materials	Massiot P, Jimonet C, European Radiation Protection Course: Basics. EDP Science (2014), ISBN: 978-2-7598-0703-1. Can be purchased on day 1 or 2 of the course.	
Practical Guide	The practical workbook will be made available on Brightspace on the second day of the course.	
Assessment	National exam in May. Resit in December. Two parts: - part 1: Multiple Choice; written closed book exam (max 33 points) - part 2: 4 problems; essay type questions, written open book exam (max 67 points)	
Permitted Materials during Tests	Part 1 (Multiple Choice part): non-programmable calculator and 'clean' dictionary. Part 2 (Essay type questions): All course materials like the textbook, notes, hand outs, etc.	
Special Information	For students from other universities: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Studyload/Week	1 day/week classes + 0.5 day/week labs + 0.5 day/week preparation	
Schedule	Note that the course starts earlier and ends later than Q3.	
Location	The detailed course program will be published on Brightspace in the second week of January. Reactor Institute Delft (building 50) Mekelweg 15 NL - 2629 JB DELFT	

AP3511	Biophysics	6
Responsible Instructor	Dr. T. Idema	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Mathematical skills as taught in e.g. the physics or nanobiology BSc programs: linear algebra, calculus, and differential equations. Thermodynamics and statistical physics at the BSc level (including the laws of thermodynamics, concept of free energy, and partition functions).	
Course Contents	Cells are built of soft matter: materials that are 'easily deformable, e.g. by thermal fluctuations or small shear forces. Fluids, gels and polymers are important examples. In this course, we study the properties of such materials with a special focus on applications in biophysics and biomaterials. Topics include macroscopic and microscopic fluid dynamics, elasticity, viscoelastic materials, polymers, mixtures, and membranes. For all of these we will build models, using the tools of statistical physics, and apply these models to biological examples, such as DNA (polymers), various tissues (viscoelastic materials), liquid-liquid phase separation, and membrane shapes and interactions. We will introduce the relevant mathematical framework where needed, using ideas from tensor calculus, differential equations and differential geometry. At the end of the course, students read and present a recent research paper on a soft matter topic, connecting what they've learned to ongoing developments.	
Study Goals	At the completion of this course, the students will be able to: Give an overview of the various topics that constitute soft matter. State, explain and apply the basic laws of fluid dynamics and elasticity theory. Explain what a constitutive equation is, and derive such equations for given viscoelastic materials. Mathematically describe polymers and membranes. Mathematically describe multi-component systems. Apply the above to various components of living cells.	
Education Method	Lectures and tutorials (until Christmas), reading, discussing and presenting research papers (in January). Video lectures covering the mathematical derivations, complemented by notes, will be provided each week. On-campus (or, if necessary, live online) sessions with the teachers will complement these lectures, focusing on concepts, with ample room for questions and working out example problems. NB: All material is covered in the video lectures, the live sessions with the teachers are meant to be useful, but if you cannot attend due to scheduling conflicts, you can still take the course. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.	
Course Relations	As of the academic year 2019-2020, this course is a joint course with the NB course on soft matter (NB4070).	
Literature and Study Materials	Required: Lecture notes & articles provided during the course. Additional literature: D. Boal, Mechanics of the cell (2nd edition), Cambridge University Press, 2012. M. Doi, Soft Matter Physics, Oxford University Press, 2013, ISBN 978-0-19-965295-2. L. D. Landau and E. M. Lifshitz, Fluid mechanics (2nd edition, 1987) and Theory of elasticity (3rd edition, 1986), Butterworth Heineman. P. M. Chaikin and T. C. Lubensky, Principles of condensed matter physics, Cambridge University Press, 1995.	
Assessment	Homework assignments (problems, 20%), paper presentation (10%), midterm exam (first part of course, 35%), final exam (second part of course, 35%).	
Permitted Materials during Tests		
Enrolment / Application	This course is merged with the MSc Nanobiology course Soft Matter (NB4070). We will use the NB4070 Brightspace page for the course, and the NB4070 code for exam registration.	
Remarks	Students interested in learning more about the maths behind polymers and (especially) membranes are encouraged to check out the elective Geometry of physics (NB4110).	
Elective	Yes	

AP3531	Acoustical Imaging	6
Responsible Instructor	Dr. D. Maresca	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Dr.ir. M.D. Verweij	
Instructor	Dr.ing. P.L.M.J. van Neer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	B.Sc.-level knowledge on wave phenomena and wave theory B.Sc.-level knowledge on Fourier transforms and signal processing. Basic knowledge of programming in Matlab or Python	
Course Contents	Applications of acoustical imaging in medical imaging, seismic imaging and non-destructive testing of construction materials. Review of Fourier and Radon transforms. The 3D acoustic wave equation and its properties. Derivation of the Kirchhoff and Rayleigh integral for acoustic waves. Forward and inverse wave field extrapolation in homogeneous and inhomogeneous media and its use in acoustical imaging. Resolution of the acoustic images and estimation of background propagation velocities. Medical ultrasound: basic physical principles, principles of ultrasound applications in medicine and neuroscience, Doppler imaging, ultrasound contrast agents. Ultrasound transducers and beamforming. The non-linear wave equation and harmonic imaging. Medical ultrasound imaging using contrast agents. Inverse scattering theory and full waveform inversion. Invited medical lectures on ultrasound imaging in cardiology and neurosurgery.	
Study Goals	Thorough understanding of the principles underlying acoustical imaging. Physical aspects and practical insights in the properties of wavefields in a medium. Understand the basic methodologies for geophysical and medical imaging, their resemblances and differences. Acquire comprehension of the potential and limits of current imaging techniques and the value of non-linear inversion techniques. Use theoretical concepts and bring them into practice.	
Education Method	Oral lectures and mandatory homework assignments in Matlab or Python.	
Literature and Study Materials	- Handouts of lecture notes - Book: The Principles of Quantitative Acoustical Imaging, by A. Gisolf and D.J. Verschuur, EAGE Publications BV (for sale at secretary office of section MI/CI, room F268)	
Assessment	Examination after successful completion of homework assignments. Oral exam if nr of students <20, otherwise written exam. For the homework assignments there is only pass/no-pass	
Remarks	The lectures are given by several lecturers: Dr. Eric Verschuur, Dr. Martin Verweij, Dr. David Maresca and Dr. Paul van Neer. Invited lecturers will cover specific topics, including medical lectures.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3582	Medical Physics of Photon and Proton Therapy	6
Responsible Instructor	Dr. Z. Perko	
Instructor	Dr.ir. D. Lathouwers	
Instructor	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Some knowledge in basic programming with Python is required. The assignments can be made in any program that you are familiar with, but the big assignment notebooks are provided in only Python.	
Course Contents	The course deals with various topics concerning radiotherapy (mainly photon and proton): Introduction to radiotherapy Clinical aspects Radiobiology Biophysical models Proton therapy physics Dose shaping and plan optimization Dose calculation using clinical and Monte Carlo schemes Theory of dosimetry and in-vivo dosimetry Brachytherapy	
Study Goals	Main learning objectives 1. The student can define and explain the main clinical aspects of radiotherapy for different treatment modalities, list and describe the steps of a typical radiotherapy workflow. 2. The student can list and describe the basic principles of radiobiology and discuss their practical consequences for radiotherapy. 3. The student can recall and explain the function of biophysical models in radiotherapy, is able to apply these models and evaluate different treatment approaches by interpreting results. 4. The student can describe the main parts of the machinery used in various modes of radiotherapy and explain their role in administering dose. 5. The student can explain the clinical treatment planning process, can apply dose optimization methods and solve simplified planning problems, and can create and evaluate different treatment plans for idealistic model geometries. 6. The student can identify possible sources of errors arising in radiotherapy, and explain and describe correction methods used in the planning process and in practice for high precision radiotherapy. 7. The student can describe and explain the basic physics related to particle acceleration and beam transport, and can perform dose distribution calculations.	
Education Method	Lectures	
Literature and Study Materials	Handouts of powerpoint presentations and questions	
Assessment	The final grade for the course is calculated according to the following formula: $\text{Final grade} = 0.4 * \text{Homework grade} + 0.6 * \text{Exam grade}.$ To earn a passing grade, Homework grade has to be at least 5.5, and Exam grade and Final grade at least 6.0. Homework grade calculation The homework assignments are mandatory. The homework grade is based on the results of 3 small and 3 big assignments, with small and big assignments making up 25% and 75% of the total points for the homework, respectively. Small assignments must be made individually. Big assignments can be made either individually, or in pairs of 2, but you can only pair up with the same person once (so you will have to work with 3 different students if you want to make each big assignment in a pair). Pairs will always receive the same grade. All homework assignments will be handled via Brightspace (BS), with well-defined electronic submission deadlines at 23:59:59 on various dates. Late submission of all homework assignments will be allowed, but is only eligible for partial grade according to the following formula: $\text{Final assignment grade} = \text{assignment grade} * (1 - 0.2 * \text{number of started days after deadline}).$ Examination is written, and resit exam can be oral (by appointment), depending on the number of students. The partial grades for the exam and the homework assignments are valid for 1 year, i.e., for the given and the next academic years.	
Permitted Materials during Tests	Non-graphical/non-programmable pocket calculator.	
Elective	Yes	
Tags	Physics	

CH3412	Biological Transport Phenomena	4
Responsible Instructor	Dr. B. Bera	
Responsible Instructor	Dr.ing. S. Kenjeres	
Course Coordinator	Dr.ing. S. Kenjeres	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	- Concepts from a B.Sc. Transport Phenomena course - Bulk Transport: mass, energy and momentum transport - Micro-scale transport: Diffusion, Conduction, Laminar Flow - Fick's Law, Fourier's Law, Newton's Law of Viscosity - Concepts of Velocity profiles and Shear Stress profiles in Momentum Transport - Derivation of Navier Stokes Equation	
Course Contents	In this course, we will deal with: - Mass Transport in Biological Systems - Momentum Transport in Biological Systems - Energy Transport in Biological Systems - Micro-scale Flows in Biological Systems - Transport at/across Interfaces in Biological Systems	
Study Goals	After completing this course, you will be able to: - demonstrate understanding of the relevance of transport phenomena in various biological systems - Apply the concepts of mass, energy and momentum transport for various examples related to plants and human body - Analyse the features of transport in human body - Apply the concepts of micro-scale transport to relevant biological systems, both natural (blood-stream to cell) and artificial (drug delivery)	
Education Method	Lectures & Tutorials. Familiarity to contemporary relevant literature through group project.	
Computer Use	No programming involved	
Literature and Study Materials	Book: Transport Phenomena in Biological Systems by G. A. Truskey, F. Yuan, D.F. Katz (4th Edition)	
Assessment	There are two parts in the assessment: a literature review (in group) and a final written exam. 40% of the Final Grade will be from the group project literature review, and 60% of the final grade will be from the written exam.	
Permitted Materials during Tests		

CH3764	Nuclear Medicine	4
Responsible Instructor	Dr.ir. A.G. Denkova	
Instructor	Ir. R.M. de Kruijff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Nuclear Medicine is responsible for the diagnosis of more than 30 million patients yearly as well as for the treatment of cancer, especially when tumours have spread all over the body. In this course we will cover the basics of NM including production of radioisotopes, preparation of radiopharmaceuticals, imaging modalities and radionuclide therapy. As part of this course several guest lectures are incorporated given by researchers working in Nuclear Medicine departments who can provide insight into real practice in the clinic.	
Study Goals	Be able to explain which properties of radionuclides are relevant for imaging and therapy Be able to explain the working principle of nuclear imaging techniques Be able to provide basic radiolabelling approaches for radiopharmaceuticals Be able to name and explain different production routes for radionuclides Be able explain the working mechanism of radionuclide therapy and distinguish between the different radionuclide therapies Understand the damage of radiation induced to cells	
Education Method	Lectures + case study	
Prerequisites	Svp invullen of bij prior knowledge	
Assessment	The final grade consists for 70 % of the case study and 30 % the exam. At least a 5 needs to be obtained for the exam to pass the course. The exam will be a written, closed book exam on campus.	

CH3771	Nuclear Chemistry	6
Responsible Instructor	Dr.ir. A.G. Denkova	
Responsible Instructor	Ir. R.M. de Kruijff	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge		
Course Contents	This course is designed for students having either chemistry or physics background who are interested in learning more about nuclear chemistry and applied radiochemistry. This course covers topics from the Bing Bang to the production of radionuclides for medical and industrial applications. This course also provides hands-on-experience through practicals where you get to work with radioactive materials. Students should complete this course with an in depth knowledge of nuclear and radiochemistry.	
Study Goals	1. Identify the factors that affect nuclear stability 2. Explain the different kinds of radioactive decay 3. Interpret a radioactive decay series 4. Understand the interaction of radiation with matter 5. Be able to calculate ranges of different ionising particles 6. Distinguish between different radionuclide production routes 7. Be able to calculate the specific activity of the produced radionuclides 8. Be able to use the principle of Nuclear analytical techniques 9. Be able to apply radiotracer principles 10. Identify the best separation techniques for a specific radionuclide 11. Be able to perform radiodating calculations	
Education Method	Oral lectures on campus if possible. Practical if allowed	
Literature and Study Materials		
Assessment	A midterm exam counts for 20 % and a final exam for 80 % of the grade. The midterm exam will be 'a closed book' written exam on campus. The duration will be 1 hour. The final exam will be a 3 hour written, closed book exam on campus. Attending the practicals is obligatory to pass the course.	
Permitted Materials during Tests		
Studyload/Week	This is a 6 ECTS course composed of 24 lecture hours, 134 self-study hours, a midterm examination of 1 hour, and a final examination of 3 hours. This class will meet twice per week for approximately two hours each day.	

NB4070	Soft Matter	6
Responsible Instructor	Dr. T. Idema	
Contact Hours / Week x/x/x/x	6/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Mathematical skills as taught in e.g. the physics or nanobiology BSc programs: linear algebra, calculus, and differential equations. Thermodynamics and statistical physics at the BSc level (including the laws of thermodynamics, concept of free energy, and partition functions).	
Course Contents	Soft matter physics is the study of materials that are 'easily deformable, e.g. by thermal fluctuations or small shear forces. Fluids, gels and biological tissues are important examples. In this course, we study the properties of such materials with a special focus on applications in biophysics and biomaterials. Topics include macroscopic and microscopic fluid dynamics, elasticity, viscoelastic materials, polymers, mixtures, and membranes. For all of these we will build models, using the tools of statistical physics, and apply these models to biological examples, such as DNA (polymers), various tissues (viscoelastic materials), liquid-liquid phase separation, and membrane shapes and interactions. We will introduce the relevant mathematical framework where needed, using ideas from tensor calculus, differential equations and differential geometry. At the end of the course, students read and present a recent research paper on a soft matter topic, connecting what they've learned to ongoing developments.	
Study Goals	At the completion of this course, the students will be able to: Give an overview of the various topics that constitute soft matter. State, explain and apply the basic laws of fluid dynamics and elasticity theory. Explain what a constitutive equation is and derive such equations for given viscoelastic materials. Mathematically describe polymers and membranes. Mathematically describe multi-component systems. Apply the above to various components of living cells.	
Education Method	Lectures and tutorials (until Christmas), reading, discussing and presenting research papers (in January). Video lectures covering the mathematical derivations, complemented by notes, will be provided each week. On-campus (or, if necessary, live online) sessions with the teachers will complement these lectures, focusing on concepts, with ample room for questions and working out example problems. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.	
Literature and Study Materials	Required: Lecture notes & articles provided during the course. Additional literature: M. Doi, Soft Matter Physics, Oxford University Press, 2013, ISBN 978-0-19-965295-2. D. Boal, Mechanics of the cell (2nd edition), Cambridge University Press, 2012. L. D. Landau and E. M. Lifshitz, Fluid mechanics (2nd edition, 1987) and Theory of elasticity (3rd edition, 1986), Butterworth Heineman. P. M. Chaikin and T. C. Lubensky, Principles of condensed matter physics, Cambridge University Press, 1995.	
Assessment	Homework assignments (problems, 20%), paper presentation (10%), midterm exam (first part of course, 35%), final exam (second part of course, 35%). There will be a single resit which replaces both the midterm and final exam. If there are very few participants for the retake, it may be given as an oral instead of a written exam. There is no retake option for the homework; if students cannot give their presentation for legitimate reasons and inform the teacher in time, they will be given an alternative option to present.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Remarks	Students interested in learning more about the maths behind polymers and (especially) membranes are encouraged to check out the elective Geometry of physics (NB4110).	
Location	Delft	

NB4160	Engineering of living systems	3
Responsible Instructor	Dr. G.E. Bokinsky	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is intended for students from Nanobiology; however students from other programs are certainly welcome. Prior knowledge of molecular biology at a basic level is assumed. Material for students with minimal biology background can be provided.	
Course Contents	Even the simplest organism is more complex than the most sophisticated man-made device. How did biological evolution build such intricate machines, and could we build such organisms ourselves? In this course, we will examine how to engineer the properties of existing organisms. In doing so, we will consider the both the successes and failures of modern synthetic biology in its efforts to apply general engineering principles to biology. Through lectures and interactive discussion of recent research articles, we will develop an integrated view of biological behavior, evolution, and the state-of-the-art of bio-engineering. We will explore how we can use modern genetic engineering tools to engineer the properties of organisms. We will study basic tools and advanced synthetic-biology approaches that can be used to engineer biological solutions for real-world problems. A background in biology will be appropriate preparation for participating students. Throughout the course you will get the opportunity to study, present, and discuss recent articles from the scientific literature on cutting-edge discoveries and techniques. Specific topics include: real-time experimental evolution, the modern genetic toolbox, feedback control and logic gates, design properties and construction principles of synthetic biology circuits and networks, and the debugging and optimization of engineered networks.	
Study Goals	By the end of the course, students will be able to: 1. Summarize how living systems work in terms of structural and functional concepts at multiple levels of biological organization. 2. Identify why living systems are engineerable. 3. Describe the limitations of the use of engineering approaches learned from other fields when applied to biology. 4. Apply design principles to create new biological functions. 5. Improve and debug the results of genetic engineering by utilizing specific genetic and evolutionary principles. 6. Critically evaluate literature papers in the fields of synthetic biology. 7. Assess the social and safety implications of genetic engineering and directed evolution projects. 8. Engage in constructive scientific discussions	
Education Method	The course will be conducted using a combination of styles to promote student involvement and interaction with the material. Lectures describing and explaining the topics of study will be supplemented with readings from ground-breaking journal articles. Students will present and lead discussions about research articles.	
Literature and Study Materials		
Assessment	Students will be assessed based upon the following: * Literature case study: oral presentation of a journal article and leading a classroom discussion that follows (40%) * Written reflection on the impact of a journal article assigned at the end of the course (30%) * Written responses to journal articles and participation in classroom discussion (30%) Retake possibility: Students who do not receive a passing grade on the Written Reflection assigned at the end of the course are given one opportunity to submit a reflection on a second article.	
Permitted Materials during Tests		
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Elective	Yes	
Location	Delft	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Physics for Instrumentation

AE4896	Space instrumentation	4
Responsible Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of this course is to provide students with an understanding of instrumentation and its role in observational strategies for missions. The course will explore the main observables of the universe and how they are measured in space. For each of the observables, the course will provide: i) The physical context of the observables and how they are generated, ii) The physics of the measure, iii) A review and explanation of the existing and under development space instruments, iv) The missions related to these observables. The course will be divided into three main parts: - The Fundamentals of Instrumentation - Remote sensing space Instrumentation - In-situ space Instrumentation.	
Study Goals	This course aims to familiarize students with the wide range of space instrumentation and observables. By the end of the course, students will be able to: - Understand and describe the relationships between instruments, missions, and observation strategies - Identify and explain the physical observables that space instruments measure, including light, ions, magnetic and electric fields, etc and related phenomena - Distinguish between the different types of space instruments, and explain how they are designed and constructed - Derive instrument requirements from scientific questions - Perform preliminary analysis and use the key driver parameters to size space instruments. Overall, this course will equip students with a comprehensive understanding of space instrumentation, enabling them to dive into space observables diversity and the complex processes involved in their measurement.	
Education Method	Lectures	
Assessment	The exam will test all theories provided in the lectures and skills practised in the provided exercises. It will be a written exam (3h). The resist will have the same settings as the first exam.	

AP3061	Acoustic, Elastic and Electromagnetic Waves	6
Responsible Instructor	Dr. F. Bociort	
Responsible Instructor	Dr. K.W.A. van Dongen	
Contact Hours / Week x/x/x/x	4/4/0/0 (only the first half of the second quarter)	
Education Period	1 2	
Start Education	1	
Exam Period	2 Exam by appointment	
Course Language	English	
Expected prior knowledge		
Course Contents	Introduction to elasticity theory; acoustic waves in fluids; elastic waves in solids; electromagnetic waves; wave propagation in heterogeneous media; Fraunhofer diffraction; Fresnel diffraction.	
Study Goals	To describe a variety of wave phenomena, notably acoustic, elastic and electro-magnetic waves, by the same mathematical concept: the wave equation. To obtain insight in phenomena like scattering, refraction, diffraction and polarisation of waves. These phenomena play an important role in imaging systems based on wave propagation. To understand the relationship between observed wave phenomena and the relevant properties of the medium in which the waves are propagating. Once the forward problem is understood, quantitative inversion of recorded wave-fields, to obtain the media properties, can be undertaken. To understand the motivation for approximate descriptions of wave propagation such as the WKB method as well as the Fraunhofer and Fresnel approximations.	
Education Method	Oral lectures	
Literature and Study Materials	The course is based on the slides from Brightspace and the following books: 'Physics of Waves', Elmore and Heald, Dover 0-486-64926-1. 'Electromagnetic Waves and Antennas', S. J. Orfanidis, 2010 Recommended reading: 'Thin-film Optical Filters', H. A. Macleod, 2001	
Assessment	Two oral exams; one on acoustics and elastodynamics, and one on electromagnetics. The final grade is the arithmetic mean of the grades of each oral exam. The minimum required grade for each exam is 5.0	

AP3082	Computational Physics	6
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.R. Akhmerov	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor physics.	
Course Contents	This course starts with an introduction into methods and techniques in computation that are viable today. Following this, several projects in the field of computational physics are executed by the students, who work in pairs. Projects are on molecular dynamics, (quantum) Monte-Carlo calculations, and a final project of own choice. The course projects are close to the research level.	
Study Goals	At the end of this course a student will be able to 1. describe standard physical simulation algorithms. 2. implement these algorithms in a well-structured computer code. 3. apply these algorithms to calculate physical properties and interpret the outcome. 4. verify correctness of the simulation outcome and use this analysis to improve the code. 5. organize and perform joint programming work. 6. present the results according to the requirements of scientific communication.	
Education Method	Before the start of the semester we offer an optional crash course of two days where you learn how to program in python, use version control and store codes online. The course itself is project-based. This means that it contains little formal teaching, but most of the knowledge is acquired hands-on by implementing, running and analyzing the results of computer simulations. Students are expected to learn the necessary background theory by studying literature and from contact with the lecturers. Another important element is code review, where the students read and comment on each other's code.	
Literature and Study Materials	The textbook Computational Physics by J.M. Thijssen (2nd edition, Cambridge University Press) is useful. Extensive use is made of online resources (course material, cloud storage portals like github/gitlab). In addition, papers from modern research literature are used.	
Prerequisites	The bachelor degree must be finished before the start of the course.	
Assessment	Assessment is based in the results of the three projects, consisting of a paper (first two projects) or oral report (final project), and the simulation code and development history. The grade of each project is rounded to two digit accuracy. The final grade is then computed from these grades, with the first project counting 40%, and the second and third 30% each. The assessment is based on 1. Presentation. 2. Contents (understanding of the physics/algorithms and results). 3. Structure and correctness of computer codes produced. 4. Contribution to the discussions, independence and learning abilities are judged by the instructors and also contribute to the final mark.	
Studyload/Week	The two-day crash course on programming during the spring break is optional and offered as a courtesy to the students. During the course, there will be 2 contact hours per week. The remaining workload is more or less evenly divided over the course, corresponding to around 8-10 hours of course work outside class. (6 ECTS = 168 hours; 168 hours / 15 week = 11 hours per week). To facilitate contact with instructors outside class, a forum will be implemented.	
Schedule	Optional two day crash course on programming during the spring break. 15 weeks with 2 contact hours.	

AP3091	Elementary Particles	6
Responsible Instructor	Dr. N. Tuning	
Instructor	Dr. E.J. Buis	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 Exam by appointment	
Course Language	English	
Expected prior knowledge	In the theoretical part, some knowledge on special relativity and quantum mechanics is expected, but the most important concepts are repeated during the course.	
Course Contents	Introduction to the theory of, and experiments in, particle physics. The theoretical part includes: 1) 4-vector calculus to understand the kinematics of relativistic particle collisions; 2) Basic concepts of quantum mechanics, scattering theory, and the Dirac equation, describing relativistic spin-1/2 particles; 3) Quarkmodel, isospin and strangeness; 4) Standard Model Lagrangian, gauge invariance, and the resulting interactions between particles; 5) Quantum electro- and chromodynamics, and e⁺e⁻ and e⁺p scattering experiments; 6) The description of massive bosons and fermions through the Higgs mechanism. The experimental part includes: 1) Accelerators; 2) Electro-magnetic interaction of charged particles with matter; 3) Working principles of gaseous 4) Working principles of semiconductor (silicon) detectors; 5) Interaction of photons with matter (e⁺e⁻ pair production, Bremsstrahlung, Cherenkov radiation); 6) Scintillators and calorimeters. Demo-experiments are shown, and include: cloud chamber, cosmic muons through scintillators, drift chamber, medipix, gridpix, positron annihilation.	
Study Goals	To master the basis of the Standard Model of Elementary Particles. The theoretical part aims at a thorough understanding of the Standard Model Lagrangian. To reach this goal, 4-vectors, spinors, gauge-invariance, and the Higgs mechanism are introduced. In addition, the quark model, scattering theory and various historical experiments are shown. The instrumental part aims at introducing all principles and measurement techniques of charged and neutral particles. This includes state-of-the-art (medical) imaging techniques.	
Education Method	Class room lectures. The course is concluded with a visit to the National Institute of Subatomic Physics, and the leading particle physics laboratory worldwide, CERN.	
Literature and Study Materials	The slides shown during the lectures are available on Brightspace. In addition, specific literature is recommended for the theoretical part: M. Thomson, "Modern Particle Physics", Cambridge University Press For supplementary reading, the student is referred to: F.Halzen and A.D.Martin, "Quarks & Leptons". D.Griffiths is followed: D. Griffiths, "Introduction to Elementary Particles", Wiley-VCH.	
Assessment	Written exam. Weekly homework exercises will contribute with 15% to the final score, if it improves the final result: Final grade = max(exam, 0.85*exam+0.15*homework). The retake exam is not scheduled. Students make an appointment for an oral exam. (Unless many students require a retake: in that case a written exam is scheduled.)	
Permitted Materials during Tests	All necessary documentation for the (re-)exam will be provided by the lecturers at the exam.	

AP3113	Quantum Optics	6
Responsible Instructor	T.H. Taminiau	
Instructor	Prof.dr. S. Groeblacher	
Contact Hours / Week x/x/x/x	0/0/2/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Physics bachelor level quantum mechanics, waves, optics, electromagnetism.	
Course Contents	Quantization of the electromagnetic field, Coherent states, Interaction between atoms and light, Coherence, Non-classical light	
Study Goals	At the end of this course the student should be able to: Describe the quantized electromagnetic field, field quadratures, quantum measurements. Have knowledge of coherent states of light, thermal light states, photon number states. Understand correlation functions, squeezing of light, parametric down-conversion, single photon sources. Be able to describe advanced effects in quantum optics, such as Hong-Ou-Mandel effect, quantum key distribution, violation of Bells inequality, etc. Present physics concepts clearly backed by derivations. Discuss and present the main experimental and theoretical principles of the topic chosen for the final project.	
Education Method	The course will be taught in an interactive way. Every chapter is partly presented by the lecturers, and the students present selected topics of problems to the rest of the class. In the second part of the class we study several more advanced topics in more detail, these topics are prepared and presented by the the students in coordination with the teachers, and discussed with the rest of the class.	
Literature and Study Materials	Textbook: C. Gerry and P. Knight, Introductory Quantum Optics, Cambridge University Press, Cambridge, UK (2005)	
Assessment	In-class presentation, midterm exam, final project presentation.	
Permitted Materials during Tests		

AP3122	Advanced Optical Imaging	6
Responsible Instructor	Dr. J. Kalkman	
Instructor	Dr. K.S. Größmayer	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	BSc level courses on analysis, systems and signals, waves, and optics. Experience in programming with Python.	
Course Contents	We explain the basic properties of advanced optical imaging systems within the framework of Fourier optics and linear systems. We treat wave propagation, diffraction, phase transformation of lenses, modulation by apertures and objects. The spatial domain optical systems description will be transformed to the Fourier domain description of coherent and incoherent optical systems. Additionally, the course will treat various advanced imaging systems such as microscopes with different contrast mechanisms, optical coherence tomography, digital holography, and adaptive optics. We will show applications of microscopy in biomedical imaging. The course consists of lectures, pen and paper assignments, and (Python) programming assignments.	
Study Goals	The learning objectives of this course are: - Students can quantify the performance of optical imaging systems by applying Fourier optics methods - Students can theoretically describe the working of advanced optical imaging systems - Students can apply computational optics methods to model and analyze imaging concepts	
Education Method	Lectures, (pen and paper) homework, and computational optics assignments.	
Computer Use	Part of the course consists of making assignments in Python.	
Literature and Study Materials	Lecture notes and online material. The book: Introduction to Fourier Optics by J. Goodman is used as reference material.	
Assessment	Final written exam, computational optics assignments, and homework. The exam grade and the assignments grade both need a grade higher than 5.0. Homework is assessed by pass-fail. A maximum of one point bonus is provided on the end grade according to the ratio #passes/6. The final grade is computed with the following formula: Final grade = 0.6*(Exam grade) + 0.4*(Average assignment grade) + (Homework). The validity of result of the computational optics assignments and homework is restricted to one academic year.	
Permitted Materials during Tests	Handwritten notes page.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-guest-student/	

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morphological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: $((0.85 \cdot EX + 0.15) \cdot AS) / 2$ or without the bonus point from the practicals: $(EX + AS) / 2$ With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3152	Optics for Lithography	6
Responsible Instructor	Prof.dr. W.M.J.M. Coene	
Instructor	Dr. Y. Shao	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	BSc level Optics	
Course Contents	In this course, the students will deep-dive into the fascinating world of optical lithography, which is the key component of integrated circuit (IC) manufacturing in the semiconductor industry. The technology of IC is ubiquitous in today's connected world with smartphones and tablets. The societal need for being connected via this technology has become more than obvious in the current COVID-19 era as working from home, on-line meetings, teaching via on-line courses, and all digital social contacts became the new normal in our life. The complete chain of technologies as used in optical lithography will be discussed. It is exciting to realize that various aspects of optics and optical imaging theory have its detailed application in such a high-end industrial tool as the lithographic scanner. Further, it is inspiring to know more about the key-technology that enabled the digital devices. A few of these technical topics that will be addressed are: - a recap of diffraction theory and its application to lithography - imaging theory and Fourier optics (aerial image formation; and image formation in photo resist) - the imaging optics (for a DUV litho-scanner (at wavelength of 193 nm); and for an EUV litho-scanner (at wavelength of 13.5 nm)) - the mask technology - optical wafer metrology inside the lithographic scanner (for leveling and alignment) - optical wafer metrology outside the lithographic scanner (for overlay and profile) - aspects of the light source, especially the laser-produced plasma source for EUV as a highlight of the course, we will guide you through the R&D process for the Next Generation High-NA EUV Lithographic Scanner Finally, we conclude the course with a visit to the Experience Center of ASML in Veldhoven.	
Study Goals	The primary ambition of this course for the student is to get in touch with a variety of optics related technologies that are needed in the semicon industry. This first contact may trigger new interesting perspectives for the student who aims at: - Starting a PhD-project in one of the related fields - Getting a job in the semicon industry in one of the leading companies or institutes in this field: ASML, Zeiss, TNO, ARC NL, IMEC, VSL, PTB, . The Learning Objectives of the course are multiple, and students will be enabled to: 1. Students are able to apply a Fourier Optics approach to the optical image formation in lithography, hereby including primarily scalar imaging and diffraction theory, and the pattern transfer from mask to wafer, which leads to an aerial image at wafer level. For vectorial imaging with high numerical aperture, students are able to calculate the point-spread function. 2. Students are able to explain the difference between on-axis and off-axis illumination, and to apply Abbe's formalism to simulate the aerial image of an extended source (spatially partially coherent imaging). 3. Students are able to describe the effect of aberrations in the lithographic imaging, especially the effects of defocus, of spherical aberration, of coma and of astigmatism, all within the Fourier Optics formalism. 4. Students are able to analyze the difference between an aerial image at wafer level and the image realized in the photo-resist layer. In particular, the students have to be able to discuss about swing curves and standing wave patterns in resist (incl. polarization). 5. Students are able to discuss about the basic concepts of optical wafer metrology in lithography, both inside and outside the scanner, incl. focus control and level sensing, alignment sensing, critical dimension metrology and overlay metrology, the latter two based on optical scatterometry. 6. Students are able to discuss about the basic concepts of optical lenses and mirrors (design of EUV mirror multilayer structure) for DUV and EUV lithography. In addition, students are able to discuss regarding metrology of the aberrations of the lithographic scanner. 7. Students are able to discuss about various concepts that are key for optical lithography, like computational lithography, Rayleigh resolution limit, Rayleigh Depth-of-Focus, Abbe and Hopkins formalisms, focus&dose process control and process window, source-mask optimization, phase-shift masks, optical proximity effects and correction, and differences between conventional photo-resist and chemically-amplified photo-resist.	
Education Method	In total 14 lectures, followed by a full-day visit to ASML (at Main Campus in Veldhoven, incl. lunch) at the end of the course, before the Exam. 10 lectures by Wim Coene and Yifeng Shao 4 guest-lecturers by ASML-experts in specific fields Structure of one on-line session with 3 time-slots: - Slot-1: 40 min: plenary teaching - Break: 10 min - Slot-2: 40 min: plenary teaching - Slot-3: 15 min: discussion about lessons learned and next assignments and/or guided exercise	
Literature and Study Materials	- Main Book: Chris Mack, Fundamental Principles of Optical Lithography, Wiley, 2007, ISBN 978-0470727300.	
Assessment	The course assessment comprises three parts. All three parts are mandatory. At the end of the course, a list of "hot-topics" for exam questions will be given. For the exam, guidance will be given about the expected extent and level of details of the answer to the exam questions. (1) Written-examination (both for Exam (1st examination) and Resit (2nd examination)): weighing 60% (2) Assignments-during-Course (given at the end of each session; results to be submitted before the next session): weighing 20% (3) Exam-Assignment (given at the end of the course; results to be submitted before the Exam (1st examination)): weighing 20%	
Permitted Materials during Tests	None.	

AP3222	Nanotechnology	6
Responsible Instructor	S. Goswami	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic physics and mathematics are required; solid state physics, materials science and physics of microelectronics are recommended. Good understanding of the typical physics approach to problems is helpful.	
Course Contents	The synthesis and study of objects with nanometer-size functional elements require dedicated fabrication methods. This course will provide the technological and scientific background needed to master the methods of nanofabrication.	
Study Goals	Subjects include: material properties, growth and etching methods; pattern definition and transfer (=lithography); self-assembly; molecular manipulation; methods of inspection, analysis and characterization. The subjects will be taught with a focus on practical applications. The student can evaluate the merits of the fabrication technologies for nanotechnological applications. The student can design the processing steps for a nanotechnological application. The student can find, understand and apply the relevant scientific literature. The student can evaluate, model and understand the functioning of nano-objects. The student can execute a simple but complete nano fabrication task.	
Education Method	Lectures, expert lectures, student presentation & discussion sessions, and lab work.	
Literature and Study Materials	1. Sami Franssila, Introduction to Microfabrication, 2nd edition, Wiley 2. Copies of the lectures 3. Scientific papers for the student presentations and discussions 4. Lab-work manual	
Assessment	Written exam (35%), home work (15%), lab work (30%), and student presentation (20%). The scores for the written exam and the lab work should be both at least 5.0.	
Permitted Materials during Tests	Ruler, pocket calculator	
Enrolment / Application	Because of limited space and support in the cleanroom, a maximum of 30 students can do the course. Enrollment will start on Monday 8 January 2024 at 8:00am via this link: https://tudelft.fra1.qualtrics.com/jfe/form/SV_1NRgv4nMpdMlnb8 Students who do not have a Physics background are asked to submit a motivation (on prior knowledge, current activities and interest in nanotechnology) by e-mail to Srijit Goswami S.Goswami@tudelft.nl. Students who do not attend the first lecture (without prior notice) will have lower priority than those present. Students who do not have their bachelor degree at the start of the second semester (12 February 2024), will have lower priority than those with the degree.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Tags	Lab Research Physics	

AP3243	Lasers and Photodetectors	3
Responsible Instructor	Dr.ir. S.F. Pereira	
Instructor	Dr. N. Bhattacharya	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Bachelor level electromagnetism, waves, optics, quantum mechanics	
Course Contents	Basics of laser theory, types of lasers, detection of light, single and multi-photon detectors	
Study Goals	At the end of this course the student should be able to: Laser part: understand how lasers work, design cavity resonators, understand light matter interaction which leads to light amplification, analyze different ways to operate a laser, and describe the working principles of various types of lasers (gas, liquid and solid state). Detector part: understand the theory of detection of optical radiation and noise in optical detection (classical vs. quantum photon detection), describe different detection schemes and different types of photodetectors and apply the theory in quantum imaging and quantum communication.	
Education Method	The course will have lectures, demonstrations and at the end of the course, the students should do a short project in small groups (2-3 students) about one specific subject related to lasers. We will meet 2x a week.	
Course Relations		
Literature and Study Materials	Textbook: Basics of laser physics 2nd edition, Karl F. Renk, Springer (available as e-book via the TU Delft library DOI 10.1007/978-3-319-50651-7) plus lecture slides and complementary reading material that will be available on Brightspace.	
Assessment	Individual exam plus a report (done in small group) on a specific subject related to the course contents. Final grade: 75% written exam and 20% report 5% individual short oral exam about the project (minimum grade written exam = 5). The project is compulsory. Resit: written exam. The grade of the project will also count on the final grade in this case.	
Permitted Materials during Tests	Calculator and formula sheet	

AP3252	Electron Microscopy Characterization of the Nanoscale	3
Responsible Instructor	S.C. Conesa Boj	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge		
Course Contents	This course provides a comprehensive and in-depth exploration of electron microscopy, specifically focusing on transmission electron microscope (TEM). You will gain a deep understanding of the underlying principles that drive these powerful instruments, and how they are used to advance our understanding of materials science. With a solid theoretical foundation, you will explore into the building blocks and main imaging modes of the transmission electron microscope, including diffraction, imaging with diffraction contrast, atomic resolution images, and spectroscopy. By the end of the course, you will have the knowledge and skills to answer the why and how of using a specific electron microscope technique, as well as the ability to critically evaluate TEM images and data from real-world applications. This course is a must-take for anyone looking to expand their knowledge in the field of materials science and deepen their understanding of electron microscopes and their practical applications.	
Study Goals	* To describe the different TEM imaging modes, their advantages and limitations. * To extract and interpret quantitative structural information from TEM images. * To interpret the crystalline structure of the specimen from the TEM diffraction patterns in terms of Bragg reflections. * To associate TEM measurements with complementary data analysis techniques such as atomic modeling and TEM image simulation tools.	
Education Method	Lectures, expert lectures, examples of TEM problem-solving, practical sessions in the laboratory, and student presentations.	
Literature and Study Materials	Transmission Electron Microscopy. M.D.B. Williams, C.B. Carter. Plenum Press, New York, (1996). ISBN: 0-306-45247-2. In addition, Lecture handouts will be added to the Course documents folder after the lectures.	
Assessment	The assessment of the course consists on the combination of formative assessments. These include: quizzes (10%), problem-solving activities (30%), laboratory reports (30%), and final presentation (30%) where the number in parenthesis indicates the weight that each assessment component has on the final grade of the course. Handing in all the course assessments within the indicated deadlines is compulsory to obtain a passing grade of the course. This includes the final presentation of the course, which has mandatory attendance for all course participants. Since the assessment of the course is formative, there will be no written final exam. Instead, you will have the opportunity to demonstrate your understanding of the material through the various assessments, with the relative weight indicated above. Submitting all the assessments within the deadline is required to obtain a passing grade. Students that fail some of the assessments will be offered an opportunity to redo them in the retake period only. If you are unable to take part in one of the assessments due to a well-justified reason, please contact the responsible instructor. In such cases, it will be possible to repeat the failed assessments in the retake week only. Participation in person in the laboratory activities is also required to obtain a passing grade in the course, and cannot be replaced by any other assignment. The final presentation will take place in person, and students are also required to attend all other presentations as part of their participation and assessment in the course. The dates of the presentations will be indicated in Brightspace.	

AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics	6
Responsible Instructor	Dr. W.G. Bouwman	
Instructor	Dr.ir. N.H. van Dijk	
Instructor	Dr. H. Schut	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	The microscopic structure and dynamics of condensed matter in physics, chemistry and biology is obtained by a wide variety of scattering techniques. Structures are determined on length scales ranging from 10⁻¹¹ to 10⁻⁵ m, dynamics on time scales from 10⁻¹⁴ to 10⁻⁶ s. In this course we discuss the different neutron, X-ray and positron techniques, illustrated by a broad range of scientific and technological applications, for which recent scientific articles will be studied. After a general introduction an overview is given of the scattering theory, both for neutrons and X-rays. Then the different techniques will be discussed in more detail. Diffraction (either by X-rays or neutrons) yields information about the structure of single crystals, powders, and liquids. Small-angle scattering (SAXS and SANS) reveals the shape and arrangement of colloidal systems, micelles or polymers in a solvent. Reflectometry is used to find the composition of layered structures like proteins adsorbed at surfaces or magnetic layers in recording materials. Polarized neutrons are used in two different ways. First, for determining magnetic structures and dynamics. Second, as a special technique (spin-echo) to study the larger length scales and time scales (polymer dynamics). Inelastic scattering measures phonons in crystals, diffusion in liquids and the vibrational spectrum of molecules. Neutron sources (fission and spallation), X-ray sources (from X-ray tube, via synchrotron to XFEL) and instrumentation will be an integral part of this course. An introduction to positron annihilation (PA) will be given. Positron annihilation lifetime, Doppler Broadening techniques are particularly suited for the detection and characterisation of (open volume) defects such as vacancies, vacancy clusters and voids in e.g. metals, alloys, semiconductors and polymers. The main application of the Angular Correlation of Annihilation Radiation (ACAR) technique is the study of the electronic structure of solids, and that of defects, precipitates and nanocrystals. A visit to the neutron and positron experimental facilities at the Reactor Institute Delft will be part of the course.	
Study Goals	Being able to explain the theoretical background of neutron and X-ray scattering and positron annihilation techniques; Being able to talk about the applications of these techniques; Being able to perform basic calculations and interpretations of the techniques; Being able to read and discuss recent articles on the application of the techniques; Being able to talk in depth about one of the topics to be chosen by the candidate in combination with a recent research article.	
Education Method	The course consists of 14 times 2 hours contact on the following chapters: I. Theoretical background neutron and X-ray scattering II. Neutron experimental techniques III. X-ray experimental techniques IV. Diffraction from crystalline materials, theoretical, experimental, application V. Diffraction from amorphous materials and liquids, theoretical, experimental, application VI. Small-angle scattering, theoretical, experimental, application VII. Reflectometry, theoretical, experimental, application VIII. Polarized neutrons, theoretical, experimental, application IX. Imaging and tomography with neutrons and X-rays, theoretical, experimental, application X. Positron annihilation, theoretical, experimental, application Each 2-hours-lecture is accompanied by a set of assignments from the website https://e-learning.pan-training.eu/. This has to be completed before the next class. In the class question about the previous lecture will be discussed as well as recent scientific articles. In turn the students have to give short presentations about the assignments and about the scientific articles. Those students who have attended at least 12 of the 14 lectures and sent in at least 11 of the 13 assignments are allowed to enter the oral exam.	
Literature and Study Materials	Lecture notes. The accompanying material with the website https://e-learning.pan-training.eu/. For mathematical background we recommend the book D.S. Sivia, Elementary Scattering Theory for X-ray and Neutron Users, Oxford University Press, ISBN 978-0-19-922868-3.	
Assessment	Having done at least 11 of the 13 times homework is a requirement for taking the oral exam. The oral exam of 45 minutes determines the final grade. The student proposes an article or scientific topic connected to the course for the first 15 minutes of the exam.	
Elective	Yes	

AP3352	Introduction to Nuclear Science and Engineering	6
Responsible Instructor	Dr.ir. M. Rohde	
Instructor	Dr. L. van Eijck	
Instructor	Dr.ir. M.C. Goorden	
Instructor	M. Schouwenburg	
Instructor	Dr. H. Schut	
Instructor	Dr.ir. A.G. Denkova	
Instructor	A.L. Smith	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Summary	This course is designed as an introduction to the broad range of topics that comprise nuclear science, including radioactive decay, radiation dosimetry, neutron and positron beams, nuclear reactor physics and designs, nuclear waste disposal, radioactivity in health sciences, and more. The course centers on the *essential concepts* rather than in-depth theory. In other words: this course shines in its broadness of subjects and is perfect for deliberately selecting more in-depth follow-up courses such as e.g. Nuclear Reactor Physics, Nuclear Chemistry, Medical Imaging Signals and Systems and Neutrons, X-rays and Positrons for Studying Microscopic Structures and Dynamics. The lectures are given by staff from all sections within the department of Radiation Science and Technology (RST). Students should complete this course with a greater understanding and appreciation for the relevance of nuclear science and technology in today's global society.	
Course Contents	Subjects include (tentative): - The history of radioactivity - Modes of decay / de-excitation - Interactions of radiation with matter - Radiation dosimetry - Applications of research reactors - Neutron beams - Neutron scattering techniques - Positron beams - Nuclear reactors - Nuclear waste disposal - Health application of nuclear science - Medical imaging - Radiotracers	
Study Goals	Sub-goal (total % for sub-goal) : Weekly exercises(10%)/Exam(75%)/Presentation(25%)* Has insight into the nature of radiation and radioactivity and its interaction with matter (37%) : 5%/37%/0% Understands, in a broad sense, how nuclear science is/will be applied in fields of energy, health, medicine industry and others (38%) : 5%/38%/0% Is able to read, understand and present a scientific paper on a nuclear science subject (20%) : 0%/0%/20% Is able to collaborate with another student, while preparing the presentation (5%) : 0%/0%/5% * The sum is larger than 100% to indicate that the weekly exercises are not compulsory.	
Education Method	Oral lectures, guided tours and presentations given by the students on a subject related to nuclear science.	
Computer Use	Some homework exercises may require basic knowledge on Matlab, Python or Fortran (or any other programming code)	
Literature and Study Materials	J. Kenneth Shultis, Richard E. Faw: Fundamentals of Nuclear Science and Engineering Third Edition (available as e-book. Link will be provided on Brightspace); Lecture slides Homework exercises, to practice every subject of the course. The exercises are meant to train you for the written exam.	
Assessment	A written exam + a presentation of a scientific paper. All students give a presentation in couples about a paper that is related to an NSE subject. The presentation should contain sufficient physics/chemistry (hence, no general/managerial talk). Grade is P. The exam is a written exam and is closed-book. The student is allowed to make a *written* cheat sheet of 1 A4 and bring that to the exam. Grade is E. The final grade is calculated as $G = (P + 3E)/4$ The grade for the presentation can be re-used for the resit in the same academic year. If you fail for this course (either with the exam or resit), you need to redo everything in the next academic year.	
Permitted Materials during Tests	During the tests and exam: calculator + pens + written cheat sheet	
Elective	Yes	

AP3382	Advanced Photonics	6
Responsible Instructor	Dr. O. El Gawhary	
Instructor	Prof.dr. H.P. Urbach	
Contact Hours / Week x/x/x/x	0/0/3/3	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Electromagnetism bachelor level, optics course bachelor level. Some knowledge about Fourier transforms and mathematics for physicists at the bachelor level will be useful.	
Course Contents	The course consists of a basic part, which provides all the background needed to study the advanced applied topics, and of a second part where we discuss advanced subjects, which are barely encountered in any textbook but that are of vital importance for any modern applications of optics. In the first part of the course we discuss some of the foundations of optics, embedded in the more general context of electromagnetic theory. Here below some more detail on the content. 1. The basic part provides rigorous (namely at a higher and deeper level than usually treated in other courses and books) treatments of the basic theoretical aspects of optics. This material can be found scattered in different references, but we also provide lecture notes where the theory is described. The material contained in those lecture notes is much more extensive than we can cover in the course. During the first lectures, we start with the electromagnetic theory of optics. We will focus on how to formulate and solve scattering problems in optics using the Lippmann-Schwinger equation. We will discuss how to solve that fundamental equation using perturbation methods (the Born series) and we will explain why this method seems to be fail in practice and what can be done about it. We will recall diffraction theory and we will make use of the stationary phase method to approximate rapidly oscillating integrals, which are often encountered when dealing with propagation problems. The plane wave expansion method to propagate optical fields through homogeneous space is derived. That method is important as it will give us the chance to understand the fundamental limit that exists for the amount of spatial information about an object that can be transferred by any monochromatic light wave (although to really understand why such a resolution limit exists we will need to discuss on inverse problems, ill-posedness, regularization techniques etc, which will do as well). Equipped with this knowledge, you will be able to also understand the true origin of classical resolution limit of optical imaging, which is instrumental to trying to figure out ways to overcome that limit. Then, depending on the applied topics which we will choose to study in the second half of the course, we discuss one or a few topics out of: a) optical beams; b) Parameter estimation and inference; c) partial coherent light and statistical optics; d) holography and phase sensitive imaging; Which of these subjects is covered depends, among other things, on the knowledge and interest of the students taking the course how long and how extensive the basic theory will be treated. 2. In the second half of the course we will apply the theory to study advanced applications. Examples are new (proposed) methods to achieve super-resolution by using meta-materials, hyperbolic materials and multiple scattering. Plasmonic waves and their role in achieving high sensitivity will be treated as well as the concepts of a perfect and super lens are explained. Furthermore, regularization methods in parameters retrieval and inverse scattering problems are discussed. Also new ways of imaging, such as lensless imaging using phase retrieval techniques and ghost imaging are discussed. Optical beams with special properties (non-diffracting beams, beams carrying an orbital angular momentum, phase singularities) may be also discussed. More advanced topics, such as non-trivial symmetries and conservation laws in optics can also be discussed. In many cases, the topics are treated by studying scientific papers together which will be made available. The particular choice of topics will also depend on the preference of the students. Finally, the course is made of tutorials (typically 1 hour a week) and normal lectures. In the tutorial we will show the detailed solution to problems	
Study Goals	1. Students acquire an in depth knowledge about the basic theory of optics. The topics are discussed starting from few common-sense assumptions and starting from those, the theory develops as a natural sequence of logical steps. 2. Students learn the hierarchy of optical models (electromagnetic optics, diffraction optics, geometrical optics) and the the modelling of coherence of light and know when these models can be applied. 3. Students practice reading scientific papers in the field of optics and develop a critical attitude.	
Education Method	oral lectures, tutorials and self-study. Lectures and tutorials are given in a blackboard-like approach (slides are used only very occasionally, to show some figure or the result of a simulation). All material treated during the lectures is then made available to the students in Brightspace.	
Literature and Study Materials	All lecture materials are handed out and will be put on Brightspace. Most of the material is a selected from lecture notes which are also available on Brightspace.	
Assessment	Written or oral exam (depends on the number of students).	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student	
Elective	Yes	

AP3391	Geometrical Optics	6
Responsible Instructor	Dr. F. Bociort	
Instructor	Dr. B. Kruizinga	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	This course consists of two parts: the larger part is about the theoretical basis of geometrical optics, the second part is an introduction to optical system design. The topics are: - Fundamentals of geometrical optics (geometrical optics as a limiting case of wave optics, the eikonal function, rays and wave fronts, ray paths in inhomogeneous media) - Ray tracing (Snells law in vector form, formalism for reflection, refraction and transfer, ray failure, aspherical surfaces) - The paraxial approximation (paraxial and finite rays, matrix formalism, characteristics of ideal imaging, principal planes, telescopic systems, aperture and field stops, pupils, vignetting, marginal and chief rays, Lagrange invariant, F number, telecentric systems) - Aberrations (transverse ray aberration, wave front aberration and the relationship between them, power series expansions for optical systems with or without rotational symmetry, rotationally invariant combinations of ray parameters, defocusing, Seidel aberrations. Aberration balancing, caustic, Design aspects: situations when some aberrations are more important than others, aplanatic surfaces, ideal placement of aspheric surfaces.) - Optical design software, local and global optimization of optical systems.	
Study Goals	- Mastery of the concepts, theories and methods listed above at an advanced academic level. - Development of the ability to perform simple lens design tasks with state-of-the-art lens design software.	
Education Method	Oral lectures and independent work with lens design software. Because of the work with software, late enrollments in this course (more than 10 days after the 1st lecture) are not accepted.	
Literature and Study Materials	1. J. Braat, Diktaat Geometrische Optica , TU Delft 1991 (in Dutch; English-speaking students should use Born and Wolf (Ref 5) instead); 2. J. Braat, Paraxial Optics Handout (on Brightspace); 3. W.T. Welford, Aberrations of Optical Systems, Adam Hilger, 1986 (or the earlier version Aberrations of the Symmetrical Optical System, 1974); 4. F. Bociort, Optimization of optical systems (can be found on Brightspace). Supplementary reading (not required for the exam, just if you want extra depth on some subject): 5. M. Born and E. Wolf, Principles of Optics; 6. R.R. Shannon, The Art and Science of Optical Design, Cambridge University Press, 1997; 7. D. Sinclair, Optical Design Software, Handbook of Optics, Chapter 34.	
Assessment	The final mark will be for 70% the mark for the theoretical part (oral exam) and for 30% the mark of the design part (submitted solutions to exercises). To pass the exam, for both parts the minimal mark is 5.0.	

AP3401	Introduction to Charged Particle Optics	6
Responsible Instructor	Dr.ir. J.P. Hoogenboom	
Instructor	Dr. C.W. Hagen	
Instructor	Dr. A. Mohammadi Gheidari	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge		
Course Contents	Electron and ion lenses, aberrations, deflectors, multipoles, spectrometers, simulation programmes, transmission and scanning electron microscopes, lithography tools, electrical and magnetic fields in vacuum; Laplace equation, Fourier analysis, numerical methods, series expansion, flux lines, equipotential planes, making sketches of these; Geometrical optics: focal point, thick lens model, matrix description, phase space, Liouville, aberrations; Calculation of trajectories: paraxially in lenses, spherical and chromatic aberration constants, paraxially in multipoles, Lagrangians, manual calculations, analytically, numerically far from the axis, adiabatic, wave character; Partial optical elements: magnetic lenses, electrostatic lenses, electron sources, multipoles, analyzers; Partial optical systems: transmission electron microscope, scanning electron microscope (probe calculations), electron beam pattern generator, ion beam pattern generator.	
Study Goals	Understand electron and ion beam instruments and be able to design basic optical components (lenses, quadrupoles)	
Education Method	Explanation of principles, self study of material, assignments, discussion.	
Literature and Study Materials	course book and material on Brightspace.	
Reader	Reader digitally available, a hard copy can be obtained via the instructors (by payment of printing costs)	
Assessment	Assignments	
Elective	Yes	
Tags	Physics	
Studyload/Week	8 hours per week	

AP3412	Experimental Techniques in Optics	3
Responsible Instructor	Dr.ir. S.F. Pereira	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	bachelor level optics, electromagnetism, waves, systems and signals.	
Course Contents	This course is meant to give students an opportunity to have hands-on experience in optical techniques. It includes several experiments and experimental techniques with as goal to achieve a good training in basic optical principles and to master optical techniques such as interferometry, reflectometry, beam shaping. There will be also experiments involving electronics for optics and single photon detection. The students will write a structured report of each experiment that will include describing the goals of the experiment, experimental realization, data analysis and conclusions. The course will consist of five experiments that will include building the setup, taking data and analysing it and writing a report on their findings. The available experiments are in the following subjects: Interferometry, reflectometry, spatial light modulation, detection of evanescent waves in total internal reflection, analogue electronics for optics, single photon interference.	
Study Goals	After taking this course, students are able to: 1. build optical experimental setups 2. take data and analyze them 3. compare the data with theory 4. write reports about their findings	
Education Method	The course will consist of 4 h a week compulsory presence on campus to perform the experiments. Experiments will be performed in groups of 3 students. In addition, after the lab sections, it is expected that the students will meet to analyse their results and write the report.	
Literature and Study Materials	instruction manual and suggested readings will be available prior each experiment.	
Assessment	evaluation of the reports, oral individual evaluation at the end of the course.	
Enrolment / Application	Maximum number of students: 15	

AP3531	Acoustical Imaging	6
Responsible Instructor	Dr. D. Maresca	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Dr.ir. M.D. Verweij	
Instructor	Dr.ing. P.L.M.J. van Neer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	B.Sc.-level knowledge on wave phenomena and wave theory B.Sc.-level knowledge on Fourier transforms and signal processing. Basic knowledge of programming in Matlab or Python	
Course Contents	Applications of acoustical imaging in medical imaging, seismic imaging and non-destructive testing of construction materials. Review of Fourier and Radon transforms. The 3D acoustic wave equation and its properties. Derivation of the Kirchhoff and Rayleigh integral for acoustic waves. Forward and inverse wave field extrapolation in homogeneous and inhomogeneous media and its use in acoustical imaging. Resolution of the acoustic images and estimation of background propagation velocities. Medical ultrasound: basic physical principles, principles of ultrasound applications in medicine and neuroscience, Doppler imaging, ultrasound contrast agents. Ultrasound transducers and beamforming. The non-linear wave equation and harmonic imaging. Medical ultrasound imaging using contrast agents. Inverse scattering theory and full waveform inversion. Invited medical lectures on ultrasound imaging in cardiology and neurosurgery.	
Study Goals	Thorough understanding of the principles underlying acoustical imaging. Physical aspects and practical insights in the properties of wavefields in a medium. Understand the basic methodologies for geophysical and medical imaging, their resemblances and differences. Acquire comprehension of the potential and limits of current imaging techniques and the value of non-linear inversion techniques. Use theoretical concepts and bring them into practice.	
Education Method	Oral lectures and mandatory homework assignments in Matlab or Python.	
Literature and Study Materials	- Handouts of lecture notes - Book: The Principles of Quantitative Acoustical Imaging, by A. Gisolf and D.J. Verschuur, EAGE Publications BV (for sale at secretary office of section MI/CI, room F268)	
Assessment	Examination after successful completion of homework assignments. Oral exam if nr of students <20, otherwise written exam. For the homework assignments there is only pass/no-pass	
Remarks	The lectures are given by several lecturers: Dr. Eric Verschuur, Dr. Martin Verweij, Dr. David Maresca and Dr. Paul van Neer. Invited lecturers will cover specific topics, including medical lectures.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3701	Submm and terahertz physics and applications	3
Responsible Instructor	Dr. A.J.L. Adam	
Responsible Instructor	Dr. J.R. Gao	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The course is intended for MSc students from different disciplines, including those from Applied Physics, Electrical Engineering, and Space Engineering, but it is also open to interested PhD students.	
Expected prior knowledge	Electromagnetism, Any introduction of Optics, and Introduction of solid-state Physics. If you miss some of them, we will also add necessary background knowledge during the course	
Course Contents	The submm and terahertz region of the electromagnetic spectrum, which spans from 100 GHz to 10 THz, has attracted a lot of attention in recent years due to the developments in new detection and generation techniques and due to the applications for astronomic observations in space and the potential applications as an imaging technique for security, communications and biomedical applications. The Submm&THz frequency range is rapidly emerging which allows unearthing new science and technology. We wish to provide a course in which students can get an introduction and learn the basic physics & principle, and will also be updated with the latest developments. The course will focus on the topics like Photonics, Time-Domain Spectroscopy, Antenna/Waveguide, terahertz detectors and sources, THz Space application. Here the titles of all the different lectures given: 1. Introduction to THz. Light Interaction with Matter 2. THz photonics (THz fibers, Photonic crystals & Metamaterials. High THz fields. Nonlinear interaction of THz waves with matter) 3. THz Imaging (THz Time Domain spectroscopy, Transmission, Reflection, Time of Flight Tomography, Near field, Tomography) 4. THz Sensing (Application to Chemistry, Security, Biology & Medicine; Pump-probe experiment) 5. Submm & THz Coupling (Waveguide, Si Lens, Antennas) 6. Sources 7. Nano-detectors and cameras 8. Submm & THz Sensing instruments from space (as a few examples) Additionally: A case study and design of an space instrument by a subgroup of students (Adam/Gao) (To be announced)	
Study Goals	Understand submm & terahertz electromagnetic radiation and propagation fundamentals, and the basics of typical terahertz detectors and sources. Distinguish different terahertz antennas with regard to their advantages and disadvantages for terahertz radiations. Explain the working principle of THz Time-Domain Spectroscopy. Build practical experience on a THz imaging setup and a heterodyne receiver system. Highlight recent advances of THz research and development from the academic and astronomic sectors. Summarize state-of-the-art THz applications, including space applications.	
Education Method	The course will be given as following: 1) one 2x45 min lecture per week 2) Home work every week on the topics following the lecture. The homework counts 30% of the grade. 3) Design an instrument concept and address a specific topics with groups and present 4) a written exam on all topics discussed during the lectures and homework We based our lectures on current research and science on THz technologies	
Literature and Study Materials	The lectures will be based on a few selected chapters of different books. For Gao's lectures: Chapter 5, 6 and 7 in the book of "Terahertz Astronomy" by Christopher K. Walker. For Aurele's lectures: Terahertz Technology, Volume 77, A. Rostami, H. Rasooli, and H. Baghban, An Overview of the Technological and Scientific Achievements of the Terahertz, pp. 189. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011. (http://link.springer.com/chapter/10.1007/978-3-642-15793-6_1) After each lecture, the teachers will provide also the powerpoint slides used during the lectures.	
Assessment	20% from the homework 30 % from case/project study report/presentation; 50 % from the written exam	
Permitted Materials during Tests		
Elective	Yes	
Tags	Building & Spatial Development Physics Projects Technology	

EE4745	Superconducting Astronomical Instrumentation	5
Responsible Instructor	Dr. A. Endo	
Instructor	Prof.dr.ir. J.J.A. Baselmans	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Successful completion of the following courses is recommended but **not required**: - EE4C05 Electromagnetics - EE4510 Advanced Electromagnetics - EE4580 Quasi Optical System - AP3701 Submm and terahertz physics and applications	
Summary	Astronomical signal detection is one of the most extreme areas of electrical engineering, involving scales ranging from nanometers to gigaparsecs in length, microseconds to gigayears in time, and $>10^{40}$ W to zeptowatts in power. Indeed, the quest to detect the faintest signals from the most distant objects in our Universe involves a continuous push in what human can observe. Exposure to such radical problems should expand the scope and imagination of enthusiastic Delft students who do not wish to stay within the limitation of current technology. Detection of light, or electromagnetic waves, is one of the most fundamental methods of sensing, yet the correct understanding of the interaction of light and matter (the detector) requires careful analysis, especially when it comes to detecting signals that are so faint that the quantum nature of light become significant. Through the first part of the course, students will learn fundamental concepts for describing the process of the emission and detection of light from astronomical sources, for example basic astronomical emission processes including black-body radiation, photon bunching, Noise Equivalent Power and noise temperature. One of the earliest yet still most active areas of application of superconductors is astronomical detectors. Nowadays, the extreme sensitivity of superconducting circuits is also being actively utilized in the field of quantum computation. Indeed, one of the most promising architecture for a quantum computer involves the integration of large numbers of superconducting microwave resonators, which were originally inspired by microwave kinetic inductance detectors (taught in great depth in this course.) Students will learn the basics of superconductivity that is relevant for microwave circuits, and get an introduction to a commonly used EM simulator Sonnet lite, helping them also visualize the functionality of distributed element circuits and the effect of superconductivity. Finally, as practical applications we will take microwave kinetic inductance detectors as an example of an incoherent detector, and hot electron bolometer mixers as an example of a coherent detector. TU Delft, in close collaboration with SRON Netherlands Institute for Space Research, are world-leading in the technology of both of these detectors, hence the students can interact with the true state-of-the art of superconducting astronomical detectors with ample live examples.	
Course Contents	1) Introduction to Millimeter-Submillimeter astronomy 2) Superconducting detectors and instruments (mm-submm wave to visible light) 3) Observation techniques	
Study Goals	The students will be able to - Explain and perform calculations regarding the fundamental physics and sensitivity in detecting photons from astronomical sources - Analyze and design simple superconducting transmission line circuits - Explain the working principle and performance (limits) of incoherent and coherent superconducting detectors, especially hot electron bolometers and kinetic inductance detectors.	
Education Method	- Lectures through PowerPoint slide projection - Exercises on calculation problems - Exercises using a free EM solver Sonnet Lite	
Assessment	The final mark will be based on the homework assignments, and a final written exam.	
	In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ME46310	Opto-Mechatronics	4
Responsible Instructor	Dr. N. Bhattacharya	
Instructor	Dr.ir. S. Iskander-Rizk	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Optical sensors and actuators are being integrated at an accelerated rate into mechatronics systems making these systems more precise, autonomous and intelligent. In this course the principles of optics, optomechanics and optical detectors are taught and discussed from the perspective of integration in optomechatronics systems for increased performance and improvement/addition in functionality. Topics included in this course are: 1) The different models of optics 2) Optical measurement methods. 3) Tolerances and sensitivities of mechanical positioning which influence optical performance of a machine. 4) Total system design where optical requirements can be translated to mechanical and mechatronical requirements. 5) Sources of light and optical detectors which can influence system performance (SNR, reliability, accuracy).	
Study Goals	At the end of this course the student can LO 1: Understand the practical applications and limitations of the different models of optics (geometric optics, diffraction optics, electromagnetic waves). LO 2: Understand the basic principles of imaging, interferometry, and spectrometry. LO 3: Understand how aberrations are related to mechanical tolerances and imaging quality. LO 4: Evaluate the function and performance of an optical system including the electronic readout module LO 5: Design an optical system for given environmental and application requirements (stress, thermal, kinematic effects) LO 6: Justify the mechatronic system components choices which can improve the performance of a given optomechanical system LO 7: Understand the influence of material choices and types of mounting optics	
Education Method	Lectures and Quizzes	
Assessment	Written exam + Quiz grade + 2 Guest lecture summary.	
Elective	Yes	
Tags	Broad Challenging Industry Mechanics Mechatronics Physics Signals	
Department	3mE Department Precision & Microsystems Engineering	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	C.S. Smith	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ir. G.V. Vdovine	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Remarks	Old course code: SC4115	
Department	3mE Department Delft Center for Systems and Control	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Track Physics for Quantum Devices and Quantum Computing

AP3082	Computational Physics	6
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.R. Akhmerov	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor physics.	
Course Contents	This course starts with an introduction into methods and techniques in computation that are viable today. Following this, several projects in the field of computational physics are executed by the students, who work in pairs. Projects are on molecular dynamics, (quantum) Monte-Carlo calculations, and a final project of own choice. The course projects are close to the research level.	
Study Goals	At the end of this course a student will be able to 1. describe standard physical simulation algorithms. 2. implement these algorithms in a well-structured computer code. 3. apply these algorithms to calculate physical properties and interpret the outcome. 4. verify correctness of the simulation outcome and use this analysis to improve the code. 5. organize and perform joint programming work. 6. present the results according to the requirements of scientific communication.	
Education Method	Before the start of the semester we offer an optional crash course of two days where you learn how to program in python, use version control and store codes online. The course itself is project-based. This means that it contains little formal teaching, but most of the knowledge is acquired hands-on by implementing, running and analyzing the results of computer simulations. Students are expected to learn the necessary background theory by studying literature and from contact with the lecturers. Another important element is code review, where the students read and comment on each other's code.	
Literature and Study Materials	The textbook Computational Physics by J.M. Thijssen (2nd edition, Cambridge University Press) is useful. Extensive use is made of online resources (course material, cloud storage portals like github/gitlab). In addition, papers from modern research literature are used.	
Prerequisites	The bachelor degree must be finished before the start of the course.	
Assessment	Assessment is based in the results of the three projects, consisting of a paper (first two projects) or oral report (final project), and the simulation code and development history. The grade of each project is rounded to two digit accuracy. The final grade is then computed from these grades, with the first project counting 40%, and the second and third 30% each. The assessment is based on 1. Presentation. 2. Contents (understanding of the physics/algorithms and results). 3. Structure and correctness of computer codes produced. 4. Contribution to the discussions, independence and learning abilities are judged by the instructors and also contribute to the final mark.	
Studyload/Week	The two-day crash course on programming during the spring break is optional and offered as a courtesy to the students. During the course, there will be 2 contact hours per week. The remaining workload is more or less evenly divided over the course, corresponding to around 8-10 hours of course work outside class. (6 ECTS = 168 hours; 168 hours / 15 week = 11 hours per week). To facilitate contact with instructors outside class, a forum will be implemented.	
Schedule	Optional two day crash course on programming during the spring break. 15 weeks with 2 contact hours.	

AP3101	The Interpretation of Quantum Mechanics	3
Responsible Instructor	Dr. A.F. Otte	
Instructor	Prof.dr. G.A. Steele	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor level Quantum Mechanics	
Summary	From the preface of the textbook: The difficulties of quantum mechanics originate from the object it uses to describe physical systems, the state vector $\psi\rangle$. While classical mechanics describe a system by directly specifying the positions and velocities of its components, quantum mechanics replaces them by a complex mathematical object $\psi\rangle$, providing a relatively indirect description. This is an enormous change, not only mathematically, but also conceptually. The relation between $\psi\rangle$ and physical properties leave much more room for discussions about the interpretation of the theory than in classical physics. Many difficulties encountered by those who tried to really understand quantum mechanics are related to questions pertaining to the exact status of $\psi\rangle$. For instance, does it describe the physical reality itself, or only some partial knowledge that we might have of this reality? Does it describe ensembles of systems only, or one single system as well?	
Course Contents	7 weeks, 7 subjects: 1. Historical perspective and present situation, realism in classical physics 2. EPR theorem and hidden variables interpretations 3. Bell theorem and understanding non-locality 4. Modified Schrodinger dynamics and Transactional interpretation 5. Many worlds interpretation 6. Consistent histories interpretation 7. Bohmian Mechanics	
Study Goals	To get a deeper understanding of the conceptual difficulties in quantum mechanics, including of course wave-particle dualism, but also concepts like physical realism, non-locality and decoherence. To get a broad overview of the different interpretations that have been proposed. To be able to discuss the logical and conceptual difficulties of the various interpretations.	
Education Method	A weeks program: The material is studied in advance of the course contact hours. 2 contact hours: (Guest) lecture on the subject + discussion with the (guest) lecturer. 2 contact hours: Discussion among students with input of the teachers/teaching assistants to make sure all questions are raised and discussed.	
Literature and Study Materials	Do We Really Understand Quantum Mechanics? by Franck Laloë, second edition, Cambridge University Press 2019, DOI 10.101/9781108569361	
Assessment	Critical essay on one of the interpretations (includes some extra reading).	
Studyload/Week	Total time spent for 3 ECTS = $3 \times 28 = 84$ hrs: $7 \times 4 = 28$ hrs preparation for class $7 \times 4 = 28$ hrs class 28 hrs for essay	

AP3113	Quantum Optics	6
Responsible Instructor	T.H. Taminiau	
Instructor	Prof.dr. S. Groeblacher	
Contact Hours / Week x/x/x/x	0/0/2/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Physics bachelor level quantum mechanics, waves, optics, electromagnetism.	
Course Contents	Quantization of the electromagnetic field, Coherent states, Interaction between atoms and light, Coherence, Non-classical light	
Study Goals	At the end of this course the student should be able to: Describe the quantized electromagnetic field, field quadratures, quantum measurements. Have knowledge of coherent states of light, thermal light states, photon number states. Understand correlation functions, squeezing of light, parametric down-conversion, single photon sources. Be able to describe advanced effects in quantum optics, such as Hong-Ou-Mandel effect, quantum key distribution, violation of Bells inequality, etc. Present physics concepts clearly backed by derivations. Discuss and present the main experimental and theoretical principles of the topic chosen for the final project.	
Education Method	The course will be taught in an interactive way. Every chapter is partly presented by the lecturers, and the students present selected topics of problems to the rest of the class. In the second part of the class we study several more advanced topics in more detail, these topics are prepared and presented by the the students in coordination with the teachers, and discussed with the rest of the class.	
Literature and Study Materials	Textbook: C. Gerry and P. Knight, Introductory Quantum Optics, Cambridge University Press, Cambridge, UK (2005)	
Assessment	In-class presentation, midterm exam, final project presentation.	
Permitted Materials during Tests		

AP3211	Advanced Solid State Physics	6
Responsible Instructor	Prof.dr. Y.M. Blanter	
Instructor	Prof. M.N. Ali	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The prerequisites of this course are: 1) Having achieved the learning objectives of BSc level Solid State Physics. 2) Being able to explain and apply BSc level Quantum Mechanics. Concepts from the Advanced Quantum Mechanics course will also be used. 3) Being able to use advanced mathematical tools, in particular the integration of functions of complex variables and of distributions.	
Course Contents	The course is organised in two parts: 1)lectures and 2)presentations. Lectures. We start with 15 lectures. 1. Introduction. Elements of band theory and thermodynamics of the Fermi gas. 2. Occupation number representation. Phonons. 3. Fermi liquids. 4. Magnetism in solids 1. 5. Magnetism in solids 2. 6. Landau theory of continuous phase transitions. 7. Scattering and transport in metals. 8. Superconductivity, phenomenology. Kondo effect. 9. Superconductivity, microscopic theory. 10. Superconductivity, Ginzburg-Landau theory. 11. Band structure theory. 12. Quantum matter 1. 13. Two-dimensional materials. 14. Topological states of matter. Presentations. The rest of the course is occupied by student presentations on scientific literature as described below.	
Study Goals	At the end of this course the students will be able to 1) Discuss modern condensed-matter physics theories and apply these tools to the analysis of the electronic properties of real materials with a particular focus on magnetic and superconducting systems. 2) Describe modern experimental techniques used in condensed-matter physics with an emphasis on spectroscopic and transport techniques. 3) Discuss, criticise and relate modern scientific literature on condensed-matter physics.	
Education Method	Lectures, presentations. Slides, notes and references to book chapters are provided.	
Literature and Study Materials	Marvin L. Cohen and Steven G. Louie "Fundamentals of condensed matter physics", Cambridge University Press (though we do not follow the book very closely). Some lecture notes will be available.	
Assessment	The total grade is calculated from the grades of a written test (T), of two presentations (P1 and P2) and of questions (Q), according to the formula $(T + P1 + P2)/3 + Q$, subsequently rounded to the closest integer or half-integer. 6 is the minimum passing grade, 5.5 is a failing grade (it will be added to OSIRIS as 5). Written test. At the end of the third quarter, a written closed book test will be scheduled. We estimate that the test preparation would require the students to spend 6 hours per lecture (including attending the lectures), reading the literature and studying the slides, making it in total (with the attendance of the test) 74 hours. The test will be given as a set of 20 multiple choice questions covering the material of all lectures. The grade for the test, T, is an integer or a half-integer (the number of correct answers divided by two). The grade of the written test must be at least 5 or higher in order to count for the final grade. If due to the COVID limitations a written test can not be held on campus, we will choose either an open book test, or an oral exam, depending on the number of students participating. Presentations. The presentations are graded as follows: each student is assigned two subjects (each a title of a presentation), accompanied by two main research papers which are related to the content of the course (but do not necessarily strictly follow the materials of the lectures) and is scheduled for two presentations (15 minutes each). A presentation is followed by questions from the audience and with oral feedback from the lecturers. Each presentation is graded according to the grading scheme (rubric available on the Brightspace) and receives an integer grade (P1 and P2). Students who are present in the audience also use the same form to provide feedback. The grades for the first presentation (P1) will be posted on Brightspace with the names of the students, to facilitate the feedback. We expect that it takes 4 hours to read each paper and 6 hours to read related literature and to prepare the presentation. Including 4 hours presence in the classroom this amounts to a minimum effort of 24 hours. It is expected that the students who want to pass the course with more than the minimum grade will also attend other presentations and ideally read the presented papers in advance. Questions. In addition to their own presentations, every student is assigned six more papers (which are scheduled to be presented by other students, three in the first round and three in the second round). The student is supposed to read these papers and prepare one question to each of the presentations of these papers. (Study load 5 hours per paper + 2 hour attendance of the presentation session = 42 hours). The questions have to be prepared even if the presentation itself is canceled or rescheduled, and need to be sent to the lecturers 24 hours before the presentation (if the presentation was rescheduled or cancelled, 24h before if was originally scheduled). They are graded as follows: - If the student did not send the question 24h before the presentation (even if the presentation was rescheduled or cancelled), or if the question is judged by the lecturers as poor, -0.25 (subtraction of 0.25); - If the student sent a good question on time but did not show up at the presentation to actually ask it, -0.1 (subtraction of 0.1); - If the student sent the question on time and showed up on the presentation to ask the question; or if the student sent the question on time and the presentation was rescheduled or cancelled, + 0.1. The sum of these grades for six questions makes the total question grade, Q (it may vary between -1.5 and 0.6).	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Schedule	Scheduling of presentations. We will schedule up to 4 presentations per day if needed. There are 14 days for presentations, and every student gives two presentations, which means all the presentations can be held in one room if there are 28 students or less taking the course, and (partially) in two rooms if there are between 29 and 56 students. The existing format of the course can not accept more than 56 students. Due to this fact, and also due to the difficulty of scheduling, we can not accept bachelor students,	

and we require commitment in the beginning that the students will follow the whole course (the students are given two weeks, at the beginning of the course, to decide whether they want to commit themselves to the full course).

Retakes and rescheduling. If a student did not show up at the written test, the retake will be scheduled individually after the last presentation of the course. It will be an oral exam, however, very similar to the written test instead of just choosing the answer, the student may be asked to explain the answer or to provide some additional information related to the question.

Students will have a chance to review the presentation schedule in advance and verify if it is compatible with their time constraints. In case of emergency, if a student can not attend their own presentation, the presentation can be rescheduled to an available slot, or individually after the course.

Questions cannot be rescheduled or retaken.

If a student failed the test or one of the presentations, they can have a retake to improve the grade if necessary, according to the above policies.

Second retakes are not possible.

AP3222	Nanotechnology	6
Responsible Instructor	S. Goswami	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic physics and mathematics are required; solid state physics, materials science and physics of microelectronics are recommended. Good understanding of the typical physics approach to problems is helpful.	
Course Contents	The synthesis and study of objects with nanometer-size functional elements require dedicated fabrication methods. This course will provide the technological and scientific background needed to master the methods of nanofabrication. Subjects include: material properties, growth and etching methods; pattern definition and transfer (=lithography); self-assembly; molecular manipulation; methods of inspection, analysis and characterization. The subjects will be taught with a focus on practical applications.	
Study Goals	The student can evaluate the merits of the fabrication technologies for nanotechnological applications. The student can design the processing steps for a nanotechnological application. The student can find, understand and apply the relevant scientific literature. The student can evaluate, model and understand the functioning of nano-objects. The student can execute a simple but complete nano fabrication task.	
Education Method	Lectures, expert lectures, student presentation & discussion sessions, and lab work.	
Literature and Study Materials	1. Sami Franssila, Introduction to Microfabrication, 2nd edition, Wiley 2. Copies of the lectures 3. Scientific papers for the student presentations and discussions 4. Lab-work manual	
Assessment	Written exam (35%), home work (15%), lab work (30%), and student presentation (20%). The scores for the written exam and the lab work should be both at least 5.0.	
Permitted Materials during Tests	Ruler, pocket calculator	
Enrolment / Application	Because of limited space and support in the cleanroom, a maximum of 30 students can do the course. Enrollment will start on Monday 8 January 2024 at 8:00am via this link: https://tudelft.fra1.qualtrics.com/jfe/form/SV_1NRgv4nMpdMlnb8 Students who do not have a Physics background are asked to submit a motivation (on prior knowledge, current activities and interest in nanotechnology) by e-mail to Srijit Goswami S.Goswami@tudelft.nl. Students who do not attend the first lecture (without prior notice) will have lower priority than those present. Students who do not have their bachelor degree at the start of the second semester (12 February 2024), will have lower priority than those with the degree.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Tags	Lab Research Physics	

AP3252	Electron Microscopy Characterization of the Nanoscale	3
Responsible Instructor	S.C. Conesa Boj	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge		
Course Contents	This course provides a comprehensive and in-depth exploration of electron microscopy, specifically focusing on transmission electron microscope (TEM). You will gain a deep understanding of the underlying principles that drive these powerful instruments, and how they are used to advance our understanding of materials science. With a solid theoretical foundation, you will explore into the building blocks and main imaging modes of the transmission electron microscope, including diffraction, imaging with diffraction contrast, atomic resolution images, and spectroscopy. By the end of the course, you will have the knowledge and skills to answer the why and how of using a specific electron microscope technique, as well as the ability to critically evaluate TEM images and data from real-world applications. This course is a must-take for anyone looking to expand their knowledge in the field of materials science and deepen their understanding of electron microscopes and their practical applications.	
Study Goals	* To describe the different TEM imaging modes, their advantages and limitations. * To extract and interpret quantitative structural information from TEM images. * To interpret the crystalline structure of the specimen from the TEM diffraction patterns in terms of Bragg reflections. * To associate TEM measurements with complementary data analysis techniques such as atomic modeling and TEM image simulation tools.	
Education Method	Lectures, expert lectures, examples of TEM problem-solving, practical sessions in the laboratory, and student presentations.	
Literature and Study Materials	Transmission Electron Microscopy. M.D.B. Williams, C.B. Carter. Plenum Press, New York, (1996). ISBN: 0-306-45247-2. In addition, Lecture handouts will be added to the Course documents folder after the lectures.	
Assessment	The assessment of the course consists on the combination of formative assessments. These include: quizzes (10%), problem-solving activities (30%), laboratory reports (30%), and final presentation (30%) where the number in parenthesis indicates the weight that each assessment component has on the final grade of the course. Handing in all the course assessments within the indicated deadlines is compulsory to obtain a passing grade of the course. This includes the final presentation of the course, which has mandatory attendance for all course participants. Since the assessment of the course is formative, there will be no written final exam. Instead, you will have the opportunity to demonstrate your understanding of the material through the various assessments, with the relative weight indicated above. Submitting all the assessments within the deadline is required to obtain a passing grade. Students that fail some of the assessments will be offered an opportunity to redo them in the retake period only. If you are unable to take part in one of the assessments due to a well-justified reason, please contact the responsible instructor. In such cases, it will be possible to repeat the failed assessments in the retake week only. Participation in person in the laboratory activities is also required to obtain a passing grade in the course, and cannot be replaced by any other assignment. The final presentation will take place in person, and students are also required to attend all other presentations as part of their participation and assessment in the course. The dates of the presentations will be indicated in Brightspace.	

AP3261	Mesoscopic Physics	6
Responsible Instructor	Prof.dr.ir. H.S.J. van der Zant	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 Exam by appointment	
Course Language	English	
Expected prior knowledge	Knowledge of Solid state physics (electrons in metals, Pauli/Bose statistics, energy bands, scattering, conductance); Quantum mechanics (wavefunctions, eigenstates, transmission); Electrodynamics; Elementary Statistical Physics and Thermodynamics.	
Course Contents	Mesoscopic physics is the area of Solid State physics that covers the transition regime between macroscopic objects and the microscopic, atomic world. It concentrates on questions related to the roles of classical and quantum mechanics in these intermediate-sized systems. The main goal of the course is to introduce the physical concepts underlying the phenomena in this field. This is facilitated by the introduction and use of transparent physical models. These models are extensively complemented by (usually very recent) experimental results, highlighting the main aspects contained in the model, but at the same time, stressing the limitations by identifying deviations found in such experiments. While thoroughly discussing the formalisms essential for understanding the subject, it avoids lengthy and highly technical theoretical derivations. Where possible, suggestions for further reading are given, providing convenient paths for an in-depth study.	
Study Goals	Reach understanding of electronic properties of meso-size conductors, appreciate/recognize the role of classical and quantum processes, being able to explain electronic transport phenomena using simple physical concepts.	
Education Method	Weekly lectures, discussion of concepts and experimental results.	
Literature and Study Materials	Syllabus. Review papers.	
Assessment	Written exam; retake oral.	
Permitted Materials during Tests		
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	

AP3281	Quantum Transport	6
Responsible Instructor	Prof.dr. Y.V. Nazarov	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 Exam by appointment	
Course Language	English	
Expected prior knowledge	We operate in presumption that a student has followed bachelor level quantum mechanics, and has got an acquaintance with elementary concepts of math, electromagnetism and solid state physics. The following notions are most important for the course: 1. Math (all courses). Vector and matrix operations. 2. Quantum Mechanics. Notion of the wave function. Quantum-mechanical basis. Eigenfunctions and energy levels. Second order perturbation theory. 3. Solid State Physics. Crystalline structure. Electron states in a crystal. Space and energy scales of typical solid state.	
Summary	The goal of the course is to present the fundamental concepts of Quantum Transport that have influenced all fields of nanoscience rather than concrete experimental techniques. Nevertheless, the intended audience are experimentalists. It is similar in approach to a modern Electromagnetism course where a lot is told about magnetic field and a little about winding the magnetic coils.	
Course Contents	Electrons as scattered waves, quantum interference, quantum mechanics vs Ohm's law, quantum circuit theory, Coulomb blockade, quantum transport in superconducting devices, Josephson effect and Coulomb blockade, weak localization and conductance fluctuations, interference in quantum dots, qubits, quantum transport and environment, dissipation decoherence and dephasing.	
Study Goals	To learn the fundamental concepts of quantum transport.	
Education Method	Lectures and problem-solving sessions.	
Literature and Study Materials	The best source of relevant information is the book Quantum Transport: Introduction to Nanoscience (Authors Yuli V. Nazarov and Yaroslav M. Blanter; isbn: 9780521832465). This book has been written on the basis of this course and suits it well. However, the book contains much more material than it is given in the course. If you do not plan to study Quantum Transport in future, you may restrict yourself to the lecture notes available on Brightspace. They are not maintained anymore and may contain occasional typos.	
Assessment	Numerical and analytical exercises (25% of the grade). Written examination (75 % of the grade) Resit: oral examination by appointment.	

AP3303	Applications of Quantum Mechanics	3
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Bachelor quantum mechanics	
Course Contents	This course covers four topics which are not taught in the Delft Bachelor quantum courses. The topics and their presentation center around applications from past and current research. The topics are (with some example applications): 1. Variational Calculus (Transmon qubit, three-body problem) 2. The WKB approximation (STM, Esaki Diode, Lennard-Jones and Morse potentials) 3. The adiabatic approximation (Qubit manipulation, D-wave quantum computer) 4. Scattering theory (Resonant scattering, imaging electron flow)	
Study Goals	This course should enable you to: LO1: Discuss the topics of the course description by clarifying assumptions and limitations of the theoretical models. LO2: Analyse applications of quantum mechanics through calculations. LO3: Evaluate relevant scientific literature by understanding and reproducing the conclusions LO4: Theorize a research problem by applying the theory of quantum mechanics.	
Education Method	Lectures, home work assignments, and presentation	
Literature and Study Materials	Textbook: Griffiths, "introduction to quantum mechanics", pearson-prentice hall. Lecture notes provided by the course team. The material is also covered in numerous other standard quantum mechanics textbooks, like Liboff, Shankar,	
Assessment	Various papers will be distributed in the course of term. The assessment in this course has the following elements: 1) One mandatory written exam at the end of the course. 2) One mandatory student group project conducted in the final part of the course. The grade for the group project is based on a final presentation by the group. 3) 3 voluntary hand-in assignments throughout the course. The course cannot be passed without doing the mandatory written exam and group project. The final grade will be calculated as $\text{Grade} = \max(0.85 * (\text{grade exam/project}) + 0.15 * (\text{grade hand-ins}), \text{grade exam/project})$ where $(\text{grade exam/project}) = 0.8 * \text{grade exam} + 0.2 * \text{grade project}$ $\text{grade hand-in} = (\text{grade hand-in 1} + \text{grade hand-in 2} + \text{grade hand-in 3})/3$ Thus the voluntary hand-in assignments can maximally contribute 15% to the final grade and only if they improve on the grade from the exam and project.	
Permitted Materials during Tests		

AP3421-PR	Quantum Information Project	2
Responsible Instructor	Dr. L. di Carlo	
Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	Different	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	This is an optional additional module for students who have taken the QIST4310 (Fundamentals of Quantum Information) course.	
Course Contents	Working in groups, students will perform an independent project (to be defined by students with approval/help from instructors). The goal is to simulate/implement one of the protocols and algorithms covered in the lecture course using one of several available simulators and quantum processors. Examples of protocols or algorithms would be: Shors algorithm on a few qubits, Bell-inequality violations, tomography experiment of a one/two-qubit state, process tomography experiment of one/two-qubit gate, implementation of a small quantum error correction decoder, etc. Possible simulation platforms include: Microsoft Liquid, QuTech QX, QuTech QuantumSim, QuTech Network Simulator, QuTip. Possible experimental platforms include: IBM Quantum Experience, QuTech superconducting quantum processor.	
Study Goals	a) to apply knowledge from the Fundamentals of Quantum Information course; b) to present and defend the content to specialists in this field of research.	
Education Method	Supported self-study in groups of three students.	
Literature and Study Materials	See QIST4310	
Prerequisites	The Fundamentals of Quantum Information (QIST4310) course is required for this project. Credits for this project are not valid without passing the QIST4310 course.	
Assessment	Assessment: Each group will (1) hand in the source code if applicable and (2) make a final presentation of the project at a final meeting including the instructors and students. The instructors will assign a grade based on quality, originality, creativity, and effort.	
Enrolment / Application	Participation is limited to a maximum of 30 students (10 groups). Selection is based on a proposal/motivation. Proposals (including names of all participants) must be received before Januari 23. The selection will be announced before Februari 1.	

AP3432	Quantum Hardware 1 - Theoretical Concepts	4
Responsible Instructor	C.K. Andersen	
Instructor	M.F. Rimbach-Russ	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Undergraduate electricity and magnetism; AP3421 Fundamentals of quantum information; Bachelor Quantum Mechanics	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to control, couple, transmit and read out the fragile state of quantum systems with great precision, and in a technologically viable way. Quantum Hardware I is focused on teaching theoretical physics concepts for understanding this Hamiltonian engineering challenge in various quantum hardware platforms. The material will be taught using example systems such as superconducting qubits, atomic qubits, defect centers, and trapped-ion qubits.	
Study Goals	This course should enable you to: L01: Discuss single-qubit and two-qubit gate dynamics, qubit measurement, Rabi oscillations, dephasing & relaxation times on an abstract level. L02: Analyse quantum hardware systems by obtaining effective dynamics using approximations to the full Lindbladian of the system. L03: Evaluate performance of quantum hardware by identifying sources of noise and quantifying the effect of these. L04: Establish various quantum hardware systems relative to each other by identifying challenges and opportunities of the systems for quantum information processing.	
Education Method	The course will consist of weekly lectures and tutorial exercise sessions. Every week there will be hand-in assignments, which are factored in the final evaluation of the course.	
Course Relations	This course forms part of the curriculum on Quantum Technologies offered at TU Delft, which at present consists of: AP3421 Fundamentals of quantum information CS4090 Quantum communication and cryptography AP3432 Quantum Hardware 1 - Theoretical Concepts AP3442 Quantum Hardware 2 - Experimental State of the Art EE4575 Electronics for quantum computation Other Related Courses: AP3112 Quantum Optics and Lasers	
Literature and Study Materials	In this course, we will make use of: 1) 0Lecture notes And some selected material from the textbooks: a) Nielsen and Chuang, Quantum computation and information, Cambridge University Press. b) Gerry and Knight, "Introductory Quantum Optics", Cambridge University Press	
Assessment	Both hand-in assignments during the course and a final written exam will contribute to the final assessment of the course. Specifically, hand-in assignments will contribute 30% and the final written exam 70% of the final grade. Feedback on the hand-in assignments will be provided on a weekly basis and should help prepare students for the final exam.	
Continuing Courses	AP3442 Quantum Hardware 2 - Experimental State of the Art AP3472 Modeling of Superconducting Devices AP3662 Special Topics in Quantum Technology	

AP3442	Quantum Hardware 2 - Experimental State of the Art	4
Responsible Instructor	Prof.dr.ir. L.M.K. Vandersypen	
Instructor	Prof.dr.ir. R. Hanson	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	This course requires an understanding of undergraduate electricity and magnetism and the following concepts from quantum information theory: - Hamiltonian of a spin-1/2 particle and of a harmonic oscillator, and tensor products - quantum states (pure and mixed states, including density matrices) - unitary evolution (generated from the above Hamiltonians) - standard quantum operations (X, Y, Z rotations, Hadamard, CNOT, CPhase, SWAP, ...) - decoherence and non-unitary operations (e.g. through the product operator formalism) - simple quantum circuits and algorithms (e.g. teleportation, Deutsch-Jozsa algorithm, Grover's algorithm) - basics of quantum error correction (3-qubit bit flip/phase flip code, notion of fault-tolerance threshold) - simple quantum communication protocols (e.g. BB84, entanglement swapping for quantum repeaters) - quantum measurement (projective measurement in different bases)	
Course Contents	Quantum hardware is what turns the novel concepts of quantum computation and communication into reality. The key challenge is to initialize, control, couple, transmit and read out the fragile stage of quantum systems with great precision and in a technologically viable way. While Quantum Hardware I is focused on teaching the underpinning theoretical tools, Quantum Hardware II will give you an overview of the experimental state-of-the-art. You will learn about several promising approaches for realizing quantum hardware, and critically assess the strengths and weaknesses of each approach. You will also get insight in the conceptual similarities and differences between the various technologies. Specifically, the course will cover general concepts and considerations of qubit hardware, trapped ions, superconducting circuits, quantum dots, impurities, cold atoms, photonic circuits, single-photon sources, single-photon detectors and quantum repeaters.	
Study Goals	- To acquire a good understanding of the requirements of quantum hardware for computing, communication and sensing, both at the conceptual level and at the practical level. - To acquire conceptual insight in the operation, opportunities, and challenges of various qubit realisations. - To obtain a good overview of the state-of-the-art.	
Education Method	In this course, lectures are combined with homework assignments as well as student presentations of recent research papers.	
Literature and Study Materials	Review papers to accompany the lecture slides.	
Assessment	Grades will be established based on homework assignments (40%), paper presentations (25%), and a final exam (35%). The lowest homework grade does not count. This means you can miss one homework without having points deducted. Two missed homeworks will affect your grade, regardless of why you missed them. In unusual cases, a retake exam--which will be oral--will be offered by appointment.	
Continuing Courses	This course forms part of the QuTech Academy curriculum on Quantum Technologies offered at TU Delft, which at present consists of: - AP3421 Fundamentals of quantum information - CS4090 Quantum communication and cryptography - AP3432 Quantum Hardware 1 - Theoretical Concepts - AP3442 Quantum Hardware 2 - Experimental State of the Art - EE4575 Electronics for quantum computation	

AP3452	Quantum Error Correction	4
Responsible Instructor	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Fundamentals of Quantum Information (QIST4310) or a similar course which covers the basic model of quantum computing (qubits, quantum circuits). Optional: Quantum Hardware I (AP3442) or a course which goes into the physical description of quantum computing.	
Course Contents	We discuss the ideas behind quantum error correction and fault-tolerance such as the threshold theorem, code concatenation, stabiliser and topological codes, code constructions (e.g. color codes, hypergraph product construction), fault-tolerant gate constructions (code deformation), decoding and bosonic codes. There will be some hands-on Python exercises of decoding.	
Study Goals	Obtain deep and clear understanding of the notion of quantum error correction and the tools and concepts that have been developed. Understand how quantum error correction applies to real noise of various sources. Understand the idea behind fault-tolerance.	
Education Method	Weekly Lecture + Weekly Exercise/Study Session. Exercises will be provided and discussed weekly (no grading).	
Literature and Study Materials	Lecture notes and slides.	
Assessment	Final oral exam (list of topics will be provided) (80%) + Hand-in of numerical work (20%).	

AP3472	Modelling of Superconducting Devices	4
Responsible Instructor	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	Course not taught in 2023-2024	
Exam Period	none	
Course Language	English	
Expected prior knowledge	QIST4310 Fundamentals of Quantum Information, AP3432/AP3442 Quantum Hardware 1 and 2	
Course Contents	In this course we discuss tools for the theoretical and numerical modeling of superconducting devices with Josephson junctions (circuit-QED). We discuss the theory of electric circuit quantization and black-box quantization which allows one to translate electric circuits into Hamiltonians and their dissipative environment. We review various of the known superconducting qubits, couplers and amplifiers obtained within this description. We use the language and tools of quantum optics to understand essential properties and dynamics of the devices. Some guest lectures will be given.	
Study Goals	Ability to model and analyze the dynamics of a superconducting qubit or device. Ability to read and understand current literature on circuit-QED.	
Education Method	Weekly Lecture + Weekly Exercise Session	
Literature and Study Materials	Lecture notes, slides, scientific publications and textbook "Quantum Engineering" by A.M. Zagoskin (Cambridge University Press)	
Assessment	60% Weekly Assignments + 40% Presentations by students	
Special Information	This course is aimed at 2nd year MSc students and PhD students working with superconducting qubits.	

AP3681	Fairy tales of theoretical physics	6
Responsible Instructor	Prof.dr. Y.V. Nazarov	
Instructor	Prof.dr. J.M. Thijssen	
Instructor	Prof.dr. Y.M. Blanter	
Instructor	Dr. A.R. Akhmerov	
Instructor	Dr. M.T. Wimmer	
Instructor	E. Greplová	
Instructor	Dr. N. Chopigala	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Selected problems of theoretical physics illustrating power and beauty of the discipline.	
Study Goals	To learn the spirit of theoretical approach to physics. To be able to solve and formulate similar problems.	
Education Method	In order to illustrate the power and the beauty of theoretical physics, seven theorists present selected problems (tales). There will be 12 lectures given, each devoted to a specific problem. In addition, there will be a problem-solving session in the end of the semester.	
Course Relations	Fairy Tales is the name of the course given at Cambridge (UK) university by prof. D. E. Khmelnitskii. He shall be credited for the idea of the course and for some study material we give. You can learn more about this course here: http://www.tcm.phy.cam.ac.uk/~dek12/Ad.html	
Literature and Study Materials	The original tales corresponding to the lectures we are going to give are to be found under Course Documents/Tales. We do not follow these tales strictly adding more material or skipping some. The actual lecture notes are to be found after each lecture under Course Documents/Lectures and video presentations. The home-work is posted under Assignments. Please remember that you will need the four well-documented solutions of the homework problems for the examination.	
Assessment	In the semester, there are two tests based on multiple choice questions concerning the lecture material. By the end of the semester each student (or a team of two) should deliver their own creative problem on a topic of a lecture (explicit solution is not required). The score on the tests and the creative problem contributes to the final grade. The final grade is set in course of oral examination by appointment. In order to make an appointment, it is required to produce and submit well-documented solutions of four problems, those are homework problems or creative problems contributed by students. The appointment for the oral examination should be made before the end of september.	
Elective	Yes	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - Insight into the differences between classical and quantum communication and cryptography - Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: - Slides - Review Articles Auxilliary: - Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Homework (weighting 70%) - Final Project (weighting 30%) Final grade calculation: $0.7 * \text{Homework} + 0.3 * \text{Final Project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: - Homework: re-submit deliverable - Final Project: re-submit deliverable	
Tags	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). - Abstract - Adventurous - Algorithmics - Challenging - Group Dynamics/Project Organisation - Information & Communication - Integrated - Intensive - Involved - Lineair Algebra - Mathematics - Physics - Quantum - Signals - Technology - Telecommunication	

QIST4300	Qubit Dynamics and Quantum Control	4
Responsible Instructor	V.V. Dobrovitski	
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	0/8/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The students will learn fundamental methods and approaches to manipulating and controlling qubits (in the absence and in the presence of noise), which are used in virtually all existing architectures for quantum information processing. The students will study * fundamental notions used for description and analysis of the qubit dynamics * typical approaches to manipulation and control of most qubits, from superconducting qubits to trapped ions and atoms; * basic concepts, such as rotating-frame description of qubits, Rabi driving, decoherence by noise, and decoherence suppression via dynamical decoupling. * basic regimes of qubit dynamics and their theoretical description, basic approaches to description and analysis of the qubit decoherence by noise and by environment, as well as basic approaches to qubit manipulation and decoherence suppression.	
Study Goals	- Recognize, explain, and illustrate the basic regimes of qubit dynamics and the basic sources of noise and decoherence relevant for different qubits; - Identify typical scenarios of free and driven qubit dynamics, assess the relevant sources of decoherence and noise, and determine relevant models for description and analysis of dynamics of single- and multi-qubit systems; - Apply these relevant models and evaluate the possible influence of noise and decoherence on the qubit dynamics in different typical regimes; - Choose appropriate models for describing different types of qubits, determine corresponding noise and decoherence models, and calculate the parameters of the qubit dynamics in the presence of noise and coupling to the environment; - Explain the basics of the dynamical decoupling protocols and the physical processes underlying this method of decoherence suppression; - Apply the knowledge about qubit control and decoherence suppression to diverse qubits, such as impurity centers, quantum dots, (partially) superconducting qubits, and other similar systems; - Choose and identify the dynamical decoupling protocols, and evaluate efficiency of different dynamical decoupling protocols under different conditions.	
Education Method	Lectures followed by seminars/tutorials, including supervised problem solving, and independent reading and problem solving. Final project to be performed during the course, and presented at the end of the course	
Assessment	Assessment will consist of the four tests given during the course, and the final project to be performed during the course and presented at the end of the course. Besides, optional (but graded) homework assignments will be given after each lecture. The final grade will be determined as $0.75 \times [\text{average grade for the tests}] + 0.25 \times [\text{final project grade}]$. The average grade for the optional homework assignments will be taken into account only if it improves the final grade; in that case, the final grade will be determined as $0.15 \times [\text{average homework grade}] + 0.6 \times [\text{average test grade}] + 0.25 \times [\text{final project grade}]$. If the student wants to improve the final grade then they can take a final exam over the content of the whole course, and the final grade will be a maximum of the two grades, the average grade over the tests/homework/final project, and the grade for the final exam.	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. L. di Carlo	
Responsible Instructor	Dr. M. Blaauboer	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	30% homework assignments, 10% online quizzes, 60% final exam. A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course.	

QIST4400	Quantum Computer Architecture	5
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Instructor	Dr.ir. T.H. Oosterkamp	
Instructor	Dr. R. Ishihara	
Contact Hours / Week x/x/x/x	0/0/4/0/	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The course will assume knowledge of fundamentals of quantum information. Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. Covered topics include: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller from the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	- Slide decks including small exercises - Videos including quizzes - Interactive lectures including activities - Homework assignments - Paper reading including processing its content	
Assessment	The final grade consists of two parts: - Part 1: Weekly homework, will be submitted electronically. Individual submission. - Part 2: Final assignment, preparation of presentation of 20 minutes with 10 minutes Q&A. Assignment will be handled by pairs of two, there will be individual grades. The global course grade is an average of each parts grade (50% part 1 + 50% part 2). Students are allowed to omit one homework. There is no resit.	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

General Electives 2023

Introduction 1

General electives are technical and science subjects that are not related to a track. A maximum of 6EC can be done in the core programme. Additional modules fit in the R&D orientation.
If you wish to include a module in the programme that is not on the lists, approval from the Board of Examiners must be obtained.

4403TGR64	Theory of General Relativity	6
Contact Hours / Week x/x/x/x	Prof.dr. A. Achúcarro (achucar@lorentz.leidenuniv.nl) This course is taught at the University of Leiden. Schedule: https://www.student.universiteitleiden.nl/binaries/content/assets/science/onderwijs/sterrenkunde/schedules/r23_msc-schedule.pdf	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	The classical theory of gravity can be interpreted as the geometry of space-time. It combines the language of field theory with the language of differential geometry on manifolds. The basic concepts of these subjects will be introduced and it will be shown how they are used to describe gravitational phenomena, from relativistic corrections to orbits of planets and stars, to gravitational waves, black holes and the evolution of the universe at large. Topics: Review of newtonian gravity Review of Special Relativity Differential geometry on manifolds, including the concepts of metric, connection and curvature Equivalence principle, geodesic motion Einstein equations, variational principle Gravitational waves Schwarzschild solution, stars and black holes Cosmology of a homogeneous and isotropic universe	
Study Goals	This course provides the basis for work in relativistic astronomy, cosmology, quantum gravity and applications. It will: - acquaint you with the concepts of field theory and differential geometry - teach you to use these concepts to solve problems of gravitational physics and cosmology - prepare you for advanced courses on gravitational physics and cosmology - allow you to access the scientific literature on these topics and understand the recent progress in experimental and observational research	
Education Method	See: https://studiegids.universiteitleiden.nl/courses/117628/theory-of-general-relativity	
Literature and Study Materials	See: https://studiegids.universiteitleiden.nl/courses/117628/theory-of-general-relativity	
Assessment	See: https://studiegids.universiteitleiden.nl/courses/117628/theory-of-general-relativity	
Enrolment / Application	Registration as a guest student at Leiden University is required for Brightspace access and registration of grades! Application form: https://fd24.formdesk.com/universiteitleiden/guest-student For Study programme specify "MSc Physics" Contact Arno Haket for further information and for this statement https://www.student.universiteitleiden.nl/binaries/content/assets/ul2student/in-en-uitschrijven/guest-student-statement-own-institution.pdf	
Special Information	This course is taught at the University of Leiden. It is not available for international (Erasmus) exchange/guest students visiting TU Delft.	

AP3652	Electronics for Physicists	3
Responsible Instructor	Ing. R.N. Schouten	
Contact Hours / Week x/x/x/x	6 full days in November and December	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	We will study electronics with a strong focus on practical applications. After reviewing the basics of passive and active components and their practical limitations, we will focus on circuit simulation, systematic troubleshooting and opamp circuits. Signals, noise and interference problems (and solutions!) will also be an important topic. We finish with an overview of microwaves and various measurement techniques, and a day on advanced use of electronic measurement equipment. Several case studies from the physics lab will be used throughout the course to make the theory come alive. In the afternoon labs you will build circuits and work with equipment like scopes, generators, network analyzers and lockin amplifiers.	
Study Goals	Understand and master electronics and equipment to be able to design and operate complex measurements in a research lab.	
Education Method	The course will be given over 6 full days (9h30 to 12h30 and 13h30 to 17h30). Course days are a combination of lectures and doing lab-experiments. Additional reading and preparation in between the course days will take another 3 to 5 days.	
Literature and Study Materials	The Art of Electronics, Horowitz and Hill, 2nd or 3rd edition, Cambridge University Press, New York, 1989. Please note that the 3rd edition of this book is available now. Students should be in possession of the book (either the 2nd or the 3rd edition) when the course starts. Extra material is handed out during the course.	
Assessment	Labs, active participation. Credits will be awarded only to those participants who attend the complete course and perform the required lab exercises.	
Enrolment / Application	https://casimir.researchschool.nl/casimir-course-electronics-for-physicists--registration-open-5863.html Registration is not yet a warranty for a seat, a waiting list might be present. Receiving a conformation of the registration makes the status clear.	
Remarks	The course is mainly intended for PhD students and postdocs (working in the experimental field) of the Casimir Research School. Motivated second year master students are also welcome to apply but should realize that registration is not yet a warranty for a seat, usually we have a waiting list.	

AP3751	Artificial Intelligence for Physicists	4
Responsible Instructor	Q. Tao	
Instructor	E. Greplová	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course is part of the Applied Physics Programme. The lectures are given by Applied Physics Teachers. In this class the most fundamental and important concepts of artificial intelligence (AI) methods will be introduced for Physics students. We will discuss multiple AI methods and offer hands-on opportunities of applying these methods in physics problems. Specifically, the class will focus on the following content: 1. Basic concepts of artificial intelligence and machine learning 2. Supervised and unsupervised learning 3. Fully connected and convolutional neural networks (CNN) 4. Generative models 5. Applications to physics	
Study Goals	At the end of the course, you will be able to: 1. Explain the basic concept of artificial intelligence (AI) and major categories of methods in machine learning. 2. Determine in which context the use of machine learning techniques is meaningful and relevant in physics, either in addition to or instead of standard computational approaches. 3. Code supervised, and unsupervised AI models and apply them in physics applications. 4. Analyze, benchmark and interpret the results of the machine learning model they built.	
Education Method	The class will be composed of two parts: Lectures and practical sessions. 1. Lectures: Frontal teaching in form of 8 lectures, 2 hours each, on the fundamentals of AI methods in the context of physics 2. Practical sessions: Throughout the semester, weekly practical sessions will be offered. These sessions serve as an important course component to gain hands-on experience in coding, analysing AI models on real data, and receive formative feedback on end project.	
Computer Use	yes	
Assessment	40% homework 60% end project	

AP3831	Systems Engineering for Physicists	3
Responsible Instructor	Dr. M. Sovago	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course is an elective course for Master's students interested in learning how to design a complex system. TU Delft Master's students at Applied Sciences (and EEMCS) Faculty are welcome to join the course. It is meant for the first year Master's students.	
Course Contents	It is an obligatory preparatory course for the Systems Engineering (Physics Design) Company Project (AP3841). During Systems Engineering (SE) course for applied scientists, you will learn and discover the art behind designing, building and managing complex systems. You will get to master SE tools and methodologies as well as management capabilities. Every week you will learn the theoretical aspects of SE during an interactive lecture. Step-by-step, you have the possibility to discover the process and the tools that we can use to design a complex system. You can apply the knowledge from the lecture to design your own system. As we find important to offer the students the opportunity to put own creativity into practice, you have the possibility to define own system that will solve a day-to-day problem or a societal challenge. A few systems will also be shown as example and inspiration for the choice of your system. Together with a team of ~ 5-7 peer-students, you will work together for 10 group project sessions on your system of choice. The steps you will follow are: 1. Define the problem you will solve and the system your will design: Problem Statement and choice of a system 2. Define the requirements of the system: Requirements specification 3. Define and interview possible stakeholders 4. Define the functions of the system: Functional Decomposition 5. Discover possibilities to fulfil the requirements: Means and Morphological Analysis 6. Decide the best designs for your system: Decision Making 7. Perform a risk analysis for your system 8. Model and Design your System: use 3D CAD modelling software We do not aim to build a prototype of the designed system, but you will design your system by using a 3D CAD modelling software (Fusion 360). This SE process will be implemented to design and build a complex system during the AP3841 Systems Engineering company project (2nd year Master).	
Study Goals	By the end of this course, you will be able to: 1. Recognize the need and relevance of Systems Engineering (SE) for every applied scientist 2. Structure the key steps in Systems/Product/Process Development 3. Systematically breakdown tasks and work in interdisciplinary/multicultural teams to design a complex system 4. Apply SE methodologies and SE tools in projects 5. Learn and apply transferable skills along the course: chair a team for 1 or 2 weeks, team management and team roles, put your creativity into practice, presentation and writing skills (based on weekly activities), self- and team-reflection.	
Education Method	The course is centred around 10 group project sessions. You will work in a group of about 5-7 students with different backgrounds and study tracks. Each session focuses on one aspect of Systems Engineering. Weekly group project cycle: 1. A plenary and interactive lecture: learn the SE tools and methodologies. You will be able to ask questions and have discussions with your lecturer 2. Team activity: you will implement SE tools to design and engineer a complex system. You will be working with your peer-students and be guided by your TA. Each week you will have a team chair (a peer-student) 3. Weekly presentation: present your weekly results and progress to your lecturer and TA and receive feedback. The chair of the week is in the lead, but everyone participates to the discussion. 4. Weekly report: write your weekly report and receive feedback from your lecturer. The content of the report and adjusted feedback can be easily used for the final report. End Q4: final presentation and final report	
Literature and Study Materials	Study material will be provided on BrightSpace.	
Assessment	The final grade is a combination of the following components: Group Grade 1. Final Presentation (30%) 2. Final Report (30%) Individual Grade 3. Weekly chair Presentation and Report (40%) Please Note: 1. You will need a grade of 6.0 or higher in all components to pass the course 2. Participation/contribution to discussions to SE lectures is expected 3. Participation/contribution to team brainstorming, preparation for the weekly presentation and report is expected 4. Participation/contribution to weekly presentations is compulsory 5. You can receive a bonus/penalty for your team contribution	
Tags	Design Group work Physics Projects	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: $0.2 * \text{Assignments} + 0.8 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Instructor	Dr. N.M. Gürel	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%) Final grade calculation: $1 * \text{Written exam}$ A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IFEEMCS4250	Statistical Learning for Engineers	4
Responsible Instructor	Dr. H.N. Kekkonen	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra, Calculus, Probability& Statistics on the level of TU Delft Bachelor; some programming skills in Python.	
Course Contents	Linear regression, classification, linear model selection and regularisation, tree based methods, artificial neural networks, and deep learning.	
Study Goals	After this course you should - be familiar with several commonly used statistical and machine learning methods. - know how to estimate unknown parameters and make predictions based on data. - be able to apply statistical learning techniques using Python. - know how to evaluate model accuracy and be aware of the strengths and limitations of different models.	
Education Method	Lectures and exercise + computer labs	
Assessment	85% Final exam and 15% homework assignments. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Enrolment / Application	Please note: students do not need to register via Osiris for this course, only for the corresponding exams. Students only have to enroll via Brightspace.	
Co-Instructor	Dr.ir. W.T. van Horssen	
Co-Instructor	Dr. N. Parolya	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations) Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimization for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Is able to categorize high-performance computer systems 2. Is able to explain and classify parallel and distributed architectures, and programming models; 3. Is able to explain the concepts of data decomposition and parallel algorithms; 4. Is able to apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: $0.5 * \text{Written exam} + 0.5 * \text{lab work}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Enrolment / Application	Via Brightspace	

LM3692	iGEM	15
Responsible Instructor	Dr. M.A. Vieira Lara	
Practical Coordinator	Ing. E. Yildirim	
Contact Hours / Week x/x/x/x	n.a.	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	iGEM, the international genetically engineered machines competition was started in 2003 at MIT, Boston. The objective of the competition is to design and build an engineered biological system using DNA technologies over the course of the summer. The engineered systems (mostly E.coli) will be constructed from standard biological parts (plasmids), called BioBricks in the iGEM community. The team chooses their own project: It can be a very fundamental project, such as biological information processing, or a more applied project, with uses for medical, environmental or manufacturing purposes. During the course of the summer you will design your experiments, construct parts (plasmids) and validate the constructs using (new) characterization assays. Besides the molecular work an important aspect of iGEM is modeling. Comparable to engineering approaches, the parts' function and interaction are modeled to predict their behavior and foster the understanding of relevant characteristics. Thereby the modeling assists the design and construction of improved parts and experimental methods. Alongside the scientific work, you will be cooperating with scientific and non-scientific communities to promote and communicate your project. The project will be publicized through public relations efforts, social media, and the teams own wiki page. Funding for the team will be raised by the students from university groups and local industries. The project will also have a policy and practice component, in which the social, legal, and ethical issues of the project are considered, through considerations of the possible risks of the project, interviews with relevant end-users and affected community members, and designing of appropriate safety and deployment mechanisms. The team will be advised by several experienced faculty members, technicians, and graduate students. A diverse group of students is optimal, with backgrounds ranging from biology and bioinformatics to chemistry, physics, and engineering disciplines. You will have the opportunity to travel to Boston (or Paris) and interact with and compete against >250 universities from all around the world.	
Study Goals	After this course the students are able to: - Develop a relevant project idea. - Plan and perform experimental work, esp. cloning (digestion, ligation, transformation). - Develop assays for the characterization of gene functions (e.g. reporter genes, enzyme activities, growth phenotypes, etc.). - Evaluate experimental observations using mathematical modeling - Test hypothesis using experimental design. - Document your project and progress (web, wiki, poster, presentations) - Communicate the project and results to the scientific and non-scientific audience. - Organize the work in a team. - Raise funding from governmental organizations and industry.	
Education Method	Participation in the TU Delft iGEM team	
Assessment	The performance is monitored during the project course. Depending on the major contribution, the following aspects will be assessed: Intellectual skills - Logic, reflection - Independent formation of opinion - Motivation/interest/enthusiasm - Application of theory from scientific literature Scientific method and approach - Independent approach to work - Systematic approach - Takes initiative - Works well with others and participates in group work - Critical attitude (in a positive sense) Competence in research and/or design - Practical skills - Analytical skills - Structured, exact, and efficient working style - Creativity Reporting - Able to document observations & results (wiki) - Logical structure - Consistency - Accurate representation of the project's progress (group meetings, wiki, small paper reports) Presentation, communication and discussion - Level of (final) presentation - Handling of critical and scientific questions - Depth and argumentation in answering questions	
Enrolment / Application	This MSc project is open for students from all MSc programs at TU Delft. To apply as member of the iGEM team, please write an application letter including: - motivation to participate in an iGEM competition - describe your favorite project idea - which role in the team would suit you most? It is MANDATORY to contact your academic counsellor prior to applying. Interviews prior to the formation of the team are part of the application procedure. In this interview supervisors will discuss with you your motivation, academic level, but also your 'soft' skills and other personal capacities, such as communication & writing skills, time-management, pro-activeness, being a team player, etc.	

	Please also include a C.V. with a list of courses in the application. It is recommended that you contact your study-advisor prior to applying. NB-students, planning the iGEM project as their minor, should certainly contact the study-advisor or program coordinator in advance.
Remarks	The iGEM team is limited to 12 students. Students are welcome from all study programs of TU Delft. For LST master students: The course grants 15 ECTS. 12 ECTS can be included in the LST master study program (by application to the board of examiners, e.g. to replace the design project). The remaining 3 EC are extracurricular. This means your exam program will be set on 123 credits in total. Excellent students are encouraged to apply for the TU Delft honours program. Please contact the director of education (Ulf Hanefeld) IN ADVANCE. The iGEM project can be performed as part of an individual honours program, under the condition that the Design project will be carried out as well! For Applied Physics master students: The course grants 15 ECTS. Upto 12 ECTS may be included in the Applied Physics master study program as a General Elective. The remaining 3 ECTS are extracurricular. Alternatively, the course can be done in an individual honours program by students qualifying for the TU Delft honours program. For bachelor students: the course grants 15 ECTS. These can be included in the TU Delft minor program DREAM-team (WB-Mi-121-12). They require the propedeuse diplom, have finished at least 30 ECTS second year courses including all practical courses, achieved excellent grades and have e.g. gained additional experience as teaching assistant. Students will be selected to enter the minor program. Contact your study advisor for approval and the supervisor of DREAM Team (Dr. ir. Wim Thijs) to set up a complete minor program of 30 ECTS. iGEM may also be used to build a free minor; please contact your study advisor to help arrange this.
Studyload/Week	Part time in Q3 Full time in Q4 and during the summer
Schedule	January, February: Team formation February April: Brainstorming meeting, Fund raising April October: Lab- & Modelling work Mid November: iGEM world-finals (Boston, USA)
Location	Delft, mainly in the Applied Sciences Building (nr. 58)

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The final grade of the course consists of the following components: - homework exercises deadline start of Q2 leads to grade G1, - take home exam deadline half of January grade G2 - written exam grade G3. The final grade is $(G1+G2+2*G3)/4$, provided that all grades are larger than or equal to 5. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework exercises: resubmit deliverable - take home exam: resubmit deliverable - written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4260TU	Scientific Programming for Engineers	3
Responsible Instructor	Dr.ir. H.X. Lin	
Practical Coordinator	Ir. C.W.J. Lemmens	
Contact Hours / Week x/x/x/x	0/0/2/0 + lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: experience with a programming language (e.g., Python or C), and linear algebra Strongly advised: experience with linux is a plus (but not a prerequisite).	
Course Contents	The course tries to bring students to a level where they are able to change algorithms from e.g., numerical analysis into efficient and robust programs. It comprises 1. Introduction to programming in general; 2. Floating point number rounding-off errors and numerical stability; 3. (Numerical) Software design; 4. Data Structures; 5. Testing, debugging, and profiling; 6. Efficiency issues in computing time and memory usage; 7. Optimization and dynamic memory allocation; 8. Scientific software sources and libraries. P.S. (i) This is not a general introductory course to learn how to write a computer program. Furthermore, the course concentrates mainly on sequential programming and only briefly introduces parallel programming (MPI and OpenMP). More advanced topics like threads or parallel (MPI/GPU) programming on supercomputers are not covered by this course (they are covered by other courses, e.g., Introduction to High-Performance Computing). (ii) This is a course aimed at master students, however, if you have followed the course Scientific Programming in the BSc CSE-minor, then you should not select this course as a Master elective course due to the large overlap.	
Study Goals	1. Is able to explain the problems related to the use of floating-point numbers in scientific computing; 2. Is able to implement, compile, and debug programs in a Linux programming environment, e.g., use of a makefile. 3. Is able to explain the impact of data structures on the performance of a program, e.g., data locality, cache and memory access. 4. 3. Is able to explain the impact of the program on the performance of a program, e.g., temporal locality, pipelining, and loop-carried dependency. 5. Is able to analyze the performance of a program, and can apply optimization, debugging, and profiling tools to improve of the program. 6. Is able to make the transition from a scientific model to a structured program;	
Education Method	Weekly there are 2-hour lectures and 3-hour lab sessions. Participation in a minimum of five lab sessions is required, with the optional homework, a bonus of max. 1 point can be obtained.	
Books	Textbook: Writing Scientific Software - A guide to good style, by Suely Oliveira and David Stewart, Cambridge University Press, ISBN-13 978-0-521-67595-6	
Assessment	1. Written exam: 3-hour exam comprising theory and practical (lab) questions. (90%) 2. Lab homework (10%) Final grade calculation: $(0.9 \cdot \text{Exam} + 0.1 \cdot \text{Lab homework})$ Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: resit in next quarter - Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see TER Art 29, sub 4).	
Tags	Numeric Methods Programming	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf . Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	The final grade of the course consists of the following components: - Two graded homeworks - a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or $0.6f+0.4h$, whichever is larger, provided $f > 5$. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - homework assignments and written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). The homework grade does not count for the resit.	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Students are expected to have: superior knowledge and skill in elementary probability (details below); some experience working with Matlab, Python, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor. Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course should be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful. Testing yourself and doing a thorough review in the first few weeks is strongly advised; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.	
Course Contents	This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components: modelling and models (and some of their pluses and minuses), learning a number of stochastic processes and some of their properties, doing Monte Carlo simulations, analyzing the generated data and drawing conclusions from them. Contents and format of this course are intended to combine the following: 1. real (possibly stylized) problems as starting points; 2. enough about modelling to make participants aware of the impact of modelling decisions; 3. identification of a number of stochastic processes present in/related to models and problems; 4. (limited) treatment of underlying theory, guided by: (a) what is necessary to ensure that participants know how to do things right and (b) what enables them to do things smartly; 5. facts, properties, and theorems: (a) sincere attempt to understand where possible; (b) focus on practical implications/uses; (c) illustrations rather than proofs/theory; 6. useful tools and techniques from statistics, probability, and stochastic processes; 7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory; 8. providing as much opportunity as feasible for participants to work on/with problems from their own field. Tentative topics/contents list: Review basic probability: most as self-study, some in class, plus mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal process. Markov chains: discrete time; continuous time; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) SDEs.	
Study Goals	Learning objectives: after following the course, participants will be able to 1. in an iid setting, carry out Monte Carlo simulations; and 2. extract and collect, summarize, and interpret the data generated; and 3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?; 4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method; 5. for small-sized practical problems: build a suitable model and simulate it; 6. extract and collect required model parameters from a problem description or background; 7. recognize where modelling choices are made; 8. make elementary modelling decisions themselves; 9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion; 10. recognize/identify the covered stochastic processes in practical settings/problems; 11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models; 12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.	
Education Method	Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in R, Matlab, or Python (or other language). Spread over the semester, five homework assignments. I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time.	
Literature and Study Materials	Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes. For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge"). Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin. Lecture slides, notes, and additional references will be posted on Brightspace. A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I	

Assessment	prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab and R. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and this should be possible and arranged). In all cases: you should have the software installed and be familiar with its use. 1. Probability test on knowledge and skills needed in/for the course (10%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed); counts for 70%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience.	
Literature and Study Materials	Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3. Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: $(1/3 * \text{homework assignments}) + (2/3 * \text{group programming project})$ but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	K. Cools	

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Orientations AP 2023

EC Program	30
Introduction 1	Combining the core programme with a 30 EC orientation completes the master programme. The orientation allows for a broadening of knowledge or for additional depth. Choose from: Research and Development (R&D) an Industrial Internship, a group design project, and/or additional electives chosen from subjects on the G-list, T-list, the list of General electives, or (with a maximum of 6 EC) Societal modules. Education (Ed1/Ed2) get a Dutch secondary school qualification. Management of Technology (MoT) consists of (either the first or) the second semester of the MSc MoT programme. Study-abroad (SA) a semester, 30 EC of modules, at a foreign university; optionally including a research project.

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Orientation Research and Development (R&D) 2023

EC Program	30
Program Structure 1	The R&D orientation is the most popular orientation, and the only orientation that allows for incorporating bridging/homologation courses in the programme. It consists of two of the following three blocks of modules: a non-academic internship (AP3911) - 18 EC. a group design project (AP3841, 15 EC) and a preparatory design course (System Engineering, 3EC) - 18EC. additional electives - 12 EC, chosen from subjects on the G-list, T-list or the list of General electives, and a maximum of 6 EC of S-(Societal) list modules or assigned homologation/bridging modules. The S- (Society) list: http://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=30279 Only subjects marked as 'Category MSc level' are accepted. If you wish to include a module in the programme that is not on the lists, approval from the Board of Examiners must be obtained.

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Non-Academic Internship

AP3911	Work Placement (Internship)	18
Responsible Instructor	Dr. A.J.L. Adam	
Project Coordinator	L. van der Elst	
Contact Hours / Week x/x/x/x	Different	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	You will be working full-time in a non-academic organization for a period of 3 months. Longer internships are not recommended. Your Applied Physics knowledge is necessary for the project. The industrial internship is guided by an internal (TU Delft, Applied Sciences) and an external (company) supervisor. During the industrial internship you are expected to get to know the organization. To do so, you need to start gathering information about the organization of your choice prior to the start and make a list of questions you would like to have answered. During the internship you need to make time to get acquainted with other departments within the organization and interview, for example, a number of people in different parts of the organization. IMPORTANT: It is desirable for the internship to have a Physics content. However, in some cases it may be acceptable if this component is lacking, provided that the work requires problem solving or design skills for a physicist. The criteria formulated above do not exclude activities in a startup company (maybe initiated by the student) to be accepted as internships. The criterion is that there should be a sufficiently experienced internship supervisor, with a real involvement in the work, to be able to act as such. In order to meet the learning objectives, it is necessary that you work at least 50% of the time on location at the company/organization. Internships that are done completely remotely are not allowed. The internship is in principle intended as a way for students to broaden their awareness of working environments outside academia. This excludes certain organisations. For a list of allowed/not allowed options, see below. For very good students who aspire a career in academia, a possibility for an academic internship is through a grant from the Justus and Louise van Effen fund, which is restricted to very good students and which provides funding for internships at top-20 engineering universities only. For more information, see https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/funding-scholarships/justus-louise-van-effen-research-grant/. The student is responsible for finding an internship position. The Internship Office TNW only has an assisting, coordinating and administrative role. We therefore advice to start looking for an internship well in advance. Students are advised to start the orientation process at least 6 months in advance if planning to stay in the Netherlands, and at least 9 months if planning to go abroad. Allowed: Business, physics content; TNO (or foreign counterparts such as Fraunhofer Gesellschaft); ECN; KNMI; NFI; patent office; non-academic hospital; consultancy (provided technical/quantitative); financial analysis (provided strong quantitative). Not allowed: university, Dutch or abroad (except with van Effen grant, see above); FOM institute (such as Amolf, NIKHEF, DIFFER) and foreign equivalents (such as Max Planck Institut, ICFO); Academic hospital (unless practical work is done similar to an internship that could be done in a 'normal' hospital, and not in combination with a thesis project in a (different) academic hospital); Nationale Denktank. If the Board of Examiners has given permission to do the thesis project outside the university, the internship can not be done at the same organization as the thesis project.	
Study Goals	The goals of the internship are: - To become familiar with a professional working environment for a physicist; - To use academic knowledge and skills, acquired in the degree course, to solve problems or be active in design related to Applied Physics, in a professional working environment. Skills should be acquired on the following topics: 1. Performance: Productivity, Independence, Practical skills, Theoretical knowledge, Creative thinking, Time management; 2. Communication: Reporting, Presentation, Interpersonal; 3. Competences: Motivation & enthusiasm, Critical attitude, Taking initiative.	
Education Method	Practical training on the job, the company supervisor is responsible for day-to-day supervision. During the internship period the student will also report to the TU Delft supervisor 2-3 times about progress made on the project.	
Literature and Study Materials	The Brightspace page (https://brightspace-cc.tudelft.nl/course/1825/stagebureau-tnw--internship-office-applied-sciences) provides more information about the industrial internship. On this page you can find the step-by-step guide explaining all procedures, forms, internship offers, experiences from other students and additional information e.g. about grants and funding.	
Assessment	You can contact the Internship Office through: InternshipOffice-TNW@tudelft.nl The mark for the internship is awarded by the two supervisors. The mark is based on a series of criteria, including an individual report of the student, in which the experiences concerning the main goals, including the experience in the working environment are discussed. For the assessment you need to submit two reports: - A scientific report using the format and guidelines of the company for your TU Delft supervisor and company supervisor - An evaluation report for the internship office Two weeks before the end of your internship, the Internship Office TNW will send your company supervisor and TU Delft supervisor the internship review form. Your company supervisor needs to assess your internship at the end of your internship period and send his review form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship within two weeks after the end of your internship and give a mark. The TU Delft supervisor will then send the assessment form (including the company supervisor review form) to the Internship Office TNW. In principle, the mark awarded by the company supervisor counts for 2/3 towards the final mark; that of the TU Delft supervisor for 1/3. The TU Delft supervisor can overrule this if he/she has good reasons for this, which should be communicated to the student and the internship office. The Internship Grading Scheme will be used as a guideline to determine the mark. Please consult the Internship Office TNW	

	Brightspace for the detailed grading scheme. Your mark will be processed as soon as the Internship Office TNW has received: The review form with an original signature from both supervisors. Your supervisors can request the review form by sending an email to InternshipOffice-TNW@tudelft.nl A digital copy of both reports. If a report is submitted to the internship office more than one month after the registered end date of the internship without good reason, 1 point may be deducted from the grade for each month of delay. When the nature of the project requires so, it is possible to censor parts of the report. Details like numbers and names may be left out of the report (or struck through), for example, as long as the report represents the achievements of the project and the TU Delft supervisor is able to assess the internship. Enrolment / Application!!!! Prior to the start, the internship must be registered at the Internship Office, and be approved by the TU Delft supervisor* !!!!! Register your internship at the Internship Office through https://tudelft-onstage.xebic.com/. Your internship file in OnStage gives a clear overview of all tasks and responsibilities within the internship process. If you don't have access to the registration website, please contact the Internship Office at InternshipOffice-TNW@tudelft.nl. Remarks*A TU Delft supervisor is a member of faculty (tenured or appointed on a tenure track position) within the faculty of Applied Sciences. When doing an internship in the Netherlands, it is mandatory that the national UNL contract is signed by the company. Without a signed contract, your internship is not valid. The UNL contract is not yet required for internships done abroad but is strongly encouraged. Check with the company in advance if they are willing to sign the UNL contract. TU Delft no longer signs other contracts. If the company is not willing to sign the UNL contract, you cannot do an internship there. See: Brightspace (https://brightspace-cc.tudelft.nl/course/1825/stagebureau-tnw--internship-office-applied-sciences) and https://www.universiteitenvannederland.nl/en_GB/nieuws-detail/nieuwsbericht/861-p-research-universities-adopt-common-internship-agreement-p.html If you have any questions about this, please send an email to internshipoffice-tnw@tudelft.nl
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Organization	Applied Sciences
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Design Project

Introduction 1

The system engineering group project is scheduled in the first quarter of the second year.
To participate in the systems engineering project a preparatory course must be done in the spring semester of the first year:
Systems Engineering (AP3831, 3EC).

AP3831	Systems Engineering for Physicists	3
Responsible Instructor	Dr. M. Sovago	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course is an elective course for Master's students interested in learning how to design a complex system. TU Delft Master's students at Applied Sciences (and EEMCS) Faculty are welcome to join the course. It is meant for the first year Master's students.	
Course Contents	It is an obligatory preparatory course for the Systems Engineering (Physics Design) Company Project (AP3841). During Systems Engineering (SE) course for applied scientists, you will learn and discover the art behind designing, building and managing complex systems. You will get to master SE tools and methodologies as well as management capabilities. Every week you will learn the theoretical aspects of SE during an interactive lecture. Step-by-step, you have the possibility to discover the process and the tools that we can use to design a complex system. You can apply the knowledge from the lecture to design your own system. As we find important to offer the students the opportunity to put own creativity into practice, you have the possibility to define own system that will solve a day-to-day problem or a societal challenge. A few systems will also be shown as example and inspiration for the choice of your system. Together with a team of ~ 5-7 peer-students, you will work together for 10 group project sessions on your system of choice. The steps you will follow are: 1. Define the problem you will solve and the system your will design: Problem Statement and choice of a system 2. Define the requirements of the system: Requirements specification 3. Define and interview possible stakeholders 4. Define the functions of the system: Functional Decomposition 5. Discover possibilities to fulfil the requirements: Means and Morphological Analysis 6. Decide the best designs for your system: Decision Making 7. Perform a risk analysis for your system 8. Model and Design your System: use 3D CAD modelling software We do not aim to build a prototype of the designed system, but you will design your system by using a 3D CAD modelling software (Fusion 360). This SE process will be implemented to design and build a complex system during the AP3841 Systems Engineering company project (2nd year Master).	
Study Goals	By the end of this course, you will be able to: 1. Recognize the need and relevance of Systems Engineering (SE) for every applied scientist 2. Structure the key steps in Systems/Product/Process Development 3. Systematically breakdown tasks and work in interdisciplinary/multicultural teams to design a complex system 4. Apply SE methodologies and SE tools in projects 5. Learn and apply transferable skills along the course: chair a team for 1 or 2 weeks, team management and team roles, put your creativity into practice, presentation and writing skills (based on weekly activities), self- and team-reflection.	
Education Method	The course is centred around 10 group project sessions. You will work in a group of about 5-7 students with different backgrounds and study tracks. Each session focuses on one aspect of Systems Engineering. Weekly group project cycle: 1. A plenary and interactive lecture: learn the SE tools and methodologies. You will be able to ask questions and have discussions with your lecturer 2. Team activity: you will implement SE tools to design and engineer a complex system. You will be working with your peer-students and be guided by your TA. Each week you will have a team chair (a peer-student) 3. Weekly presentation: present your weekly results and progress to your lecturer and TA and receive feedback. The chair of the week is in the lead, but everyone participates to the discussion. 4. Weekly report: write your weekly report and receive feedback from your lecturer. The content of the report and adjusted feedback can be easily used for the final report. End Q4: final presentation and final report	
Literature and Study Materials	Study material will be provided on BrightSpace.	
Assessment	The final grade is a combination of the following components: Group Grade 1. Final Presentation (30%) 2. Final Report (30%) Individual Grade 3. Weekly chair Presentation and Report (40%) Please Note: 1. You will need a grade of 6.0 or higher in all components to pass the course 2. Participation/contribution to discussions to SE lectures is expected 3. Participation/contribution to team brainstorming, preparation for the weekly presentation and report is expected 4. Participation/contribution to weekly presentations is compulsory 5. You can receive a bonus/penalty for your team contribution	
Tags	Design Group work Physics Projects	

AP3841	Systems Engineering Company Project	15
Responsible Instructor	Dr. M. Sovago	
Contact Hours / Week x/x/x/x	Scheduled full-time in Q1.	
Education Period	1 4	
Start Education	4	
Exam Period	1	
Course Language	English	
Course Contents	The Systems Engineering (Physics Design) company project is specially designed for TU Delft Master students at Applied Sciences and EEMCS Faculty in their second year. Systems Engineering is a must-have skill for every applied science and engineering student. Systems Engineering is an interdisciplinary field of engineering and engineering management that focuses on how to design and manage complex systems over their life cycles. The brain behind every complex system is the Systems Engineering; and that can be you! Here you will develop creative and innovative thinking, gain experience while working with an industrial partner, collaborate in teams and develop strong personal skills. The project includes working in an industrial project defined by a company. It is similar to an R&D internship with an advantage of working in a motivating team. You will be supervised by an academic (systems engineering and personal skills development) and an industrial part-ner (guidance project). The programme focusses on finding your role in a multi-disciplinary team and team building, entrepreneurship, presentation and writing skills, cultural intelligence, self- and team-reflection. Join us for this fun project!	
Study Goals	Participating in this project will ensure you a must-have skills and competences luggage, pre-paring you for starting your future career. At the end of this project, you will be able to: 1. Apply your applied science and engineering knowledge to solve systems engineering prob-blems 2. Have experience with industrial, real-life systems engineering and research cases 3. Think creatively and innovative. Develop entrepreneurial skills 4. Effectively collaborate in a multi-cultural and multi-disciplinary team 5. Present and write your results in a clear and appealing manner 6. Critically reflect on your performance and interaction with your team	
Education Method	Project team work with end presentation and end report.	
Prerequisites	Passing the preparatory Systems Engineering (AP3831, Q3-Q4) course is required to be admitted to the Systems Engineering Company Project.	
Assessment	Assessment is based on the final report, final presentation and the project defence during the final presentation for the academic committee (staff TU Delft) and the industrial supervisor(s). The evaluation will be done by an evaluation committee and is divided over: Theory and concepts (40%) = quality of systems engineering Implementation (30%) = group performance, entrepreneurship, use of creativity tools, innovativeness, concrete results Communication (30%) = progress meetings, project management, report, presentations	
Enrolment / Application	In order to enrol for the project, please send an email to: Maria Sovago m.sovago@tudelft.nl. Please indicate your name, faculty, student number and the top 2 preferred projects you would like to work on. We will distribute you in a team depending on your preferences and background.	
Studyload/Week	The project takes place on September - half-November. This is a full-time activity (40h/week). Please follow the BrightSpace for the exact dates and planning.	
Schedule	Systems Engineering Course AP3831: February-June Q3+Q4 In order to participate to the Physics Design Pro0ject, students from Applied Physics should enrol and follow first the Systems Engineering course AP3831. Systems Engineering Industrial Project, September -half-November (2.5 months) All participating Master's students will work in teams for a full 2.5 months to solve an industrial systems engineering problem. You will work both at the company and at TU Delft, depending on the company case (to be establish during the kick-off meeting in June). Physical meetings as well as online teamwork will take place, depending on the situation. Final Presentation and Final Report Systems Engineering company project: half-November The teams will present the results and their report to the evaluation committee (TU Delft staff) and to the industrial partner. The date will be established together by the team and the company. Please follow the BrightSpace for the exact dates and planning.	
Location	You will work both at the company and at TU Delft, depending on the company case (to be establish during the kick-off meeting in June). Physical meetings as well as online teamwork will take place, depending on the situation. You will be meeting regularly with your supervisors. You can always contact Maria Sovago for questions: m.sovago@tudelft.nl; 061 375 8335; office at building 22, F.232.	

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Electives

Program Structure 1	12 EC, chosen from subjects on the G-list, T-list or the list of General electives, and a maximum of 6 EC of S-(Societal) list modules or assigned homologation/bridging modules. If you wish to include a module in the programme that is not on the lists, approval from the Board of Examiners must be obtained.
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Societal (S-list) Modules

Program Structure 1

The S- (Societal) list contains a.o. electives offered by the faculty of Technology, Policy and Management:
http://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=30279
Only subjects marked as 'Category MSc level' are accepted.

AS3111	ATHENS	2
Course Coordinator	Drs. E. Jacobs	
Contact Hours / Week x/x/x/x	-	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Twice a year you have the opportunity to follow a one-week intensive course, organised by several European Universities. This programme is called the ATHENS programme (Advanced Technology Higher Education Network, Socrates). More information: https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/university-exchange-partners-networks/athens/athens-outgoing-students/ and http://athensnetwork.eu/	
Study Goals	Recommended for 2nd year MSc students (only)! The courses allow you either to deepen your knowledge within your area of studies or to discover a new discipline. The courses are held in English and/or French. Each course includes about 30 hours of scientific course work. Another 10-15 hours go towards European Dimension activities; you will not only discover the university and the city, you will also meet other international students and have a great time.	
Education Method	7 day intensive course - study abroad (BME Budapest, KU Leuven, IST Lisbon, UPM Madrid, PM Millano, TUM Munich, CTU Prague, AUT Thessaloniki, NTNU Trondheim, TU Vienna, WUT Warschau, ITU Istanbul, UPB Bucharest, ParisTech)	
Practical Guide	https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/university-exchange-partners-networks/athens/athens-outgoing-students/	
Assessment	No grading. The result is indicated by a V ("Voldaan" = Pass), if a certificate is obtained at the foreign university and handed-in at the international-office of the faculty of Applied Sciences. It's not possible to earn credits more than once. No credits are awarded if you return to your home country.	
Enrolment / Application	Application Deadline: 19 September 2023 / 31 January 2024	

AS3132	Art, Empathy and Ethics	5
Responsible Instructor	Dr. E. Mendes	
Instructor	Dr. F. Santoni De Sio	
Contact Hours / Week x/x/x/x	5/5/X/X	
Education Period	1 2	
Start Education	1 2	
Exam Period	1 2	
Course Language	English	
Course Contents	Art is a sophisticated activity binding the conscious and unconscious, the rational and emotional counterparts of our species. It makes a unique contribution to society by offering various means of expression and exchange and it is a powerful investigation tool for complex human issues. It plays an important societal role in expanding the public imagination by engaging them in a challenging sensorial - intellectual dialogue, increasing the perception of themselves as individually and collectively. This course invites you to experience and produce art rather than simply broaden your knowledge in contemporary art or history of art. Using various tools and media, you will develop an art project around an ethical or a societal problem related to the design and use of technology. You will be led to perceive and portrait some relevant aspects of a chosen societal problem, questioning in a completely open and creative way, the consequences, limitations, advantages or any other aspect associated with a chosen technology, as perceived by those who experience the problem or the (mis)use of a that (imagined) technology. You will examine the situation through the eyes of your inner artist, connecting with society through an empathic pathway. As examples of tools and media we can cite, not exclusively, the case of visual arts: painting, printing, photography, installations, graffiti, films, etc. As examples of global societal problems, we can cite boat people, misuse of big data, consequences of genetically modified organisms, facial recognition, etc. This empathic and aesthetic exercise is to be complemented by a systematic conceptual reflection on the chosen ethical or societal problem, both carried out in parallel. That is achieved via lectures, debates and self-study of proposed material where you will explore the ethical, cultural and other societal aspects of problems related to new technologies. You will then write an Ethics essay on a topic of your choice that is coupled to your art trajectory and Final Art Assignment. No previous art experience is necessary. Course attendance is compulsory.	
Study Goals	After having completed the course, students can: - recognise their own way to develop a concept or reasoning that has an emotional or empathic origin - reflect on the role of art in expressing and questioning risks associated to new technologies - understand how people immersed in societal problems perceive that problem in their daily lives - communicate via own artistic production, emotions, questions, ideas related to the risk associated to new technologies - reflect on and anticipate the emotional reaction of society or specific social groups to new technologies - empathically participate in the emotional-moral debate associated to state-of-the-art science and technologies - make use of divergent thinking strategies that incorporate creative (artistic) approaches - interpret various aspects of avant-garde art in order to challenge their personal responsibility (position) as engineers	
Education Method	Compulsory interactive art and ethics sessions and self-study of ethics material	
Assessment	Art part (60% of grade) - Active participation during the working art group sessions, via art assignments, artistic experimentation, brainstorming, etc. (30%) - Your artistic output in the form of a Final Art Assignment (including how it reflects the emotional-moral process that you went through and the critical thinking emanating from the emotional art experience. (30%) Ethics part (40% of grade) - A written ethical essay coupled/inspired by the art trajectory Minimum Grade - A minimum of 5.0 has to be obtained in each of the individual parts above (Art and Ethics)	
Enrolment / Application	Just email directly to E. Mendes (e.mendes@tudelft.nl) with a motivation letter (max 1/2 A4), at max two weeks before the course starts because places are limited. PLEASE, USE EMAIL SUBJECT: "Enrolment AS3132 : Art, Empathy and Ethics"	

TPM001B	Sociotechnology of Future Energy Systems	5
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr. A.F. Correlje	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	1/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students need to have sufficient prior knowledge in the fields of energy technology, economics and policy. They should be familiar with concepts such as intermittency, Carnot efficiency, BEVs, energy vs power, network externalities, sunk costs, fixed and variable costs, ROI and NPV.	
Course Contents	In this module the changing technological and societal characteristics of future energy systems are dealt with. Contemporary energy systems are confronted by technical developments and socio-economic concerns, such as maturing renewable technologies and ICT based control innovations or energy scarcity and environmental concerns. These changes will reflect on operational roles and responsibilities, market structures and actor interests, and business models for energy services. Moreover, they may necessitate new forms of governance to ensure a reliable operation that meets socio-economic performance criteria. In this light, the course discusses the technical and socio-economic aspects of future, mostly sustainable, energy systems, touching upon issues like the integration of new forms of distributive solar PV or large-scale off-shore wind into existing networks, various forms of new motor fuels, other more experimental renewable energy sources and storage methods, and the dynamics of transition processes.	
Study Goals	At the end of this module the students know: - Which technical and societal developments can be expected in the energy sector. - How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements. - How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios. At the end of this module the students can: - Formulate descriptions of the technical and societal characteristics of future energy systems. - Make a conceptual design of an energy system, given certain demands. - Formulate a balanced advice on issues in the energy sector.	
Education Method	Interactive sessions, self-study, online & class exercises, class discussion. Reading material: scientific papers.	
Assessment	Individual written exam (on the computer, on campus) consisting of essay questions. Any reference material on paper and/or in pdf format on memory stick is allowed; internet access other than to the examination site is not allowed. Sufficient participation in the online exercises (one exercise may be missed) is a prerequisite for taking the exam, but not for the resit.	
Elective	Yes	

TPM008A	Summer School Health Entrepreneurship	3
Module Manager	P.B.M. Vandekerckhove	
Instructor	Dr.ing. V.E. Scholten	
Instructor	A. Boru	
Responsible for assignments	P.B.M. Vandekerckhove	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/24 during Summer	
Education Period	Summerschool	
Start Education	5	
Exam Period	none	
Course Language	English	
Required for	The ECTS points are accepted by the Master Annotation Program: Health Innovation and Entrepreneurship.	
Expected prior knowledge	Interest in entrepreneurship in the health care is essential, CV and Motivation letter should be submitted to: https://summerschool.eithealth.eu/registration/	
Summary	The EIT Health Entrepreneurship Summerschool aims facilitate new ventures in health care by collect ing challenges from the healthcare providers, and take participants/students through a robust development process of analysis, ideation and financing, in order to develop targeted solutions, leading to business plan propositions and ultimately new start-ups. Participants will work in international and interdisciplinary teams to develop a minimum viable business proposition. Central steps are: problem/ opportunity identification, market analysis, customer involvement and collaboration, business models, go-to market strategies, financing, funding, pitching and team-building. Personal development is important and in the course of the program students will learn to think and act as an entrepreneur.	
Course Contents	The EIT Health Entrepreneurship Summerschool aims to collect challenges from the healthcare providers, take them through a robust development process of analysis, ideation and financing, in order to develop targeted solutions, leading to business plan propositions and ultimately new start-ups. Students will work in international and interdisciplinary teams to develop a minimum viable business proposition. Central steps are: problem/ opportunity identification, market analysis, customer involvement and collaboration, business models, go-to market strategies, financing, funding, pitching and team-building. Personal development is important and in the course of the program students will learn to think and act as an entrepreneur.	
Study Goals	By the end of the programme, students should be able to: - Understand the basic principles of entrepreneurship in health care; - Analyze a case study topic, and define the potential market segments and develop market research and segmentation for start-ups, through interaction with potential clients. - Understand and experience what it takes to be an entrepreneur in health care - Develop insights into what their own motivations, ambitions and objectives are as (potential) entrepreneurs - Work in an entrepreneurial team.	
Education Method	The lectures will be given by experts in the health care area. The lectures will consist of exposition and discussion of material by the lecturers. In each week there is a single lecture block focusing on a specific subject. Each day consists of a lecture on the specific subject and the workshops on applying the subject on the case (in house-or out of the building). Subjects and Assignments include; - Ideation on cases (Day 1) - Defining a customer problem (Day 2) - Customer involvement (Day 3) - Solution/ Prototyping (Day 4) - Go to Market Strategies (Day 5) - Business Models (Day 6) - Marketing (Day 7) - Financing (Day 8) - Funding/ Pitching (Day 9) - Pitching to the Jury (Day 10)	
Literature and Study Materials	There is no mandatory reading list, as handouts will be provided.	
Assessment	50% of the final grade will result from the final pitch and the final business plan. The other 50% will result from the assignments to be completed throughout the course.	
Enrolment / Application	This course will only be offered when at least 30 students are interested. Since this course is part of a EIT Health (an EU initiative). Registration (requires CV and Motivation letter) and should be done via: https://summerschool.eithealth.eu/registration/ Please also send your cv and motivation letter to A.Boru@tudelft.nl and R.A.Kievit-Brinkman@tudelft.nl	
Remarks	The Summer School is a EIT Health program, co-organized by Delft Centre for Entrepreneurship at TU Delft, la Caixa Foundation, IS Global, University of Barcelona, YES!Delft incubator, Philips Healthcare and Achmea Healthcare Insurance.	
Elective	Yes	
Targetgroup	The course is intended for who have a background in engineering, business or the health disciplines.	
	Completing a bachelor programme is must, master and PhD students or participants from industry are welcome.	
Category	MSc level	

TPM023B	Cost-Benefit Analysis: Theory and Application	4
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	N. Mouter	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	The course TPM023B focuses not only on private (e.g. project, firm, industry) costs and benefits but also on social costs and benefits. While the private CBA only considers financial aspects, the social CBA considers both financial and non-financial aspects. For instance, in the construction of new highway, besides the financial aspects such as toll revenues or construction costs, we also assess the non-financial aspects such as safety benefits (lives saved, accidents avoided). Hence, the course actually has a much broader view than the one of the private CBA. This offers students a higher level of practical applications.	
Course Contents	Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of a proposed policy are identified, and then converted into monetary units in order to make them comparable with policy alternatives; certain decision rules of CBA help determine whether the project is desirable from society's standpoint. Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, un-priced environmental damage). Nevertheless, the application of CBA is fraught with difficulties and limitations that need to be taken into account. What rates of interest are appropriate for determination of present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How can we express e.g. environmental damage, or human health and life risks, in monetary units? How can we adjust market prices to account for external effects? How can we use alternative measures of welfare to identify the benefits and costs of a policy proposal? How can we ensure that politicians who aspire to use the results of CBA in their decisions can actually do so?	
Study Goals	At the end of this course, students will be able to: understand why CBA can be used to appraise projects with an objective to improve social welfare understand the strengths and limitations of CBA in project appraisal criticize and defend the CBA methodology from an ethical perspective explain the advantages and disadvantages of applying a CBA methodology in a political process with multi-stakeholders gain an understanding of the fundamentals of CBA perform steps or approaches to evaluation financial and economic values of a project interpret the results of a cost-benefit analysis	
Education Method	The lectures are on campus. In case of any unforeseen circumstances (e.g. the previous COVID-19 measures), further arrangement will be announced.	
Literature and Study Materials	* Textbook: Anthony E. Boardman, David H. Greenberg, Aidan R. Vining & David L. Weimer: Cost-Benefit Analysis: Concepts and Practice, 4th International Edition, Pearson, 2013. * Selected articles (available in Brightspace)	
Assessment	There are two types of the course assessment: 1. An individual essay (30%) 2. A written exam (70%) Both types of assessment are mandatory and the passing grade for each assessment is 5.5. Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations). For both types of assessment, students who do not pass in the first attempt are given the second attempt. Students may review their works for each type of assessment. This will be held physically according to the time scheduled by the lecturer(s). Rules for the individual essay only: For the individual essay, please be well informed that the maximum grade for the second attempt for the essay is a 6.0. Note that the second attempt for the essay is only meant for students who do not pass and NOT intended for students who have passed nor new students who would like to enrol in the course after the first attempt. Further details about the individual essay will be provided later in Brightspace.	
Category	MSc level	

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/3/0/3	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. The course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/0/3/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Course material will be announced on Brightspace	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrollments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

TPM401A	Technology Entrepreneurship and Innovation	5
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	P.B.M. Vandekerckhove	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/2/2/0	
Education Period	1 2 3	
Start Education	1 2 3	
Exam Period	1 2 3	
Course Language	English	
Summary	This course offers theoretical fundamentals of entrepreneurship and technology innovation. You create a case study of an existing start-up analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an high-tech start-up to academic research on the topics of technology entrepreneurship and innovation.	
Course Contents	Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, dynamics and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to:	
	LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Describe and apply the high-tech start-up growth process. LO 4 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 5 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 6 Analyse an existing high-tech start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Enrolment / Application	Enrollment via Brightspace	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft is needed.	
Category	MSc level	

TPM402A	Technology Entrepreneurship and Health	5
Module Manager	P.B.M. Vandekerckhove	
Instructor	A. Boru	
Instructor	H. Khodaei	
Responsible for assignments	P.B.M. Vandekerckhove	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	----- Due to some hick-ups, probable related to course code change, you cannot find the course in Brightspace or Timetables. This course used to carry the name MOT9614 - Basic Health Entrepreneurship. We would like you to enroll in the course TPM414 for Quarter 1. We will follow the schedule of TPM414, but acknowledge your interest in Health. ----- This course has a focus on innovation and entrepreneurship in health care systems. Using methods and theories to analyse startups, you will gain a deep understanding of their early growth and success. You will analyse a case study of an existing health start-up and analyze its entrepreneurial journey, how the company has found an opportunity to improve healthcare and how it navigated the entrepreneurial ecosystem in healthcare. This course helps you prepare for a wide array of career in health entrepreneurship and the implementation of new solutions for health care.	
Course Contents	Throughout the course you will develop the skills to develop and act on innovative business opportunities in health systems. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies in healthcare systems and analyse the value of new technology for the various stakeholders in healthcare systems. The course is organized into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, discussions of cases, literature and assignments, and presentations.	
Study Goals	Upon finishing the course, you will be able to: 1. analyse healthcare challenges and reflect on the entrepreneurial opportunities for startups and established firms. 2. Describe and apply opportunity identification in healthcare by making use of the opportunity analysis canvas. 3. Analyse the growth process for academic spin-offs in healthcare from idea to entry in healthcare. 4. Recognize the importance of the entrepreneurial ecosystem and the main stakeholders in healthcare and how they influence and contribute to the start-up process. 5. Explain and apply methods for developing business models for new healthcare startup ventures and analyse how they can change over time. 6. Analyse an existing healthcare startup case and present your findings about the dynamics of its early growth. 7. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	The course is organized in 8 online sessions (8 weeks) following 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.	
Assessment	The assessment is as follows: - Team: three case analyses of a startup company (30%) and a final report combining the individual cases analyses plus reflection and recommendation and presentation of the findings (20%) - Individual: Oral questions (30%) and theory quizzes (20%) In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Instructor	F. Delgado Medina	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade. 3. Individual: Theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Master students from all master programs can apply.	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	H. Khodaei	
Instructor	F. Delgado Medina	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Elective	Yes	

TPM405A	Patent Law and Patent Policy	5
Module Manager	Dr. L. Hartmann	
Instructor	mr.dr. E. Visscher	
Instructor	M. Westerik	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	Intellectual Property (IP) Rights are increasingly important in today's fast-moving, hyper-connected world. Awareness of the different types of IP is of vital importance, not only for large companies and startups, but also for university researchers and even individuals. Knowing what Intellectual Property is, who owns it, how to protect it, how to trade it, and how to enforce an IP Rights is essential to ensure freedom of operation and keep imitators at bay. In addition to the fundamentals of IP Rights, IP Law and IP Policy are central topics of this course. Whenever you create something, whether it is an invention, a book, a song, a painting or even a website, chances are that you will have to deal with Intellectual Property and the Rights that are associated with it. There are two sides of the medal: the IP Rights that you own and may have to enforce, and the IP Rights that are not yours and that you will have to deal with. This course is intended for entrepreneurial engineers in every discipline, whether it be the hard sciences (quantum computing or nanobiology) or the more creative sciences (architecture of industrial design), especially when it concerns innovative products, processes, services and design. Given the increasing demands placed by society on companies and universities to innovate and improve life and society at large, IP rights have become synonymous with successful innovation and startups. Intellectual property rights also play a key role in today's scientific community. Understanding how the legal framework affects both the science and the companies in the business of creating high-tech and innovative solutions for today's problems, is of paramount importance. In this course you will become familiar with the basics of European IP law and the most common types of IP law (including the requirements to obtain these rights). You will explore the process by which patents are granted, how patent rights can be exercised in court and how they can be used as a business tool. Modern-day IPR issues such as the patentability of computer-implemented inventions and biotech inventions will be addressed. Additionally, software protection (including open source) works and the related issues will be discussed. The topics will be presented on the basis of real-life cases. During the course, participants will analyze and present a case in which a university spin-off is confronted with a multitude of IP issues.	
Study Goals	Get to know the basics of European IP Law Understand the common types of IP law Understand the requirements to obtain IPR Understand the process of granting patents Understand how to exercise patent rights in court Understand how to use patent rights as a business modeling tool Understand patent rights in different industries	
Education Method	Lectures in class Group assignments	
Assessment	Exam Assignments	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	There are ten mandatory lectures. As this is a collaboration between Delft University of Technology and Leiden University, the first five lectures will be in Delft and the last five lectures will be in Leiden. For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM406A	Corporate Entrepreneurship and Start-ups	6
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course examines how large corporations engage startups to tap into new markets, technologies and business models. You will develop knowledge and skills to analyse strategic situations of existing corporates and how they innovate through engaging startups. The course provides an eye on corporate startup engagement from both the corporate and startup perspective.	
Course Contents	Large international organizations often find themselves in a luxurious position compared to small start-ups. They have more financial capital, access to markets, and more technological knowledge. However, large international organizations are confronted with increasing external turbulence, technological advancement and globalization, which puts their luxurious position in jeopardy. At the same time the luxurious position constrains them to depart from present activity and focus on radical new innovations. Hence, corporates try to renew themselves through internal entrepreneurial behaviour, but also by initiating start-ups and acquiring start-ups to experiment with new technology. Many times, these start-ups originate in academic environments. Therefore we pay specific attention to the emergence of academic start-ups and how corporate organizations provide incubation facilities and venture capital support to engage startups to renew themselves. This course examines the various forms of corporate entrepreneurship and the relationship with corporate strategy in particular. You will develop knowledge and skills to analyze strategic situations and design the most appropriate solutions in terms of corporate entrepreneurship strategy and organization.	
Study Goals	Understand the concept of corporate entrepreneurship; the contexts, the forms, and the relationship with strategic management Analyze strategic situations and design appropriate corporate entrepreneurship strategies and organizations Analyse the contribution of internal and external ventures in the renewal of large corporates Understand the emergence of academic spin-offs and how they relate to the technology development of large established organisations Identify the various key actors that facilitate and support the emergence of new business ventures and academic spin-offs	
Education Method	In case of Covid-19 restrictions for on-campus education we will offer this course in 14 online sessions. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Each student will be expected to be present during lectures (two times a week), actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Literature and Study Materials	Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Practical Guide	Given the interactive nature of this course, attendance at classes is mandatory. Only very serious grounds for absence (such as illness accompanied with doctors note) are acceptable. Holidays and (paid or voluntary) work obligations do not count as acceptable reasons for absence. This is a full-time course with scheduled holidays. Unjustified non-attendance will result in expulsion from this course.	
Assessment	The assessment is Covid-19 proof and includes three components. 1) The first component is a team level online presentation of a case analyses of an existing company. Each team will analyse a practical company case provided to you and based on literature of a specific lecture session you will present your findings online in class (40%) 2) The second component is a written report of a large firm in which you will analyse with your team the startup engagement activities this firm has put in place to renew itself. The report is an analyses of the company activities w.r.t. corporate venturing and an reflection of these activities by making use of literature. The report accounts for 40% of the final grade. 3) The third component is an individual analyses of a startup that is involved in the corporate startup engagement activities of your team case company. It reflects on the motivation for the startup engagement from both startup and corporate perspective. It accounts for 20% of the final grade.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft are needed.	
Special Information	Each student team will present during the lectures, actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. T.L. Dolkens	
Instructor	E. Çelik	
Responsible for assignments	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course used to carry the name MOT9612 - Business Development Lab. In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using computer sciences. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. Topics are discussed using case studies. You will present your work, provide peer-feedback as well and receive advise from various coaches. The course ends with a jury pitch and a two-pager of the business proposition. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2013.	
Assessment	The assessment of this course is Covid-19 proof. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and peer evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM412A	Idea to Start-up Health & Life Sciences	5
Module Manager	A. Boru	
Responsible for assignments	A. Boru	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/2/0/2	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, validate it, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one. During the eight teaching sessions we investigate societal challenges, how technology can address these and the way in which we can create viable entrepreneurial business opportunities. As a framework for this, we use the business planning process and students develop the essential parts of a business plan and investor pitch as to achieve this goal. Students will work in teams to develop a minimum viable business proposition. Central are the methodologies to conduct opportunity identification, market analysis, technology viability assessment, competitive positioning, team-building, product life-cycle planning, marketing strategy, sales channel analysis. Personal development is important and in the course of the program students will learn to think as an entrepreneur and act as a salesperson; a champion that has a specific eye for longevity of the business, sustaining growth and responsible innovation. Enrollment is through Brightspace (Osiris is not needed).	
Course Contents	We will make use of various methods, tools and techniques to develop a sound business plan: - Opportunity canvas - Value proposition development and customer empathy - Market segmentation - Value chain analyses - Stakeholder mapping - Competitive advantage analysis - Minimum viable product development - Key resource and capability analysis	
Study Goals	After completing this course you have a firm understanding of 1. Define the potential market segments and develop market research and segmentation for start-ups, through interaction with potential clients. 2. Develop a stakeholder map and recognise potential supporters and opponents and turn them into your favour (Include Health organizations, Insurance companies and other important players in the health care) 3. Reflect on leading a medical product design process how to manage actors and stakeholders to achieve their business goals 4. Identify the potential characteristics of your end user by developing persona through empathy map. 5. Ethics of collaboration with Doctors and Patients. 6. Design a medical product launch strategy and define steps for action for a specific National Systems 7. Define several potential launch strategies country depending on the needs of your product. (FDA, Insurance,) 8. Identify and develop your ideas into start-ups and analyze different factors to identify opportunities based on opportunity analysis canvas. 9. Identify and describe the main challenges or critical junctures that your company might face in the initial growth stages and describe the main activities that your company should follow to mitigate them. 10. Describe and apply the generic pattern of development and diffusions for your radically new high-tech products and identify the position of your innovative product in the pattern. 11. Identify your beach head market and calculate the TAM, SAM, SOM size for beachhead market. 12. Develop a stakeholder map and recognise potential supporters and opponents and turn them into your favour. 13. Conduct a competitor analysis and identify the main competitors that offer the alternatives and compare and analysis these alternatives against your proposition. 14. Describe the business model canvas that outlines the business case for your new product/service. 15. Calculate the Lifetime value of an Acquired Customer and calculate the Cost of Customer Acquisition 16. Define your initial Minimum Viable Business Product (MVP) 17. Make a first overview of your cash flow projections for the first 2-3 years based on the costs of product development and production, costs of marketing and sales effort and expected income generated	
Education Method	We will work in teams which are formed in the first sessions of the course. Topics are discussed using case studies. Students present their work, provide peer-feedback as well as receive advice from various coaches. The course ends with a jury pitch and a two-pager of the business proposition. During the course various assignments are given that will be discussed and presented in class.	
Assessment	The course has 3 grading components: - Active participation - In-class assignments - A business plan report In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed final assessment will be modified as follows: - Active participation in online lectures will be noted - the final (group) presentations will be online the other submissions (in-class assignments, final business plan etc. are already online so no changes will be made.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as osiris are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	

Targetgroup	Master students from all master programs can apply.
Category	MSc level

TPM413A	Idea to Start-up Energy & Sustainability	5
Module Manager	Dr. L. Hartmann	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	----- This course is NOT connected to the CleanTech Challenge. (A previous course Sustainable Business Study was linked to that competition, but this course is not in any way connected.) -----	
Course Contents	A sustainable business idea is a product, a process or a service that enables sustainability, where sustainability is defined as the capacity for the biosphere and human civilization to co-exist indefinitely. The course stimulates and assists students in creating a sustainable business idea and assessing the feasibility and the market potential thereof by testing it against prospective customers and key players in the specific industry of application. Students work together in teams which will be self-formed or if needed formed by the instructor with the aim to create optimal diversity. Each team needs to come up with an original sustainable startup idea, or a sustainable business idea connected to pressing sustainability challenges of our day. Each sustainable business idea must be technology-based we do not accept apps and definitely no consulting of any kind. The course consists of two interconnected parts: 1. Students attend lectures and workshops. 2. Teams carry out a feasibility study for their sustainable business idea.	
Study Goals	- Students learn how to assess the market potential of a sustainable product, process, or service. - Students gain basic knowledge on the commercialization of a sustainable business idea. - Students learn about different business models and gain insight in how to choose a market opportunity in the value chain. - Students gain practical knowledge about starting up a business. - Students experience real business practice by feedback from both lecturers, external advisers and experienced entrepreneurs in particular. - Students learn how to give a convincing business presentation. - Students learn to be critical about their own business case and those of their fellow students. - Students learn how to make a proper evaluation of the feasibility of their idea and business plan.	
Education Method	We offer a mix of lectures, case studies, and coaching sessions. The class is fast-paced and interactive. Many of the classes are in the form of coaching sessions and team exercises, therefore class attendance is mandatory. Lectures and online content will provide students with general guidelines for each business model milestone. Depending on availability, we will bring guest speakers, usually entrepreneurs, with experience in the energy and sustainability industry. Team coaching will help students to validate their ideas and develop their entrepreneurial skills.	
Literature and Study Materials	Mandatory books: ----- Aulet Disciplined Entrepreneurship: 24 Steps to a Successful Startup MacKay Sustainable Energy - Without the Hot Air Recommended books: ----- Scott Berkun The Myths of Innovation Goldstein Planetary Improvement More literature and reading suggestions will be provided during the course.	
Prerequisites	A BSc is required. Admission to Master students only.	
Assessment	The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course.	
Enrolment / Application	Enrol for the course on Brightspace	
Elective	Yes	
Category	MSc level	

TPM414A	Idea to Start-up Deep Tech	5
Module Manager	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology venture, whether you want to build or join one. While the starting point is tackling the societal and market problems and unmet needs, the solution ideas must be based on substantial scientific advances and cutting-edge engineering that can be used to build competitive advantage in the market (via patent for example). Feel free to bring your own ideas and technologies, but you do not need one to enjoy this course. All you need is curiosity about building a business on solid technological foundations. Fast-paced and interactive, the class consists of mostly coaching sessions and team exercises, therefore class attendance is very important. Online content and short in-class lectures provide you with general guidelines for each business model milestone. There will be weekly assignments, building blocks to a full-fledged business model. Final deliverables include: a logbook of primary market research (yes, you'll do interviews!), business model report, and a pitch presentation in front of a jury of startup professionals. You start the course with forming teams around a market problem or unmet need. Then, you gradually build a business model, mainly by answering questions such as: Who are my customers and what value do I create for them? How big is this market? How should my minimum viable product look like? How do I build a competitive advantage in the market (e.g. by patenting)? How will I market and deliver this product to the customer? Can this be a profitable business? You seek answers through customer, stakeholder, and expert interviews, by developing step-by-step interview protocols and executing them.	
Study Goals	- Develop a business idea based on a relevant economic problem or unmet need. - Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment - Develop a blueprint of your product and map its position in a competitive landscape. - Identify relevant stakeholders in your startups ecosystem. - Formulate a business model and a go-to-market strategy. - Perform initial assessment of the profitability of the business idea. - Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. - Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by modifying the business model.	
Education Method	The course offers a mix of online lectures, case studies, and coaching sessions. Depending on availability, we will bring guest speakers with relevant industry and entrepreneurial experience. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Assessment	The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course. Also, to pass the course, participation and regular submission of class assignments is necessary. Active participation and peer evaluation are assessed for each student, the rest is evaluated on a group level. The assessment of this course is Covid-19 proof.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM415B	Idea to Start-up - Climate Action	5
Module Manager	H. Khodaei	
Instructor	F. Delgado Medina	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In the ACT Climate Action course, you will work in a team to identify a start-up opportunity within the context of Sustainable Development Goals (SDGs) 13, 14, and 15. That is climate action, life below water, and life on land, respectively. You will analyze a global/regional/national/local context and look at the social, economic, environmental, political, and entrepreneurial ecosystem, identify a challenge related to the above SDGs, and develop an innovative startup idea for tackling the framed challenge. Hence, you will design your own start-up, test the idea's validity, and build a viable sustainable or circular business model. In this course, you will roadmap from an idea to a sustainable innovation-driven company and learn the benefits and barriers of developing a business in the global/regional/national/local sustainability-related system. The main focus of this course is to deep-dive into the field of sustainable entrepreneurship and develop sustainable/circular business models that can generate positive social impacts and contribute to future food security and wildlife conservation. You will focus on creating innovative solutions to some of today's most pressing issues. Transforming supply chains, rural farming, working with rural and indigenous communities, education on ecosystem services, and monitoring non-native species are some topics you will dive deep into during the course. The course narrative is built around entrepreneurial tools such as value proposition development and customer and market segmentation, value chain and stakeholder analyses, and sustainable and inclusive business modeling to analyze opportunities for new development-oriented start-ups within entrepreneurship. This contributes to food security, poverty reduction, and sustainability for the future of natural ecosystems.	
Study Goals	This course teaches students how to develop a business idea and turn it into a sustainable start-up focused on climate action and wildlife conservation. Students will learn about essential entrepreneurship and apply it to the global/regional/national/local context, which requires understanding a different cultural context and designing for sustainable change. By the end of this course, you should be able to: By the end of this course, you should be able to: Macro Level: Learning block 1 - Creation of Shared Value & Introducing the product to the market LO1: Define the basic principles of entrepreneurship and identify stakeholders in the entrepreneurial ecosystem and explain their role in supporting sustainable entrepreneurship and start-ups LO2: Define how to create shared value with startups to boost the societal transition for food systems and natural ecosystems, which is required to combat climate change, biodiversity loss, and its impacts. LO3: Analyze the competitive landscape (entrepreneurial ecosystems), stakeholders, and local communities roles and devise your products user advantage; for different geographical and cultural contexts LO4: Identify your potential market, beachhead market, and the size of the market. LO5: Identify and apply market research techniques to rigorously test and validate your business idea. Meso-level - Learning block 2 - Ideation & Prototyping LO6: Deliver ideas, products, and services in response to sustainability challenges. LO7: Acquire skills and use tools that can be used to facilitate an entrepreneurial process. Micro-level - Learning block 3 - The sustainable&/circular business models & human behavior LO8: Explain and apply methods for creating sustainable/circular business models for your startup idea LO9: Analyze and identify what are the factors for sustainable technology innovation adoption for the specific customer segment of your startup idea product/service LO10: Explain and apply human behavior strategies and marketing mix to your business model.	
Education Method	A merger of lectures and guest lectures with serial and inspiring entrepreneurs and practitioners, fieldwork visits, workshop activities, group pitch deck presentations, and in-class discussions. Coaching sessions along the ideation process are highly encouraged and must be coordinated in advance with the lecturer. The course is organized into weekly interactive lectures wherein theories of that week will be explained. Moreover, guest lectures will be given by speakers such as co-founders, food technology experts, and nature & conservation practitioners, sharing their experiences, barriers, and learnings about starting a company. Lastly, workshops will be given during the lectures, wherein the students can apply the content to their specific start-up and work on their ideas.	
Assessment	The final grade for this course consists of: Learning block peer-review assignments: 20% Executive Report: 20% Infographic Poster (e.g., Gigamap): 20% Final Pitch Presentation: 30% Active participation (Lectures & coaching sessions): 10% A minimum grade of 5.75 is required for each deliverable to pass the course.	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

TPM420A	Ready to Start-up	6
Module Manager	Ir. B.R. Joseph	
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	This course is offered twice in a year; First starting and ending in Q2 and second starting mid Q3 until end Q4	
Course Contents	Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders. RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.) The course has the prerequisite that teams should be formed and have business ideas that are analysed and validated. It is meant for students who plan to start a company. After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.	
Study Goals	At the end of the course, you will be able to - Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models. - Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors). - Plan your core operational activities such as company foundation, IP protection, product development strategy and team development - Devise financial plans and appropriate fundraising strategies - Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans. - Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.	
Education Method	The course is spread out over 8 sessions (online in case of persistent Covid-19 restrictions for on campus education) in the afternoon (13h30-17h30) where the core elements of a startup plan are presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course. In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practitioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a jury pitch and an executive summary.	
Literature and Study Materials	Handouts and literature are available trough in Brightspace.	
Prerequisites	Enthusiasm, entrepreneurial spirit and a good business idea.	
Assessment	The course has 3 grading components that are Covid-19 proof. - Individual: Active online participation and constructive peer review feedback on other business plans (30%) - Team: class-based assignments (20%) - Team: final startup plan report (40%), supported by a jury pitch (10%)	
Enrolment / Application	Application process you need to register in advance to the course. Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter. All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM425B	Start-up Challenge	3
Module Manager	Dr.ing. V.E. Scholten	
Instructor	Dr. A. Giga	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This course is a part of a TU Delft startup competition. If you would like to participate in this course, you have to be a part of the contest. This means you have to have an idea for a business that you would like to develop, and apply for the competition (either individually or as a team). Also, note that the competition (and course) runs through periods 3 and 4. It usually starts around the beginning of March and ends in the first week of June. The timeline for the competition is published well in advance.	
Course Contents	During the course/competition, you will focus on identifying a customer for your product or service and building a case for how your product addresses a customer problem or an unmet need. You will also perform a market analysis for this business idea. More specifically, you will assess the competitive advantage of your team and your product and the market size and segmentation, among other things. The focus of the course is on customer discovery and business model development. You should be talking to a lot of customers and stakeholders, thereby aggressively learning about the market you plan to enter. Throughout the course you will have an opportunity to present your project idea to industry professionals.	
Study Goals	After this course, you will have learned how to: - assess the problem-solution fit of your idea; - test, modify, and develop your idea by talking to customers and learning about the marketplace; - present your project idea; - emulate effective entrepreneurial behavior.	
Education Method	The course offers a mix of lectures, recommended online materials and books, coaching sessions, and meetings with professionals with industry and entrepreneurial experience. Meetings with industry experts will help you validate the idea and occasional individual coaching will aid your personal entrepreneurial development.	
Assessment	This is a pass/fail class. To pass the course, you must: - Submit intermediate and final deliverables of decent quality. - Show evidence of customer discovery and market learning via interviews - Attend lectures and meetings with professionals.	
Enrolment / Application	You need to apply for the competition to be eligible to participate in this course. The application deadline for the competition is in mid-February and the competition starts in March. E-mail dce@tudelft.nl to get the application form and all the other necessary information.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	This is a MSc elective. Master students from all master programs can apply.	

WM0203TU-Eng	Oral Presentations	2
Responsible Instructor	Drs. R.J. Hoffmann	
Module Manager	Drs. R.J. Hoffmann	
Module Manager	L. Meester	
Module Manager	I.M. Schouten	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	In job advertisements for engineers applicants are always required to possess excellent communication skills. This means among other things that they should be able to present their designs and research results fluently, for example during meetings with clients or with managers in their own company. A persuasive presentation does not only require thorough preparation of content, but also good style. It takes quite a bit of skill to come across understandable for any particular audience and to stay in control of the situation. For this purpose the course Oral Presentations offers students the possibility to gain experience in giving presentations under professional guidance in a small group.	
Study Goals	At the end of the course you should be able to: Present complex material clearly to both expert and non-expert audiences; Apply structural elements that are offered in the course; Use presentation techniques such as: - Using various hand gestures - Using vocal variety - Keeping a good posture; Create attractive slides that support your presentation; Interact with your audience and respond to their reactions and questions professionally.	
Education Method	The first meeting will be an introduction to several presentation principles. The second, third, fourth, fifth and sixth meeting will be dedicated to practicing presentation skills. Students will be asked to prepare short (group and individual) presentations for this purpose. For the seventh meeting students will be asked to give a final presentation about a subject that is related to their study. All presentations will be recorded on video. This will offer participants the possibility to evaluate their own presentations. Attendance is compulsory; no more than one lesson may be missed.	
Literature and Study Materials	Van der Laaken, M. & van der Laaken, B. (2013). Presentation Techniques (Rev. ed.). Coutinho.	
Assessment	Students receive a final mark from their teacher based on the final presentation. Students who fail, will be given the opportunity to try again during an extra session, or to compensate for this by means of an assignment respectively.	
Enrolment / Application	Please enroll for a group in Brightspace: wm0203tu-eng. After enrollment you can subscribe in a group. Note: before subscribing check if you get ECTS for this course or if Oral Presentations (eng) is needed. For this course there are limited spots available.	
Elective	Yes	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. There is an option to take this course as a self-study programme with a final test + a speaking test.. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Intermediate 1 (WM1117TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test + speaking test. Contact the secretary's office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether this course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as AI tools	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	This is a pass/fail course. There is no final exam. Assessment is based on the six written assignments and on the in-class discussions and presentations.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Orientation Study Abroad 2023

Contact for students	International office Applied Sciences, Physics building, Room A217, InternationalOffice-TNW@tudelft.nl.
In association with the University of	see http://partneruniversities.tudelft.nl and http://brightspace-cc.tudelft.nl/course/1827/international-office-applied-sciences
EC Program	30
Prerequisites	GPA 7.0 or higher. MSc (planning) without study delay. Selection/ranking of students who apply for each destination is based on academic achievement (BSc results) and MSc study planning.
Introduction 1	Note: the application deadline for the spring semester is September 15! Deadline for the autumn semester is December 15. This programme consists of a semester at a foreign university. It is tailored for a.o. (Dutch) students who will do part of their master education abroad. It consists of one semester, a project and/or courses. For completing the master programme within two years, it fits best in the spring semester of the first year. As an alternative: complete the core programme including the thesis project in 1.5 year and finish the master programme (with a research project) at a foreign university. Permission to do the orientation in (the first semester of) the second year is only granted if the courses from the core program (42 ec) have been successfully completed in the first year. For more information see: http://tudelft.nl/studyabroad-tnw and https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/outgoing-exchange-application-procedure .
Program Structure 1	A package of 30 EC of courses (optionally including a research project of 20 EC or less) must be done. The programme has to be approved in advance by the board of examiners. A maximum of 6 EC of language/societal courses can be done within the programme.
With Honours Regulation	Grades obtained abroad are not converted into Dutch grades. Results obtained abroad are not included in the calculation of the GPA (a.o. for graduating with honours).

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Orientation Education 2023

EC Program	30
Program Start (Study Year)	This orientation fits best in the spring semester of the first year. Contact the coordinator of the MSc Science Education and Communication well in advance to enroll in this orientation. The deadline for registering for a start in February is November 15. A teacher training internship at a school cannot be guaranteed if you register too late. Deadline for a start in September is April 30.
Contact for Students	For the planning of the education modules in the Applied Physics programme: Arno Haket (A.J.W.Haket@tudelft.nl). For questions about the education modules and registering for teacher training internships at schools: MSc Science Education & Communication programme management (Info-SEC@tudelft.nl).
Introduction 1	The educational programmes are aimed at Dutch-speaking students only, because they are oriented towards the Dutch school system and because it includes internships (Schoolpracticum) at Dutch schools. Consequently the modules of educational orientation are taught in Dutch. The programme consists of Basisdeel/Ed1 (30 EC) and Verdiepingsdeel/Ed2 (30 EC). The minor Education (Basisdeel/Ed1) can be done during the bachelor programme and leads to certification as a tweedegraads secondary school teacher with limited qualification (beperkte bevoegdheid). If a student has done the minor Education, only the Verdiepingsdeel/Ed2 of 30 EC remains for the master programme orientation. The combination of the minor Education and Ed2 orientation leads to certification as a fully-qualified eerstegraads (grade-one) secondary school teacher. The certificate will be attached to the master diploma. Students that did not take the minor Education can follow the Basisdeel/Ed1 orientation as part of their master programme and then do the Verdiepingsdeel/Ed2 as a post-master course in order to become fully qualified. Both Ed1 (basis) and Ed2 (verdieping) are scheduled twice a year. The programme must be approved by the programme coordinator of the MSc Science Education and Communication (Info-SEC@tudelft.nl). You should contact her before starting this orientation.

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Orientation Education - Basis (Ed1)	
Program Coordinator	M. Hajivandi

SL3462	Educational Sciences		6
Responsible Instructor	S. Schut		
Module Manager	Dr. J. Van Keulen		
Contact Hours / Week x/x/x/x	Periode 1: 2 uur per week, periode 2: 3 uur per week.		
Education Period	1 2 3 4		
Start Education	1 3		
Exam Period	1 2 3 4		
Course Language	Dutch		
Course Contents	Onderwijskunde (in deze context) is gericht is op het begrijpen en verbeteren van onderwijs en legt de theoretische basis en verdieping onder de onderwijspraktijk. Het heeft betrekking op leerprocessen en interactie binnen en buiten de klas, op de docent, de schoolorganisatie, maar ook op beleid en bestuur van het onderwijs. De functie van onderwijskunde in de opleiding van leraren is dat de docenten in opleiding (DIOs) zich eigen maken wat toepasbaar is uit deze wetenschappelijke discipline bij het lesgeven van alle dag en om hen te helpen een onderzoekende houding aan te nemen. Vanuit de startbekwaamheidseisen focust dit vak zich op de pedagogische rol van de docent en zoekt daarnaast de verbinding met de vakinhoudelijke en vakdidactische rol die centraal staat in de andere vakken binnen de lerarenopleiding, de educatieve minor of de educatieve module. De docent heeft in de pedagogische rol rekening te houden met de verschillende contexten die van invloed zijn op functioneren, groei en ontwikkeling van leerlingen. Die contexten zijn: de docent, de klas, de leerlingen, de school en de samenleving. In navolging van het ecologisch model van Bronfenbrenner worden deze contexten van micro- naar macroniveau behandeld in deze cursus.		
Study Goals	Je kunt: - jouw visie op de pedagogische verantwoordelijkheid van leraar en school expliciteren en die met argumenten onderbouwen; - de ontwikkeling van jouw pedagogische professionaliteit in termen van: pedagogisch bewust zijn (het kunnen zien van het belang van een leerling in een specifieke context), pedagogisch legitimeren (het kunnen verwoorden en bespreken van de pedagogische idealen die ten grondslag liggen aan het onderwijzend handelen) en pedagogisch handelen (het vorm kunnen geven aan pedagogische idealen bij het maken van onderwijs) beschrijven; - op systematische wijze gegevens verzamelen en analyseren voor de beantwoording van een leervraag over het eigen pedagogisch handelen in de praktijk; - jouw eigen functioneren zien als onderdeel van de interactie tussen leraar en leerling(en) en daarop reflecteren aan de hand van gegevens over die interactie; - het functioneren van (groepen) leerlingen analyseren vanuit een sociale systeembenadering; - basiskennis van de ontwikkelingspsychologie in de fase van adolescentie en basiskennis van zorg voor leerlingen vertalen naar jouw lespraktijk; - in jouw professionele handelen rekenschap geven van de maatschappelijke verantwoordelijkheid van het onderwijs in het algemeen en ten aanzien van burgerschapsvorming in het bijzonder.		
Education Method	Colleges, zelfstudie, interactieve sessies, groepswork, koppeling met Schoolpracticum, PLG en Vakdidactiek		
Prerequisites	Geen		
Assessment	Aanwezigheid bij de colleges is verplicht. Onderwijskunde wordt afgesloten met 1 eindresultaat. Het eindresultaat is gebaseerd op twee groepsopdrachten en een individuele opdracht. Alle deelopdrachten moeten minimaal voldoende zijn. Zie het toetsplan voor de uitgebreide informatie over de deelopdrachten, herkansingen, en/of aanvullende opdrachten.		

SL4201	Pedagogy of STEM education	4
Responsible Instructor	Dr. L. Koopman	
Responsible Instructor	Dr. W.T.M. Caspers	
Contact Hours / Week x/x/x/x	4 uur per week	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	Dutch	
Course Contents	Bij vakdidactiek gaat het om een analyse van de lesstof. Enerzijds vanuit het vakgebied en het daarvan afgeleide schoolvak en anderzijds vanuit leerlingdenkbeelden en voorkennis. Deze analyse wordt gebruikt voor het ontwikkelen van een aantal lessen waarin vaktaal, voorkennis en de inhoud van het schoolvak belangrijke aspecten zijn. Bij Inleiding Bètdidactiek ga je aan de slag met het ontwikkelen van algemene didactische competenties. De volgende zaken komen aan bod: 1. lesvoorbereidingsmodel met aandacht voor (a) leerdoelen & succescriteria; (b) voorkennis en preconcepten; (c) formatieve toetsing; (d) werkvormen; 2. enkele activerende werkvormen, waaronder het directe instructiemodel; 3. enkele vormen van formatieve toetsing; 4. opstellen en verhelderen van leerdoelen en succescriteria; 5. theatertechnieken training.	
Study Goals	De student kan: 1. Vanuit vakdidactische leerdoelen haar eigen onderwijs voorbereiden, uitvoeren, evalueren en bijstellen. Hierbij ligt de focus op het leren van de leerlingen. Er wordt gebruik gemaakt van theoretische handvatten en verzamelde data. 2. Een les voorbereiden, uitvoeren en evalueren waarbij: a. Een optimaal leerrendement van de leerlingen wordt nagestreefd. b. Gebruik wordt gemaakt van theoretische handvatten en verzamelde data. c. De aandacht van de toehoorders wordt verkregen en behouden met behulp van effectieve presentatietechnieken en activerende werkvormen.	
Education Method	Het vak bestaat uit twee onderdelen: een hoor- en een werkcollege. Het hoorcollege wordt door alle studenten, van alle schoolvakken gevolgd. Het hoorcollege richt zich vooral op de vakdidactische basiskennis die voor alle schoolvakken nodig is. Het hoorcollege reikt daarvoor de gereedschappen aan die alle studenten minimaal dienen te beheersen voor het geven van goed onderwijs. Het hoorcollege wordt gegeven in de vorm van instructievideo's en voorbereidende opdrachten op BrightSpace (ca. 2 uur per week). De student is zelf verantwoordelijk voor het bekijken en maken hiervoor voorafgaand aan het betreffende werkcollege. Het werkcollege (2 uur per week) wordt in een groep van ca. 12 studenten gevolgd. Het werkcollege wordt gegeven door een student-assistent. In het werkcollege komt vooral het oefen van vaardigheden aan bod en het leren hanteren van de gereedschappen die in het college zijn aangereikt. Om je de gereedschappen eigen te maken en je te leren ontwikkelen als docent werken we, net als in de stage, met een PDCA-cyclus (Plan, Do, Check, Act). Je kiest daarin zelf een vakdidactische leervraag. Daarnaast leer je de in het hoorcollege aangereikte gereedschappen toe te passen in de stage door het maken van lesplannen.	
Literature and Study Materials	Activerende werkvormen voor bètadocenten, Martin Bruggink (2017) Activerende werkvormen, Flokstra J.H. (2006) online	
Assessment	Leerdoel 1 wordt aangetoond door middel van twee keer een PDCA-opdracht op basis van een vakdidactische leervraag. De leervraag is naar keuze van de student. De PDCA-opdracht wordt schriftelijk vastgelegd (circa 3 à 4 A4, exclusief bijlagen). Zie voor uitleg over de PDCA-opdracht ook de BrightSpace-pagina van Schoolpracticum Basis. Leerdoel 2 wordt aangetoond door middel van twee lesplannen en bijbehorende lesevaluaties. Er zijn twee inlevermomenten voor summatieve evaluatie (voldoende of onvoldoende): één halverwege kwartaal 1, één aan het eind van kwartaal 1. Bij elk inlevermoment wordt dus één PDCA-opdracht en één lesplan ingeleverd. Tussentijds is er de mogelijkheid om werk in te leveren voor formatieve evaluatie. Voor de afronding van het vak dienen beide PDCA-opdrachten en beide lesplannen met evaluaties met een voldoende afgerond te worden. Daarnaast is actieve deelname voor het werkcollege vereist. Er kan maximaal één werkcollege gemist worden, met kennisgeving vooraf. Let op: ook bij Schoolpracticum Basis wordt er beoordeeld op basis van PDCA-opdrachten. Deze zijn echter gericht op pedagogisch-didactische leervragen. Bij deze twee vakken kunnen derhalve niet dezelfde PDCA-opdrachten ingeleverd worden. Eindcijfer: Voldoende / onvoldoende Alle onderdelen moeten voldoende worden afgerond.	

SL4202	Professional Learning Community	1
Responsible Instructor	S. Schaap	
Instructor	Drs.ir. T. Landgraf-Koster	
Instructor	Ir. E.M. Holla	
Instructor	Dr. W.T.M. Caspers	
Instructor	Dr. J.G. Spandaw	
Instructor	R. Ockhorst	
Instructor	S. de Haan	
Instructor	Dr. P.J.J.M. Dekkers	
Instructor	Ir. D.P. Lans	
Instructor	Drs.ir. M.A. Botke	
Instructor	Dr. L. Koopman	
Instructor	Drs. M.J. den Otter	
Contact Hours / Week x/x/x/x	Periode 1: 2 uur per week, periode 2: 1 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	De PLG is ondersteunend aan de andere vakken die in het eerste semester gegeven worden. Bij de PLG worden de PDCA-opdrachten voor Inleiding Bèta didactiek en de beide schoolpractica uit het eerste semester besproken. Het gaat dan om verhelderen van de opdrachten en het bespreken van tussentijdse uitwerkingen van de studenten. In de bijeenkomsten leggen studenten vast wat ze de komende week gaan doen, proberen en waarom. Dit houdt de PDCA-cyclus op gang.	
Study Goals	De student kan werken met de PDCA-cyclus: leren eigen leerdoelen stellen, manieren identificeren om in de praktijk te werken aan deze leerdoelen, feedback te verzamelen, evalueren van de leerdoelen en reflecteren op het resultaat en formuleren van verbeterpunten op basis van theorie en verzamelde data.	
Education Method	Interactieve sessies	
Prerequisites	Geen	
Assessment	Aanwezigheid en actieve deelname.	
	PLG wordt beoordeeld als voldoende/onvoldoende.	

SL4220	Physics Teaching Methodology	4
Responsible Instructor	A.J. Boer	
Instructor	R. Ockhorst	
Instructor	Dr. P.J.J.M. Dekkers	
Instructor	G. Bakker	
Instructor	Dr. L. Koopman	
Contact Hours / Week x/x/x/x	4 uur per week	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	Dutch	
Expected prior knowledge	Geen, zie ook Opleidingsspecifieke bijlage van het OER.	
Course Contents	Het cursusaanbod stelt de student in staat effectief hedendaags onderwijs in het eigen vak te verzorgen binnen het gehele bevoegdheidsgebied en zich daarin te blijven ontwikkelen. Centraal staat het ontwikkelen van een lessenserie/project waarin vaktaal, voorkennis en de inhoud van het schoolvak belangrijke aspecten zijn. In de lessenserie/project zijn toetsing, verschillende werkvormen en een onderbouwde onderwijsaanpak essentieel en daarom zal theorie over deze onderdelen aangereikt worden.	
Study Goals	Verbreiding van vakdidactiek. 1 De student kan een visie op het vakgebied en het schoolvak formuleren en onderbouwen. 2 De student kent het Nederlandse landschap behorend bij het schoolvak, zoals de vakvereniging, de tijdschriften, de conferenties en scholierenwedstrijden. 3 De student kent de samenhang tussen de onderdelen van de schoolstof en (globaal) hun samenhang met toepassingen. 4 De student kent verschillende lesmethoden en kan aan de hand van deze verschillende methoden een les en een lessenserie/project voorbereiden. Verdieping van vakdidactiek. 5 De student kan aangereikte vakdidactiek gebruiken in een lesvoorbereiding. 6 De student kan verschillende uitlegstrategieën gebruiken in een lesvoorbereiding en kan zijn keuzes onderbouwen. 7 De student kan de vaktaal correct hanteren en maakt leerlingen bewust van vaktaal. De student heeft daarbij aandacht voor pre- en misconcepties. 8 De student kan verschillende werkvormen gebruiken in les een waarmee leerlingen worden geactiveerd en de samenwerking tussen leerlingen wordt bevorderd. Ook kan de student zijn keuzes voor deze werkvormen onderbouwen. Lessenserie/project ontwerpen en evalueren. 9 De student kan leerdoelen (van een lessenserie/project) opstellen. 10 De student kan een uitvoerbare lessenserie/project voorbereiden, rekening houdend met de voorkennis, interesses, mogelijkheden en beperkingen van de leerlingen. De student kan daarbij een onderbouwde onderwijsaanpak kiezen. 11 De student kan een lessenserie/project voorbereiden gebruikmakend van andere bronnen naast een lesmethode. 12 De student kan een valide en betrouwbaar toetsing, inclusief beoordelingsmodel opstellen. De student kan een gegeven beoordelingsmodel correct hanteren.	
Education Method	De bijeenkomsten zullen bestaan uit korte instructiemomenten, intervisiebijeenkomsten en gastcolleges door experts.	
Assessment	Bij dit vak word je regelmatig gevraagd om werk in te leveren, zodat je formatieve feedback kunt ontvangen van je medestudenten en je docent op basis waarvan je je kunt ontwikkelen. De summatieve beoordeling van VD Basis bestaat uit: Een cijfer voor de opgeleverde lessenserie/project waarin aangetoond wordt dat aan de bijbehorende leerdoelen van VD Basis is voldaan en een uitgebreide reflectie op de stappen in je ontwerp- en leerproces. Een cijfer voor de ontwikkelde toets inclusief een toetsmatrijs, correctiemodel en evaluatie. Een cijfer voor het correct hanteren van vaktaal en inspelen op misconcepties. Dit kan getoetst worden aan de hand van een gegeven mini les, lesvoorbereidingen of via de opgeleverde toets. Er geldt bij dit vak een aanwezigheidsplicht. De onderdelen van het dossier worden beoordeeld op basis van een rubric. Voor ieder onderdeel is een score van minimaal 5,5 benodigd, voor het complete dossier een score van minimaal 5,75 uit 10. Het eindcijfer is het gemiddelde van $2 \times A$, $1 \times B$ en $1 \times C$.	

SL4225	Physics Foundation Teaching Placement	15
Responsible Instructor	Dr. P.J.J.M. Dekkers	
Responsible Instructor	G. Bakker	
Responsible Instructor	Dr. L. Koopman	
Responsible Instructor	A.J. Boer	
Responsible Instructor	R. Ockhorst	
Practical Coordinator	M. Hajivandi	
Practical Coordinator	S. Schut	
Contact Hours / Week x/x/x/x	In overleg met je stageschool, waarbij er een urennorm wordt gehanteerd. Hierover meer informatie op stagehandleiding.net	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Course Contents	Tijdens dit schoolpracticum ga je over van observeren naar eerst onder begeleiding lesgeven en later zelfstandig lesgeven. Tijdens het schoolpracticum komen de volgende activiteiten aan bod: Het observeren van lessen. Het individueel begeleiden van leerlingen. Het onder begeleiding van de schoolpracticumdocent voorbereiden en geven van losse lessen. Het geven van een lessenserie/project waarbij vakdidactische kennis en vaardigheden worden toegepast. Een lessenserie is een reeks op elkaar aansluitende lessen waarin een samenhangend stuk stof wordt behandeld en wordt afgerond met een toetsing. Het zich oriënteren op (en zo mogelijk participeren in) begeleidings- en organisatietaken binnen de sectie en de school. Het reflecteren op het eigen functioneren als docent in opleiding.	
Study Goals	Er zijn twee hoofddoelen voor de stages. 1. De student kan vanuit haar/zijn persoonlijke leerdoelen binnen het bevoegdheidsgebied het eigen natuurkunde onderwijs voorbereiden, uitvoeren, evalueren, bijstellen (PDCA-cyclus doorlopen) met focus op het leren van de leerlingen en gebruik makend van verkregen van theoretische handvatten. 2. De student is: a) Vakinhoudelijk bekwaam b) Vakdidactisch bekwaam c) Pedagogisch bekwaam De drie bovengenoemde bekwaamheden zijn gebaseerd op het Besluit Bekwaamheidseisen Onderwijspersoneel.	
Education Method	Stage en intervisie In overleg met de instituutsbegeleider en de begeleider op school (vakcoach) kies je de klassen waarin je onderwijs verzorgt. In dit Stagepracticum dienen dat vooral onderbouwklassen binnen het bevoegdheidsgebied te zijn. Advies is om ook een aantal bovenbouw lessen te geven. Er dient een minimum aantal lessen en lessenseries gegeven te worden: zie voor de details de Brightspace site bij dit vak. Daarnaast vinden er intervisiebijeenkomsten plaats met je medestudenten en instituutsbegeleider, met aanwezigheidsplicht.	
Prerequisites	Geen	
Assessment	Beoordeling vindt plaats op basis van prestaties in de stage (waaronder de 3 bekwaamheden) en (tussentijdse) opdrachtuitwerkingen van onder meer het doorlopen van de PDCA-cyclus. Het eindcijfer voor het vak is gelijk aan het cijfer voor de beoordeling van de 3 bekwaamheden, mits de andere onderdelen als voldoende zijn beoordeeld. De specifieke beoordelingscriteria staan beschreven in de Brightspace site voor het vak.	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Orientation Education - Verdieping (Ed2)	
Program Coordinator	M. Hajivandi
EC Program	30
Prerequisites	minor Education / Education Basis
Contact for Students	For the planning of the education modules in the Applied Physics programme: Arno Haket (A.J.W.Haket@tudelft.nl). For questions about the education modules and registering for teacher training internships at schools: MSc Science Education & Communication programme management (Info-SEC@tudelft.nl).
Introduction 2	The educational programmes are aimed at Dutch-speaking students only, because they are oriented towards the Dutch school system and because it includes internships (Schoolpracticum) at Dutch schools. Consequently the modules of educational orientation are taught in Dutch. The programme consists of Basisdeel/Ed1 (30 EC) and Verdiepingsdeel/Ed2 (30 EC). The minor Education (Basisdeel/Ed1) can be done during the bachelor programme and leads to certification as a tweedegraads secondary school teacher with limited qualification (beperkte bevoegdheid). If a student has done the minor Education, only the Verdiepingsdeel/Ed2 of 30 EC remains for the master programme orientation. The combination of the minor Education and Ed2 orientation leads to certification as a fully-qualified eerstegraads (grade-one) secondary school teacher. The certificate will be attached to the master diploma. Students that did not take the minor Education can follow the Basisdeel/Ed1 orientation as part of their master programme and then do the Verdiepingsdeel/Ed2 as a post-master course in order to become fully qualified.

SL3012	Personal Professional Development	3
Responsible Instructor	Drs.ir. M.A. Botke	
Contact Hours / Week x/x/x/x	1 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Course Contents	Als docent ben je nooit uitgeleerd. In dit vak hebben we het over professionele ontwikkeling van jezelf als docent in een school. Je doet enkele opdrachten die inzicht geven in je professionele identiteit. Daarnaast werk je aan het realiseren van een innovatie op je (stage)school. Daarbij ligt de nadruk op het proces van innoveren en de samenwerking met je collegas. Binnen dit vak is veel ruimte voor uitwisseling over onder meer professionele identiteit, het vormgeven van innovaties en richting geven aan je eigen ontwikkeling.	
Study Goals	De student kan aspecten van zijn/haar eigen professionele identiteit beschrijven. De student kan de professionele identiteit van collega-docenten beschrijven op basis van een interview. De student kan reflecteren op het proces van innoveren in de school en dit relateren aan de professionele identiteit van zichzelf en de collegas die betrokken zijn bij de innovatie. De student is in staat constructief bij te dragen aan een professionele leergemeenschap met peers.	
Education Method	Deelname aan de colleges, opdrachten of professionele identiteit, werken aan innovatie in de (stage)school.	
Prerequisites	Om deel te kunnen nemen aan vakken in het verdiepingsdeel van het beroepsgerichte jaar, moet het basisdeel afgerond zijn.	
Assessment	Toetsing vindt plaats op grond van Aanwezigheid en actieve deelname in de colleges De opdrachten over professionele identiteit Verslaglegging over het realiseren van een innovatie in de (stage)school, met focus op het proces en de samenwerking. Het vak wordt beoordeeld met een voldoende als alle bovenstaande punten als voldoende zijn beoordeeld. Er geldt bij dit vak een aanwezigheidsplicht.	

SL4205	Advanced Professional Learning Community Professional Learning Community	1
Responsible Instructor	S. Schaap	
Course Coordinator	M. Hajivandi	
Instructor	Drs.ir. T. Landgraf-Koster	
Instructor	Dr. W.T.M. Caspers	
Instructor	Dr. J.G. Spandaw	
Instructor	R. Ockhorst	
Instructor	S. de Haan	
Instructor	Dr. P.J.J.M. Dekkers	
Instructor	Ir. D.P. Lans	
Instructor	Drs.ir. M.A. Botke	
Instructor	Dr. L. Koopman	
Instructor	Drs. M.J. den Otter	
Contact Hours / Week x/x/x/x	1 uur per week PLG verdieping wordt gegeven binnen de uren die ingeroosterd zijn voor het Vakdidactiek Verdieping. De docent zal de precieze inrichting hiervan bekend maken.	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	De PLG is ondersteunend aan de andere vakken die in het eerste semester gegeven worden. Bij de PLG worden de PDCA-opdrachten voor Inleiding Bèta didactiek en de beide schoolpractica uit het eerste semester besproken. Het gaat dan om verhelderen van de opdrachten en het bespreken van tussentijdse uitwerkingen van de studenten. In de bijeenkomsten leggen studenten vast wat ze de komende week gaan doen, proberen en waarom. Dit houdt de PDCA-cyclus op gang.	
Study Goals	De student kan werken met de PDCA-cyclus: leren eigen leerdoelen stellen, manieren identificeren om in de praktijk te werken aan deze leerdoelen, feedback te verzamelen, evalueren van de leerdoelen en reflecteren op het resultaat en formuleren van verbeterpunten op basis van theorie en verzamelde data.	
Education Method	Interactieve sessies	
Assessment	Aanwezigheid en actieve deelname. PLG wordt beoordeeld als voldoende/onnvoldoende.	

SL4301	Implementation of Education	9
Responsible Instructor	Dr. P.J.J.M. Dekkers	
Responsible Instructor	A.J. Boer	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Expected prior knowledge	To participate in the advanced courses the basis courses must have been completed.	
Course Contents	Methodology directed at the pedagogical interpretation, practical implementation and iterative optimisation of beta-technical educational materials and approaches in secondary education.	
Study Goals	Developing knowledge and skills of finding, interpreting, implementing and optimising of educational ideas in the beta-technical pedagogical and professional literature. Developing the skills to find appropriate sources given a problem from practice, and to develop effective practical applications based on inspiring sources.	
Education Method	A combination of lectures, workshops and working in small groups with live and online feedback in the preparatory phase. Tasks involving actual implementation are carried out during teaching practice.	
Assessment	Production of an iteratively developed desing for implementation, based on PCK analysis, justified plans for implementation, reports of implementation and evaluation.	

SL4320	Advanced Physics Teaching Methodology	5
Responsible Instructor	Dr. P.J.J.M. Dekkers	
Instructor	A.J. Boer	
Instructor	R. Ockhorst	
Instructor	G. Bakker	
Instructor	Dr. L. Koopman	
Contact Hours / Week x/x/x/x	3 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	2 4	
Course Language	Dutch	
Course Contents	Het cursusaanbod stelt de student in staat effectief hedendaags onderwijs in het eigen vak en het vak NLT te verzorgen binnen het gehele bevoegdheidsgebied, en zich daarin te blijven ontwikkelen. Standaardthemas omvatten summatieve en formatieve toetsing, leerstofanalyse en -problematiek, vaardigheden, gebruik van ICT, STEM-, vakoverstijgend, project-, modelleer- of ontwerpgericht onderwijs. Facultatieve themas omvatten bijvoorbeeld; differentiatie, demonstratiepractica, en opties voor professionele ontwikkeling. Per thema wordt een route aangeboden bestaande uit inventarisatie, interpretatie en implementatie, die voor standaardthemas altijd en voor facultatieve onderdelen soms geheel doorlopen wordt.	
Study Goals	Startcompetentie m.b.t. natuurkunde en NLT De student is competent in het vorm geven van effectief eigentijds onderwijs in natuurkunde en NLT binnen het gehele bevoegdheidsgebied. Ontwikkelingscompetentie De student kan een selectie van huidige ontwikkelingen in het onderwijs van het eigen vak benoemen en beschrijven, zichzelf daarover nader informeren en daarin verder ontwikkelen, en met collegas die ontwikkelingen in het eigen onderwijs vormgeven (bijvoorbeeld op basis van een PDCA cyclus).	
Education Method	Interactieve sessies en colleges.	
Prerequisites	Om deel te kunnen nemen aan vakken in het verdiepingsdeel van het beroepsgerichte jaar, moet het basisdeel afgerond zijn.	
Assessment	De student levert een dossier in waarin wordt aangetoond dat: 1 - op basis van een passend product dat ieder thema op professionele wijze is geïnventariseerd; 2 - op basis van zelf ontwikkeld onderwijsmateriaal dat minstens vijf van de behandelde themas professioneel zijn geïnterpreteerd voor de eigen onderwijscontext; 3 - op basis van rapportage, reflectie en evaluatie dat minstens acht van de themas in minstens 300 minuten eigen (stage)onderwijs professioneel zijn geïmplementeerd. 4 Voor de module NLT worden studenten beoordeeld op de inhoud van het tijdens de cursus ontwikkelde invoeringsplan van de geanalyseerde lesmodule. De onderdelen van het dossier worden beoordeeld op basis van een rubric. Voor ieder onderdeel is een score van minimaal 5,5 benodigd, voor het complete dossier een score van minimaal 5,75 uit 10. Er geldt bij dit vak een aanwezigheidsplicht. Het cijfer is het gemiddelde van de vier hierboven genoemde deeltijfers	

SL4325	Advanced Physics Teaching Placement	12
Responsible Instructor	Dr. P.J.J.M. Dekkers	
Responsible Instructor	G. Bakker	
Responsible Instructor	Dr. L. Koopman	
Responsible Instructor	A.J. Boer	
Responsible Instructor	R. Ockhorst	
Practical Coordinator	M. Hajivandi	
Practical Coordinator	S. Schut	
Contact Hours / Week x/x/x/x	In overleg met je stageschool, waarbij er een urennorm wordt gehanteerd. Hierover meer informatie op stagehandleiding.net	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Course Contents	In Schoolpracticum Basis heb je aangetoond dat je in staat bent lessen(series) te ontwerpen en te geven. In Schoolpracticum Verdieping doe je meer ervaring op en verdiep je je competenties. Je gaat werken aan de leerpunten zoals je die voor jezelf formuleert op basis van eerdere stage-ervaringen en op basis van nieuwe doelen die je jezelf stelt in deze nieuwe onderwijsomgeving. Ten opzichte van Schoolpracticum Basis zijn er de volgende verschillen: Het zwaartepunt van deze stage ligt in de bovenbouw. In het bijzonder moet je ook vakinhoudelijk competent zijn op eerste-graadsniveau. Er is meer aandacht voor het examen en het schoolexamen. Er is meer aandacht voor gevorderde didactische vaardigheden, zoals ontwerpgericht onderwijs, differentiatie en groepswerk.	
Study Goals	Er zijn twee hoofddoelen voor de stages. De student kan voor het gehele bevoegdheidsgebied vanuit haar/zijn persoonlijke leerdoelen het eigen natuurkunde onderwijs voorbereiden, uitvoeren, evalueren, bijstellen (PDCA-cyclus doorlopen) met focus op het leren van de leerlingen en gebruik makend van verkregen van theoretische handvatten. De student is: a) Vakinhoudelijk bekwaam b) Vakdidactisch bekwaam c) Pedagogisch bekwaam De drie bovengenoemde bekwaamheden zijn gebaseerd op het Besluit Bekwaamheidseisen Onderwijspersoneel.	
Education Method	Stage en intervisie In overleg met de instituutsbegeleider en de begeleider op school (vakcoach) kies je de klassen waarin je onderwijs verzorgt. In dit Stagepracticum dienen dat vooral bovenbouwklassen binnen het bevoegdheidsgebied te zijn. Advies is om ook een aantal onderbouw lessen te geven. Er dient een minimum aantal lessen en lessenseries gegeven te worden: zie voor de details de Brightspace site bij dit vak.	
Prerequisites	Om deel te kunnen nemen aan vakken in het verdiepingsdeel van het beroepsgerichte jaar, moet het basisdeel afgerond zijn.	
Assessment	Beoordeling vindt plaats op basis van prestaties in de stage (waaronder de 3 bekwaamheden) en (tussentijdse) opdrachtuitwerkingen van onder meer het doorlopen van de PDCA-cyclus. Het eindcijfer voor het vak is gelijk aan het cijfer voor de beoordeling van de 3 bekwaamheden, mits de andere onderdelen als voldoende zijn beoordeeld. De specifieke beoordelingscriteria staan beschreven in de Brightspace site voor het vak.	

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

Orientation Management of Technology (MoT) 2023

Contact for students	Arno Haket (A.J.W.Haket@tudelft.nl)
In association with the Faculty of	TBM
EC Program	30
Introduction 1	This orientation is offered by the faculty of Technology, Policy and Management. The programme consists of either the first semester or the second semester of the MSc Management of Technology. In general the second semester fits much better in the schedule of the AP master programme. A mixture of courses from both semesters is not possible.
Maximum number of participants	Due to an increasing number of non-MOT students taking the Management of Technology courses, the number of students that can take the orientation Management of Technology is limited. In the fall semester there are five places available for Applied Physics students taking the MOT orientation. In the spring semester there are eight places for Applied Physics students. Signing up, show interest, questions, etc : Arno Haket, A.J.W.Haket@tudelft.nl. Application deadline for the Spring semester is November 15th. Deadline for the fall semester is July 1st. If necessary, there will be a lottery. After the deadline, participation is only possible if places are still available.

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

1st semester MoT Modules

MOT111A	Financial Management	5
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	A.A. Ralcheva	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	MOT STUDENTS HAVE PRIORITY. This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. Bachelor students are excluded from participation.	
Expected prior knowledge	None	
Summary	This course is about how corporations make financial decisions. It is designed to equip technical/engineering students with fundamental financial management skills.	
Course Contents	The course covers topics such as financial statement analysis, types of financing and investment decisions, and financial consequences of those decisions. While investment decisions involve spending money, financing decisions involve raising it. Important to highlight is that corporations finance their investments by borrowing, by retaining and reinvesting cash flow, and by selling additional shares of stock to the corporations shareholders. Hence, there will be among others three broad financial questions discussed in the lectures. First, what investments should the corporation make? Second, how should it pay for those investments? Third, what plans can the firm make to ensure that it stays financially strong and healthy?	
Study Goals	At the end of the course, students are expected to acquire the ability to: evaluate financial performance of companies through financial statement analysis evaluate investment opportunities using investment appraisal techniques such as internal rate of return (IRR), net present value (NPV), payback period, profitability index discuss different types of long-term (debt and equity) financing such as venture capital, initial public offering, corporate debt perform valuation of bonds and stocks analyze choices of capital structure and portfolio investment evaluate investment and capital structure decision risks (e.g. financial distress) understand and analyse dividend payout policy	
Education Method	The lectures and workshops are on campus.	
Literature and Study Materials	Lecture slides and textbook. The fourth edition is also acceptable	
Books	Jonathan Berk and Peter DeMarzo (2020). Corporate Finance: The Core. Fifth Edition, Pearson. ISBN: 9781292304151	
Assessment	The course assessment is a written individual exam through Digital exam (using Ans) on campus. The exam is open questions and closed book. Only a calculator and a pen are allowed. Dictionary is not allowed. There will be a resit exam scheduled for those who have not succeeded in the regular exam. For detailed information and schedule about the regular and retake exams, please see OSIRIS. For both regular and retake exams, students may review their exams. The exam review will be held physically according to the time scheduled by the lecturer(s). To pass the course, students have to score at least 5.75. The final grade of the exam is computed as bonus points (that students can earned from the workshops, max. bonus points 0.5) plus the exam grade. Hence, final grade = bonus points + exam grade. Note: 1. Bonus points are voluntary and one-time attempt (during each of the workshops). In other words, students who miss a workshop or workshops do not have a second opportunity to re-do the bonus points. 2. If a student does not do any of the bonus questions during the workshops, the students final grade will simply be based on the exam grade. 3. For the re-exam, the bonus points will only be added to the grades of students that are very close to the passing grade of 5.75. In other words, for the re-exam the bonus points are no longer applicable for students who score higher than 5.75. Only the final grade will be entered in OSIRIS.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://qfreeaccountssjc1.az1.qualtrics.com/jfe/form/SV_2hjb2VfUQhJJIq2 Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT112A	Economic Foundation	5
Module Manager	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. C.W.M. Naastepad	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Bachelor students are excluded from participation	
Summary	Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works, why it sometimes does not work as well as we would like, how its performance can be improved by regulation and policy, and how over time it changes and develops. The course does not assume that the student has any prior knowledge of economics. The course has two parts: (1) microeconomics, which covers the standard topics of supply and demand, forms of market competition, imperfect competition, efficiency, labour markets and technological progress by firms; and (2) macroeconomics, which examines aggregate demand, growth, welfare, inflation, unemployment and globalization. In both parts, special attention is given to (economic factors explaining) the extraordinary technological dynamism and productivity growth of market economies and to governments role in improving micro- and macroeconomic performance. Microeconomics (2,5 ECTS) focuses on markets, and takes both the firm as the government as principal actors. How are firms' attempts to realize their objectives (in terms of profits, sales or market shares) influenced by demand, technology and market structure? How can public regulation improve market performance which falls below norms considered acceptable? We identify important market failures, due to which markets under-perform, and specifically investigate firm behaviour in (realistic) oligopolistic markets. Macroeconomics (2,5 ECTS) deals with the workings of the economy as a whole, why the market system often underperforms, and how macroeconomic performance can be improved by fiscal and/or monetary policy. Attention will be given to conflicts between achieving different goals of macroeconomic policy at the same time, and to the ways in which globalization affects the capacity of governments to regulate their economies. The answer to this question holds major lessons for public policy.	
Course Contents	Microeconomics: demand and supply; consumer behavior; market competition; perfect competition; monopoly; oligopoly; market failures; externalities; public goods; regulation; efficiency; welfare; production; profit maximization; choice of technique; technological progress; game theory; dominant strategy; labour markets; employment; worker-management relations. Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; neoclassical economics; Keynesian economics; unemployment; fiscal policy; monetary policy; money creation; crowding out; stabilization policies; labour market regulation; technology policy; foreign direct investment; financial markets.	
Study Goals	At the end of this course, students will be able to: 1. analyze firm behavior in markets of perfect competition, monopoly, and oligopoly as well as the impact of firm behavior on economic welfare. 2. identify and appraise the rationale, scope and limits of public market regulation (in situations of "imperfect markets" and/or "market failures"). 3. derive and document the impact(-s) of regulation on the static efficiency and dynamic efficiency of markets. 4. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness. 5. critically examine and reflect on the existing alternative theories and models. 6. understand how money is created and how the financial sector works, and how this affects the real economy.	
Education Method	Lectures. Students are expected to prepare for each class by reading the indicated literature.	
Books	Lecture Notes, Question sets and papers/articles - all made available on Brightspace.	
Assessment	In normal circumstances, the exam is a closed-book electronic (in-person, on campus) multiple-choice examination of 3 hours. The on-campus exam will be made in Ans. The passing grade for the exam is 5.5. Students will be offered two opportunities to do the exam in an academic year. In case of unforeseen circumstances (such as due to COVID-19), the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours.	
Special Information	None	

MOT121A	Leadership and Technology Management	5
Module Manager	Dr. R.M. Verburg	
Instructor	Dr. N. Pachos-Fokialis	
Co-responsible for assignments	Dr. N. Pachos-Fokialis	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Leadership and management in the context of high technology firms and digital business are the key theme of this course. The core of successful management in technology-based environments is the recognition of the value of employees for organizational success. Selection, development and performance management of employees are core activities of personnel management. Such practices are no longer seen as tasks that are the responsibility of a separate personnel department, but they become part of top managements personnel strategy. Line managers are increasingly expected to play an active role in implementing management policies and practices. Leadership appears to be a critical determinant of organizational effectiveness whether we are discussing a football team, an army, a street gang or a multinational corporation. Leadership is the subject of intense academic research and debate and journalistic commentary. In this course we will highlight the critical roles of leadership on different levels for the success of technology-based companies.	
Study Goals	At the end of this course, students are expected to competently: analyze the nature of leadership management within advance technology organizations provide arguments how leaders contributes to organizational performance recognize knowledge processes in organizations recognize management practices and analyze the success of these within different organizations show the need for alignment between management practices show the need for alignment between management practices and business strategy recognize best practices with regard to the selection and performance management of employees qualify scientific research in the area of Leadership and Organizational Behavior recognize the current and future challenges of technology firms in leading and managing people	
Education Method	The course consists of weekly lectures. The lectures are dynamic and participative including: - Theory presentations and interactive lectures by the staff (in class and online) - Presentations by a guest speaker from practice - Presentations by students who work together in expertise groups - Case write-ups	
Assessment	Individual final examination The final exam will be a closed book exam on the content of this course. Hence, students who take the course seriously and structurally prepare the literature and assignments will be very likely to pass the exam. The exam will consist of Open Questions. A minimum grade of 5.75 in the exam is necessary for a passing this course. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: Take-home assignment (open book), i.e. an Individual assignment based on open-ended question(s), typically to be delivered within period of a few days	
Enrolment / Application	Individual digital final examination. The final digital exam will be a closed book exam on the content of this course (text book, slides, and articles) and will be administered through Ans. Samples of these essay questions will be provided both on Brightspace and during the course. Hence, students who take the course seriously and structurally prepare the literature and assignments will be very likely to pass the exam. The exam will consist of Open Questions. A minimum grade of 5.8 in the exam is necessary for a passing this course. A mandatory group assignment is also part of the course. Although this assignment is not graded it is a prerequisite for passing the exam. Students who did the assignment but wish or need to re-take the exam in the next academic year(s) are exempted from doing the assignment again.	

MOT131A	Emerging and Breakthrough Technologies	5
Module Manager	Dr. J.R. Ortt	
Responsible for assignments	Dr. J.R. Ortt	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	The course focuses on innovation processes in a high-tech environment. Firstly, innovation processes are studied on the micro-level, distinguishing different groups of project approaches to innovation. Secondly, the course will discuss the development and diffusion of breakthrough technologies on the macro-level. The pattern, of the development and diffusion of such technologies, the actors and factors that explain the evolution of the pattern, and strategies that companies can adopt during the pattern are discussed. Thirdly, the course will discuss how technologies in different patterns can affect each other and interact. This may refer to disciplines (defined as combination of related technologies) to competing or to complementary technologies. The three levels of analysis (project, pattern, multi-technology) together form a comprehensive perspective on innovation processes.	
Study Goals	Describe innovation processes on three levels: (1) project level, (2) level of the pattern of development and diffusion of technologies and (3) multi-technology level. Distinguish alternative theoretical perspectives on each level and the assumptions of these perspective. Be able to explore and analyse an innovation opportunity or problem, choose the level of analysis and apply the perspective on that level that fits the context of the opportunity or problem.	
Education Method	lectures and assignment. Normally lectures are face-to-face. If needed, due to circumstances, lectures will be on-line	
Assessment	Exam and one main assignment complemented with possible smaller assignments In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: 1. The exam will administered on-line using maple-ta or a similar program. 2. The assessment will remain the same.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://qfreeaccountssjc1.az1.qualtrics.com/jfe/form/SV_5cFual2gB7w6Xk Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT141A	Research Methods	5
Module Manager	Dr.ir. G.A. de Reuver	
Instructor	Dr. L. Rook	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Student learn how to design and execute research, as well as how to interpret and critically assess the outcomes of research. Both quantitative and qualitative research methods are taught. To do so, students learn the fundamentals of research design as well as quantitative and qualitative research methods. This course is designed to help students develop and execute research steps from the initial research problem until drawing conclusions. These steps include conceptualization of research ideas/problems, development of a relevant theoretical framework and related hypotheses, defining and operationalizing core concepts, and developing an appropriate research design. Students learn how to scrutinize the findings and draw implications drawn from statistical analysis and qualitative analysis of data. Topics that will be covered among others: - define research problem, research objectives and questions - conduct research process of scientific investigation - develop a conceptual model - formulation of hypotheses - operationalize and develop measurement of concepts - develop research design, as well as make motivated decisions on data collection and data analysis methods - evaluate different types of quantitative techniques (hypothesis testing, non-parametrics, regression analysis, multivariate analyses) - interpret and critically reflect upon findings from statistical analysis - qualitative data analysis, based on coding principles - interpretation and presentation of results - reflection on reliability and validity of research results - research ethics	
Study Goals	1. Create a research design for a research problem relevant to the context of MOT To do so, sub-goals are to: 1a. Understand core concepts of research design (i.e. theory, population, sample, hypotheses, research strategy, operationalization, data collection) 1b. Apply these core concepts to design a research project (i.e. from a broad topic towards research questions, conceptualization, research strategy, research design, data collection approach, sampling, operationalization, and selection of analysis methods) 2. Critically evaluate outcomes of qualitative and quantitative research in the MOT context To do so, sub-goals are to: 2a. Interpret statistical analysis findings 2b. Scrutinize / Reflect upon findings / conclusions from both qualitative and quantitative research (i.e. validity, reliability, traceability, ethics) 2c. Suggest alternative / future research directions	
Education Method	Course contains three main parts: 1. Research design 2.1. Qualitative research methods 2.2. Quantitative research methods Part 1 and 2.1 consist of: - Lectures, which are a mix of instruction and formative assignments. - Summative assignments (weekly, group of 2 students) in which a research design is created Part 2.2 consists of: - Lectures and practice	
Literature and Study Materials	To be announced via BrightSpace. At least one textbook on research methods for business.	
Assessment	Summative assignments (15% = 0.75 EC): Group of 2 students, weekly assignment, week 1-5 of the course. The summative assignments cover part 1 and 2.1 of the course. Digital exam (85% = 4.25 EC; at least 5.0 to pass the course). The exam covers part 1, 2.1 and 2.2 of the course. The exam contains open questions and is closed book. Administered via ANS. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed exam will be modified as follows : Timed digital (ANS) open book exam The two grade components (i.e. summative assignments; digital exam) are registered separately into Osiris. This implies that they stay valid in next academic years.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3ObBU6LgvAgQ0Hs Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT142A	Epistemology and Ethics of Business Analytics	5
Module Manager	M.B.O.T. Klenk	
Instructor	M.B.O.T. Klenk	
Instructor	Dr. J.J. Casey	
Co-responsible for assignments	Dr. J.J. Casey	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	As prospective managers of technology, you will have to make difficult decisions regarding the development and deployment of innovation. You will increasingly rely on a data-oriented decision-making approach in doing so. That means that you will need to be able to answer two crucial questions: What can you really conclude from data? And what should you or your company be allowed to do with that data? This course will provide you with an advanced introduction to theories and concepts from the philosophy of science and moral philosophy (ethics) to help you answer these questions and, ultimately, become more informed and reflected about the management of technology. In short: The main aim of the course is to enable you to recognise and understand the normative dimension of gathering and using data in the management of technology. Part 1: Philosophy of Science You want to base your decisions on data. But what does the data really tell you about the world? How do you distinguish good from bad arguments? And what is a reasonable criterion for reliable, scientific claims? The first part of the course will provide an introduction to core topics in the philosophy of science that help you answer these questions. We will focus on this guiding question: What should we conclude from data? We discuss the following topics: - The basics of natural language argumentation. - Commonfallacies that make arguments misleading - Translating natural language arguments to propositional calculus. - Testing the validity of arguments using truth tables. - Falsificationism & verificationism in the philosophy of science. - The difference between inductive & deductive inference. - Evaluating scientific models from the perspective of realism & antirealism. - Limitations of what scientific data and data from business analytics can teach about the world. Part 2: Ethics Companies do and must make increasing use of the data available to them, using a variety of data-sources, from purchasing histories to the usage of services (location, behaviour etc). Naturally, companies use these data to improve their services and attract more customers, but there is a pressing issue with the ethics of gathering and using data to influence people. After learning, in the first part, what you can learn from data, we will discuss in the second part of the course what you should be ethically allowed to do with data. You will be introduced to fundamental concepts and theories in moral philosophy. Specifically, we will discuss the following topics: - A selection of fundamental ethical theories & perspectives on the objectivity of ethics. - Reasoning and argumentation about morality. - Answering moral questions through normative reasoning. - Ethical implications of data gathering during non-traditional experiments (such as A/B Testing & field experiments). - Understanding important concepts like privacy, autonomy, consent, and manipulation. - The ethical implications of behavioural change technology.	
Study Goals	After having finished this course, the student: LO1. Is able to understand informal fallacies in natural language argumentation. LO2. Is able to analyse natural language arguments, translate them into formal arguments in propositional calculus, and test them for validity using truth tables. LO3. Is able to analyse arguments for positions such as verificationism and falsificationism about knowledge generated by science. LO4. Is able to evaluate arguments for and against scientific realism and anti-realism in light of innovations in the use of AI in inquiry. LO5. Is able to understand the core aspects of ethical arguments, based on an understanding of validity in propositional logic. LO6. Is able to apply core arguments for objectivity in ethics to the management of technology. LO7. Is able to analyse core questions concerning the ethics of data gathering and the ethics of influence in the management of technology. LO8. Is able to evaluate business analytics projects by addressing core questions concerning the ethics of data gathering or the ethics of influence.	
Education Method	The module is taught through a mix of lectures and tutorials. Possibly, some lectures may be online. Students will be notified at the beginning of the semester if that is the case, and which lectures this may concern. All tutorials will be on campus. In lectures, the students will be presented with the theoretical background of the course contents and will be able to discuss these with the lecturer. In tutorials, students will apply the theoretical knowledge from the lectures by working on concrete cases, individually or in groups.	
Assessment	Summative assignments: (1) Six pre-lecture assignment (PLA) --individual assignment --written --submitted digitally (presumably before each weekly lecture, see study guide for details) --each pre-lecture assignment is graded as pass/fail/distinction --At least 4 out of 6 assignments need to be graded with 'pass' to pass the PLA assignment. --Passing the PLA assignment is required to pass the course. --BonusGrade: Receiving at least one distinction in the PLA counts toward a BonusGrade (see BonusGrade in the calculation of the final grade below.) (2) Two Reflections --individual assignment --written --submitted digitally (presumably, one Reflection to be completed during the first part of the course, another to be completed during the second part of the course, see study guide for details). --graded as pass/fail	

--'pass' on both reflections required to pass the Reflection assignment.
--Passing the Reflection assignment is required to pass the course.

(3) Exam 1

--individual
--closed book
--on campus
--covers the first part of the course (Epistemology of business analytics, see study guide for details)
--Graded on a scale from 1-10.
--Passing grade is 5.75.
--Exam 1 grade contributes 50% to the Final Grade.
--Passing Exam 1 is required to pass the course.

(4) Exam 2

--individual
--closed book
--on campus
--covers the second part of the course (Ethics of business analytics, see study guide for details)
--Graded on a scale from 1-10.
--Passing grade is 5.75.
--Exam 2 grade contributes 50% to the Final Grade.
--Passing Exam 2 is required to pass the course.

Calculation of the Final Grade:

Calculated Grade = IF (PLA = (Pass OR Distinction) AND Reflection = Pass AND Exam 1 $\geq$ 5.75 AND Exam 2 $\geq$ 5.75) THEN (0,5 * Exam 1 Grade + 0,5 * Exam 2 Grade) ELSE 5.

Bonus Grade = IF (Calculated Grade $\geq$ 5.75 AND PLA = Distinction) THEN (IF (10 - Calculated Grade $\leq$ 0.5) THEN (10 - Calculated Grade) ELSE 0.5) ELSE 0.

Final Grade = Calculated Grade + BonusGrade

A Final Grade of 5.75 is required to pass the course. The maximum Final Grade for the course is 10.

Re-Testing rules:

The results of PLA and Reflection assignments are valid only for the current year and are not transferable to the next academic year. In addition, re-take is not allowed for these assignments. A resit will be offered for Exam 1 and Exam 2. If a student fails the course, grades for Exam 1 and Exam 2 are transferable to the next academic year. Grades for Exam 1 and Exam 2 cannot be repaired - only a retake is possible.

Year	2023/2024
Organization	Applied Sciences
Education	Master Applied Physics

2nd semester MOT Modules

MOT113A	Technology Dynamics	5
Module Manager	Dr. C. Werker	
Responsible for assignments	Dr. C. Werker	
Co-responsible for assignments	Dr. C.C.R. Penna	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	In this course we discuss technical innovations and technological development as the result of human choices. Therefore, you learn how to identify the factors that determine these choices, i.e. institutions, behaviour and choices of innovative agents, their relationships, knowledge and collaboration, economic value as well as legitimacy and societal values. Based on that you will figure out how processes of socio-technological change take place as well as study how innovative agents can steer new innovations into a societally responsive direction. You study these issues by using the following theory-based topics: 1) The framework of innovation systems by discussing how innovative agents and institutions drive and shape them on the sectorial/industrial/technological level as well as on the regional, national or global level, in particular by figuring out which role management and policy can play in influencing innovation processes and technological change in different kinds of innovation systems. 2) Ways to integrate societal values into systems, particularly to find out how to make innovation responsible by using the responsible research and innovation systems approach.	
Study Goals	Students are expected to: 1. analyse innovations as the result of human choices, 2. identify the actions and interactions that are relevant to innovation processes, 3. analyse innovations in a specific technological and geographical context, 4. apply different concepts and methods for specific innovation processes, 5. reflect on the societal implications of innovation and technological change and the challenges that society poses to technology and industry, 6. give recommendations for steering an innovation process into a more societally responsive direction, 7. collect, identify and read relevant literature to do the above mentioned analysis. 8. write up the results of the analyses in a concise and clear way, 9. present these results to fellow students and lecturers, and 10. give feedback to other students.	
Education Method	We organize this course in the form of several workshops. Those workshops will contain lectures, mostly given in the form of blended learning, and interactive groups work. Students have to prepare the lectures and the group work in advance. We will also invite a few guest lectures.	
Assessment	The assessment will take place in the form of a written digital exam. The mark of the digital exam is the final mark for Technology Dynamics. Students can only take the written exam if - they are present during the workshops and actively participate in them, - they take the two mock exams and correct the mock exams of their fellow students assigned to them. We will organize the mock exams and their correction during the workshops. Students who do not meet the above criteria receive an extra-assignment in written form that they have to defend orally.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_1FcBgaGLXnjIP0G Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations.	

MOT122A	Digital Business Process Management	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr. J. Ubacht	
Instructor	Dr. F. Kleiman	
Instructor	Dr. Y. Ding	
Co-responsible for assignments	Dr. Y. Ding	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course is part of the MOT Master programs, but can be used as an elective. MOT students have priority when there are too many enrollments. Bachelor students are excluded from participation.	
Course Contents	Business processes are at the core of high-tech organizations and consist of a large number of activities aimed at creating value for customers. These processes can be supported or executed by humans or by digital technology. In the digital age business processes are supported by, enabled and changed by technologies like the Internet of Things, Artificial Intelligence, Blockchain, Robotics and so on. Business processes are situated at the point where the business side meets the IT side and where technology should fulfill customer needs. If business processes fail or do not adapt to the needs of customers the whole business might fail. Business Process Management (BPM) is necessary for ensuring that processes are operating in concert and that these processes are adaptable to changes in the overall environment of an organization. In this course the relationship between strategy, technology and business processes will be illustrated, explained and discussed. Methods and tools for business process analysis and improvement will be presented, including a number of statistical and architecture design methods. New solutions will be architected for high-tech organizations. The extent to which the nature of business processes is affected by contemporary technology constitutes the core of this course. In particular to analyze technologies and their impact on business process management and to architect new solutions in the organizational context of the firm. Topics - Aligning strategy and processes, value perspective, supply chains - Workflow, resources, BPM elements and control - Business process modeling (concepts, BPMN) + practical work - Process measurement, Balanced Score Card, Activity-based Costing (ABC) + practical work - Business process improvement strategies and approaches, path dependencies, capabilities, resource-based view, business process maturity - Improvement methodologies (LSS, lean, six sigma, theory of constraints, technology ..) - Statistical analyses and improvement + practical work - Compliance by design, build-in-controls in processes and technology - Business process automation and technology (BPM, ERP, workflow, SOA,...) - Business processes automation (case-based, process -based, use-based) - Agility and adaptability of BPM systems and the users, business process and agile enterprises - AI, Business and knowledge rules - Internet of Things, data-driven organizations	
Study Goals	The focus is on realizing business processes enabled by digital technology for high tech organizations. By the end of the course, you should be able to improve business processes using technology to realize organizational strategies. Learning objectives: 1. Analyze the relationship between business processes, strategy and technology 2. Model a business process using BPMN 3. Employ improvement methodologies to the selected business process 4. Assess if new technologies improves the chosen business process 5. Propose a new business process including technological and process improvements	
Education Method	-lectures -business process management game -practical work -assignments	
Computer Use	Planning tool. There are suitable planning tools like Trello and Asana BPMN modelling tool. There are several modelling tools which are appropriated like Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/) Activity Based Costing (ABS) calculations and simulations using a spreadsheets, like MS Excel.	
Literature and Study Materials	List of articles (containing classical papers and showing the state-of-the-art) Reader (containing an overview of main elements and explaining some methods) Exercises + answers (for practicing)	
Assessment	Students will collaborate in groups of 4-7 students to make an assignment which should contain 1) the analysis 2) improvement and digitalization of a business process and 3) implementation aspects. Students are free to select a process that is critical or a company and can be improved by ICT. The assignment consists of three parts 1. Analysis: Students select a company, analyze their strategy and select a critical business process. The existing process should be modeled (in BPMN) and the resulting product, stakeholders roles, customer needs and technology used should be analyzed. 2. Improvement: Based on the improvement methods and instruments a proposal for business process improvements should be made. 3. Realization: how the proposed business process should be realized by means of digital technology. The assignment should be handed in as a report. The report should adhere to the APA reference standards. The report cannot be updated for a higher grade after having handed in the report. The students are expected to prepare two presentations and some of the groups will be asked to present these during the lectures. Based on these presentations feedback will be given, which are relevant for all groups. Presentations will be assessed using a pass or fail. The report will be graded; this grade will reflect the final grade.	
Elective	Yes	
Tags	Business Design Information & Communication Software Technology	

MOT123A	Inter- and Intra-Organizational Decision-Making	5
Module Manager	J. Lieu	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. W.W. Veeneman	
Instructor	Dr. H.G. van der Voort	
Instructor	N.V. Prozorova	
Instructor	H.A. Sarpong	
Responsible for assignments	J. Lieu	
Co-responsible for assignments	Dr. L.D. Virla Alvarado	
Co-responsible for assignments	Dr. W.W. Veeneman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Organizations, whether large or small, public or private, old or new, make decisions on a daily basis. Organizations are comprised of individuals, and individuals can make around 35,000 decisions a day! What are some of the bases of individual decisions, how does this impact organisations and how does the wider societal context impact individual and organisation decisions? We may wish to believe that organisation decisions are well-planned and based on sufficient information. However, information is often incomplete and contested, and competing interests are involved, which make the decision process appear chaotic. The decision-makers, or individuals in organisations, also make these decisions based on their positionality, that is their social position and power. Decisions are also impacted by individual and societal biases which impact organisational decision-making This course aims to help you realize that individual and organisational decisions whether internal or among them cannot be made purely with a rational problem-solving approach. The teaching team will introduce you to the theories that describe how decisions are made in practice, and why they are made this way. You will learn strategies to manage multiple actors in the decision process, while consider wider societal implications.	
Study Goals	1. Explain key concepts and perspectives on decision-making within networks; 2. Identify individual and collective decision biases; 3. Reconstruct and evaluate the decision process surrounding a complex real-world problem; and 4. Recognize and design strategies used in multi-actor decisions	
Education Method	The course lecture and assignments are designed to facilitate reflective thinking and reflexivity (activating effortful or systems 2 thinking) surrounding decision-making. In the first hour of the lecture, we present core concepts based on the readings and there is time during and after the lecture to ask questions. It is important to do the required readings before the lecture as it builds the foundation for the materials discussed in class. The lecture will not cover all aspects of the required readings and students are expected to go in greater depth via the readings. After the lecture, there is short break, where you can rest your mind a bit as well as digest what was learnt. After the break, we reconvene and briefly discuss any remaining questions. The class is then divided into in different groups in different classrooms to work on a group assignment related to the lecture. In each classroom, you will work in smaller groups to answer the assignment question together. There is one facilitator in each classroom to guide you through the assignment and to help bring the different group insights together. The assignments provide more time to apply and make links between theories taught in the course and application in a real-world context. After completing the assignments in groups, you are asked to provide a short personal reflection and submit to bright space for a pass/fail course. For those who are unable to make it to the lectures/group assignment, we provide an alternative individual assignment.	
Computer Use	Students are required to submit an individual assignment at the end of each lecture period. This can be done via individual laptops or written and later uploaded onto Brightspace.	
Assessment	1. Short individual reflection: 15% to assess theoretical knowledge and motivate active participation and application of course materials (5 in-class group assignments worth 3% each). After each lecture you have time to discuss the lecture content through a group assignment with your peers (4-5 people). You are then asked to submit an individual short reflection based on your discussion with your peers. This is to be submit by the end of the day for a pass/fail grade. There will be a total of 6 assignments opportunities offered and only 5 will be counted. If you attend the 6th session you can get a bonus of 1%. For those who are unable to attend the group assignment, they can submit an individual reflection but this is not a pass/fail and is graded up to 3%. 2. Individual assignment (take home exam): 35% towards the end of the course, you are asked to combine your cumulative learning process in an individual assignment that critically reflects on how the course material has/failed to help you understand individual, intra- and interorganizational decision-making. Students have an option to re-take the individual assignment. 3. Group paper: 50% to assess the ability to apply theory to a real-world problem and to write a well-argued research paper in a team environment. The group sizes are between 4-5 students.	

MOT132A	Technology, Strategy and Entrepreneurship	5
Module Manager	Dr. G. van de Kaa	
Instructor	Prof.dr. C.P. van Beers	
Instructor	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Prof.dr. C.P. van Beers	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation	
Summary	The course Technology, Strategy and Entrepreneurship focuses on formulating and implementing technology strategy for large firms and entrepreneurs. However, before a manager can formulate a successful strategy, it is needed to understand the specifics of the external economic and societal environment in which the strategy will be implemented. Throughout the course, students will be acquainted with a variety of academic perspectives which are utilized by strategy and entrepreneurship researchers.	
Course Contents	Greater product and technology complexity and increased internationalization of innovation activities have increased costs and risks for innovators such that the boundaries of the firms have changed and are changing. Specifically, we explain the typical patterns of technological innovation in order to provide a foundation for the formulation of a technology strategy. Furthermore, we focus on identifying the reasons why most industries adopt a dominant design and why a particular firms technology is adopted as the dominant design. We will examine the factors that affect the outcome of standards battles. These include e.g. timing of entry strategies, collaboration strategies, and appropriability strategies. We will provide a basic background of entrepreneurship. Specifically, we will pay attention to networking strategies for new firms (entrepreneurs). How do these firms attempt to achieve success with their new technology, in a new market? As the boundaries of the firms are getting more transparent due to greater complexity of technical projects, it is required to decide whether to collaborate with external partners, and with whom. This can be a good option for entrepreneurial firms as they lack the necessary complementary assets to successfully commercialize a new technology. We will explore collaboration strategies for entrepreneurial firms. Technological innovation is a much more uncertain activity for a firm than other activities. Strategic thinking and formulation require knowledge on the boundaries of the firm and the markets in which the firm operates, its value chain and its core competencies. Also, strategy formulation aims at controlling the external environment of the firm, which is characterized by uncertainties. A distinction can be made in changes in the external environment in time or by location. General Purpose Technologies affect strategy formulation in time and frugal or inclusive innovations by location. After formulating the sources and innovation strategies we will pay attention to the resources firms have for innovation processes and specific valuation techniques, like real options, conjoint analysis and data envelopment analysis.	
Study Goals	After the course students are able to - understand the theoretical background of technology and innovation strategy. - position a variety of strategy topics thereby taking into account sectoral and country context differences. - understand the theoretical background of entrepreneurship. - apply key technology strategy models in different sectoral and country contexts. - understand the relations between those models. - analyse the industry dynamics of technological innovation. - analyse companies technology strategy. - analyse and explain a companies technological innovation strategy vis-a-vis its strategic or entrepreneurial posture - analyse and explain the organization and structure of new product development teams based on the firms technological innovation strategy	
Education Method	This is a blended course and the course includes regular sessions, games, and teaching cases. Each week will start with a spark which will be communicated through brightspace. On the first day the teacher will discuss the spark with you. On the fourth day of the week, a regular interactive lecture is planned. On the fifth and final day of the week quizzes are available. Each regular session will consist of lectures. In the teaching case sessions students discuss questions posed in assigned teaching cases that relate to the topics studied in the regular sessions. For the regular sessions students study the literature and for the teaching case sessions students prepare the teaching cases. In the teaching games students apply the theory by implementing and evaluating various strategies in a real life situation.	
Literature and Study Materials	Book: Schilling, M. A. 2019. Strategic Management of Technological Innovation. New York, USA, 6th edition Journal articles: Will be posted in Brightspace. Slides presented in class: Will be posted in Brightspace.	
Assessment	The final grade for this course is determined as follows: 1. Individual digital (ANS) exam on book, literature and lectures (70%). 2. quizzes (30%) It is possible to retake the quizzes. If the exam is not passed, the grade obtained for the quiz remains valid.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3EjZgyQcSYv7Fk Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT133A	High Tech Marketing	5
Module Manager	Dr. L. Rook	
Instructor	Dr. A.C. Smit	
Co-responsible for assignments	Dr. J.R. Ortt	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Companies and institutions in high-tech markets face constant uncertainties caused by market dynamics, technological progress and volatile customer preferences. In order to survive under such highly dynamic circumstances, companies must keep on innovating, and carefully select -- or modify -- appropriate high-tech marketing strategies and tactics. This course familiarizes students with several theoretical and methodological approaches to high-tech marketing, and captures recent developments -- i.e., the growing role of artificial intelligence, electronic and mobile commerce, product ownership vs. sharing economy, grassroots movements, and pressure groups. These issues are discussed under reference to new marketing realities determined by the interplay of technology, globalization and social corporate responsibility. Cause-related and social marketing approaches to high technology will be presented as alternative approaches to high-tech marketing.	
Study Goals	At the end of the course, students are expected to have gained knowledge on principles of high-tech marketing and will be able to recognize, analyze, reflect upon, and creatively respond to, high-tech marketing problems and challenges with regard to: - Marketing strategies and tactical marketing plans to deal with uncertainty in high-tech environments; - Customer segmentation, targeting, and positioning for business-to-business (B2B) and business-to-consumer (B2C) markets; - Product and service bundles development, (offline and online) word-of-mouth, advertising and promotion, incl. modern online tools; - Some qualitative and quantitative marketing research techniques (various methodologies) aimed at analyzing customer preferences for regular as well as persuasive marketing communications and technologies.	
Education Method	The course consists of lectures and marketing analytics workshops based on: - Theory discussions and interactive lectures based on the course materials; - Empirical analysis and evaluation in small project groups of existing (corporate) marketing strategy problems for a set of case brands in a societal or industry sector of choice. Analysis will be based on strategic assessment of the business case, and should be grounded in a customized marketing plan. The collection and use of qualitative and quantitative marketing methodologies is an obligatory component of the assignment, and should result in a creative proposal for a business-to-business (B2B) or business-to-consumer (B2C) marketing campaign to tap changing customer preferences. Recommendations should be made as to how the case brand is expected to cope with (some of) the most pressing high-tech marketing challenges.	
Literature and Study Materials	The materials for this course are: - A course book (t.b.a.). - Academic articles from leading journals in Marketing Research (Marketing Science, J. of Marketing, J. of Marketing Research, J. of Consumer Research) and in the Automation, Computing Machinery (ACM) research community to be accessed via the TU Library online repositories.	
Books	t.b.a.	
Assessment	Students will be graded on the quality and rigor of a marketing research project to be executed in group setting. Specifically: - A substantial group paper will (a) critically review the relevant literature on high-tech marketing to a relevant marketing case. At the heart of the group paper lies (b) empirical (qualitative and/or quantitative) analysis of a specific marketing problem, and (c) the adequate discussion thereof; analyses will inform (d) the development of a creative marketing campaign intended to solve the business challenge identified. - Each group paper must be supplemented with contributions (theoretical reflections) from each of the individual group members. - The research project may include peer review so as to determine the weight of individual contributions to the group work. - The assessment of the marketing research project will be based on a detailed grading rubric shared in advance. - A minimum final grade of 5.75 is a requirement for passing the course.	

MOT143A	Business Analytics	5
Module Manager	Dr. A.C. Smit	
Responsible for assignments	Dr. A.C. Smit	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Many businesses have large amounts of data at their disposal nowadays. This data can be structured or unstructured and derives from various sources, such as enterprise systems (e.g., customer data), social media (e.g., online reviews), mobile devices (e.g., sensor data), or physical ecosystems created by interconnected devices (e.g., smart thermostats). Harnessing this data to create a competitive advantage is imperative. Many have labelled data as the 'new oil' for businesses in this respect. Unleashing value from data remains a challenge for most businesses, however. Whereas data traditionally was used to support existing business activities, its more extensive volume and diversity have made it offers opportunities for value creation for businesses and customers alike. Consider, for example, the strategic value of online product review data generated by customers for new product development. Or think about the cost reductions a business can achieve by better predicting equipment failure. Dealing with the data that one needs for such applications does not only pose technical challenges (i.e., dealing with large data volumes, data quality issues, and complex data structures), it also requires a shift in the mind- and skillset of businesses when it comes to working with data. So, like oil, data must be unlocked and refined before it can live up to its potential. In this respect, the emerging discipline of business analytics offers valuable tools and techniques. This course offers a broad perspective on business analytics in the specific context of technology-driven companies. We will go through diverse business analytics projects starting from several business-related issues. In each project, the focus is on (1) problem understanding, (2) harnessing, processing, and storing (un)structured data, (3) selecting and using a machine learning technique that fits the problem, and (4) effectively presenting the knowledge and actionable insights that derive from the analytics project. Each project contains elements students can use in their business analytics group project. Simultaneously with describing the projects, aspects that touch on analytics projects will be discussed, such as data types, "big" data, data infrastructures, types of machine learning techniques, typical business analytics roles, responsible analytics, and visualization principles. We use the R programming language for executing each project and the group assignment. Business analytics is a broad and fast-developing discipline that is impossible to cover in one introductory course. Hence, the overall goal of this course is for students to get an idea of the overall business analytics landscape and lay a broad foundation by getting hands-on experience with several techniques. This foundation can then be further developed independently or through more advanced courses.	
Study Goals	After having followed this course, students can: - Describe common business analytics projects and select suitable analytic techniques; - Obtain, explore, and clean the data needed for a business analytics project; - Perform common analytic techniques conducted in a business context and interpret the results; - Effectively present the knowledge and insights that derive from a business analytics project.	
Education Method	Lectures, knowledge clips, tutorials (flipped classroom)	
Literature and Study Materials	Shmueli, G., Bruce, P. C., Gedeck, P., Yahav, I., & Patel, N. R. (2023). Machine Learning for Business Analytics. Concepts, Techniques and Applications in R. Hoboken, NJ: John Wiley & Sons, Inc.	
Assessment	- Final group assignment (80%). The grade for this assignment should be at least a 5.0. - Individual quizzes (20%): Four individual quizzes (1 hour each) take place. These quizzes are organized in weeks 4, 5, 6 and 7 of the course. The final grade for the quizzes is the average grade of the four quizzes. This final grade should be at least a 5.0. The course grade will be determined as the weighted average of the final grade for the quizzes and the grade for the group assignment. Students pass the course when this weighted average is at least 5.8. Partial grades (i.e. the final grade for the quizzes or the grade for the group assignment) can be transferred to the next academic year if they fulfil the requirement of being at least a 5.0.	

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